

Smoothed Picard Hamiltonian Monte Carlo

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Abstract

We develop a new low-accuracy sampler, called *smoothed Picard Hamiltonian Monte Carlo*, which combines Gaussian smoothing, Picard iteration, and higher-order discretization. For a log-concave target $\pi \propto \exp(-V)$ in dimension d satisfying $0 \prec \alpha I \preceq \nabla^2 V \preceq \beta I$, with condition number $\kappa := \beta/\alpha$, smoothed Picard HMC returns a sample with $\sqrt{\alpha} W_2(\cdot, \pi) \leq \varepsilon$ using $\tilde{O}(\kappa^2 + \kappa^{7/6} d^{1/6} / \varepsilon^{1/3})$ gradient queries. We also prove stronger W_q bounds, and then develop an algorithmic framework, the recursive warm start generator, to upgrade these W_q bounds to stronger divergence guarantees. This produces a warm start for the proximal bouncy particle sampler, introduced in a companion work, leading to a high-accuracy log-concave sampler with complexity $\tilde{O}((\kappa^{7/6} d^{1/6} + \kappa^{1/2} d^{1/4}) \text{polylog}(1/\varepsilon))$.

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1 Introduction

We study the problem of sampling from a target probability density $\pi \propto \exp(-V)$ over $\mathbb{R}^d$, where $V \in C^2(\mathbb{R}^d)$ satisfies, for $0 < \alpha \leq \beta$,

$$\alpha I \preceq \nabla^2 V(x) \preceq \beta I, \quad x \in \mathbb{R}^d, \quad \kappa := \frac{\beta}{\alpha}. \quad (1.1)$$

We design a new sampler that combines Gaussian smoothing of the density with a high-order discretization of Hamiltonian dynamics via Picard iteration and Chebyshev–Lobatto quadrature.

This combination gives a stable unadjusted Hamiltonian step even though the original potential is assumed to have only two derivatives.

Our first result is a low-accuracy sampler with Wasserstein guarantee.

Theorem 1.1 (W_2 guarantee). *Assume (1.1), let $0 < \varepsilon \leq 1/2$, and suppose that we are given a point x_{ref} with $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{\alpha d}$. Then, the smoothed Picard HMC algorithm (Algorithm 3.2) returns a sample from a probability law $\hat{\pi}$ satisfying*

$$\sqrt{\alpha} W_2(\hat{\pi}, \pi) \leq \varepsilon.$$

The expected number of gradient and proximal queries is at most

$$O\left(\left\{\kappa^2 + \frac{\kappa^{7/6} d^{1/6}}{\varepsilon^{1/3}}\right\} \log^4 \frac{\kappa d}{\varepsilon}\right).$$

We find the dimension dependence in this result to be quite surprising. Prior to this work, under (1.1), the best dimension dependence was $d^{1/3}$. This was achieved with the randomized midpoint discretization of the underdamped Langevin, with rate $\tilde{O}(\kappa d^{1/3}/\varepsilon^{2/3})$ in W_2 [SL19], and also the shifted ODE method [FLO21]. With the development of the shifted composition framework [ACZ26] and accelerated entropic hypocoercivity [Lu26, LL26], the rate was improved to $\tilde{O}(\kappa^{5/6} d^{1/3}/\varepsilon^{2/3})$ in KL. The closely related Poisson midpoint discretization of underdamped Langevin achieved an incomparable rate of roughly $\tilde{O}(\kappa^{7/6} d^{1/3}/\varepsilon^{1/3} + \kappa^{11/8} d^{3/8}/\varepsilon^{1/4})$ in W_2 [SN26]. More recently, exact simulation of the underdamped Langevin diffusion via FORS achieved $\tilde{O}(\kappa^{2/3} d^{1/3} \text{polylog}(1/\varepsilon))$ complexity [CCRZ26]. In contrast, Theorem 1.1 achieves dimension dependence $d^{1/6}$.

In a companion work [CCLZ26a], we introduce a new high-accuracy sampler, called the proximal bouncy particle sampler or *proximal BPS*, with $\tilde{O}(\kappa^{1/2} d^{1/4} \text{polylog}(1/\varepsilon))$ complexity from a warm start. One key application of smoothed Picard HMC is to provide an implementable warm start for that sampler.

Toward this end, we develop a novel algorithmic framework which we call the recursive warm start generator. It is designed as a wrapper around smoothed samplers—that is, samplers which target the Gaussian smoothed distribution $\pi * \mathcal{N}(0, \eta I)$ —and it upgrades transport metric guarantees into stronger divergence guarantees. Namely, with negligible overhead, it upgrades W_2 bounds into KL bounds; W_q bounds into mixed TV and Rényi bounds; and W_{ψ_2} bounds into genuine Rényi bounds. We apply this to smoothed Picard HMC by extending the arguments for Theorem 1.1 to produce stronger W_q guarantees, leading to the following warm start result.

Theorem 1.2 (Warm start). *Assume (1.1) and that we are given x_{ref} satisfying $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{\alpha d}$. For every $0 < \delta < 1/2$, the algorithm of Section 3.4 returns a sample from a law $\hat{\pi}_\delta$ with the following guarantee. There exists a law $\hat{\pi}_\delta^\dagger$ satisfying*

$$\text{TV}(\hat{\pi}_\delta, \hat{\pi}_\delta^\dagger) \leq \delta, \quad \text{R}_2(\hat{\pi}_\delta^\dagger \parallel \pi) \leq 1.$$

Moreover, the expected number of gradient queries is at most

$$O\left(\kappa^{7/6} d^{1/6} \log^7 \frac{\kappa d}{\delta}\right).$$

In Theorem 7.1, we prove a stronger statement that relaxes the assumption of strong log-concavity to a log-Sobolev inequality, and provides a KL divergence guarantee. Note that the recursive warm start generator also removes the κ^2 term from the complexity.

Finally, by using this warm start result to initialize proximal BPS [CCLZ26a], we obtain a new high-accuracy sampler which enjoys the following guarantee.

Theorem 1.3 (Warm start + proximal BPS). *Assume (1.1) and suppose that we are given a point x_{ref} with $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{\alpha d}$. For every $0 < \varepsilon < 1/4$, the warm start algorithm followed by the proximal BPS algorithm returns a sample from a law $\hat{\pi}$ satisfying*

$$\text{TV}(\hat{\pi}, \pi) \leq \varepsilon.$$

The expected number of gradient evaluations is

$$O\left(\kappa^{7/6} d^{1/6} \log^7 \frac{\kappa d}{\varepsilon} + \kappa^{1/2} d^{1/4} \log^{11/4} \frac{\kappa d}{\varepsilon}\right).$$

Prior to this work, the best high-accuracy sampler was exact diffusion simulation [CCRZ26], which achieved a complexity of $\tilde{O}(\kappa^{2/3} d^{1/3} \text{polylog}(1/\varepsilon))$. Hence, our new high-accuracy sampler has state-of-the-art dependence on the dimension, but not on the condition number.

Remark 1.4 (Further consequences). The proximal sampler reduction [LST21, CCSW22] converts general log-smooth sampling problems into well-conditioned log-concave subproblems. By invoking this framework, we could also state corollaries in other settings, such as when π is only assumed to satisfy a Poincaré inequality. Since this reduction is standard, we omit these extensions for brevity.

We give an overview of the main new ideas—both algorithmic and analytic—introduced in this work in the technical overview (Section 3).

Related work. This work advances the study of the complexity of log-concave sampling; an introduction to the subject and an overview of existing approaches can be found in [Che26].

Our use of Gaussian smoothing is reminiscent of score-based diffusion models [SDWGM15, SE19, HJA20, SSDK+21] and stochastic localization [Eld13], which also rely on score functions for Gaussian convolutions of a target distribution. Here, we do not learn score functions from data: we use queries to the unsmoothed score to construct nearly unbiased estimators of the smoothed score, and control their bias and fluctuations in the sampling analysis.

Hamiltonian Monte Carlo was introduced as hybrid Monte Carlo in lattice field theory [DKPR87]; see [Nea11] for its development in statistics. Early analyses quantified its favorable dimension dependence through asymptotic scaling results for product targets [BPR+13]. Subsequent works established non-asymptotic convergence guarantees for ideal HMC [LSV18, BREZ20, MS21, CV22]. More recent work has focused on accelerated convergence of randomized HMC and related piecewise deterministic processes, including convergence in relative entropy [LW22, MSWW26, MW26].

A substantial literature studies numerical discretization, both for unadjusted HMC [MV18, MS19, BRE23, BRM25, GLBMM25, BRMW26, ZAC26] and for Metropolized HMC [CDWY20, LST20, CGJ26, ZAC26]. Representative results achieve $d^{1/4}$ dimension dependence under quantitative third derivative bounds, such as structured or Frobenius-Lipschitz Hessian assumptions [MV18, CGJ26, ZAC26]. Higher-order integrators and polynomial collocation yield still better dimension dependence under stronger smoothness and structural assumptions [MS17, LSV18]. Our departure is to improve the dimension dependence *without* imposing higher-order smoothness on the original potential: Gaussian smoothing provides the regularity needed for accurate integration, while we control the error from estimating the smoothed score using queries to the unsmoothed score.

The recursive warm start framework builds heavily on the proximal sampler [LST21, CCSW22]. The main analytic mechanism for warm start generation is regularization along the heat flow, leading to reverse transport inequalities. Such inequalities are dual formulations of dimension-free Harnack inequalities, originating in Wang’s work [Wan97, Wan04]; see also [BGL15, AC25] for the transport formulation. Related regularization arguments have been used to generate algorithmic warm starts

with underdamped Langevin [AC24] and HMC [ZAC26], and to establish KL divergence guarantees via the shifted composition framework [AC26, ACZ26]. Coupling-based regularization has also yielded total variation bounds [BRE23] and divergence guarantees for unadjusted HMC [BRMW26]. These works analyze regularization for particular sampling dynamics. Our framework instead applies heat flow regularization to the output of any smoothed sampler, yielding a general route to KL, mixed TV and Rényi, or genuine Rényi warm starts.

AI usage. The main ideas underlying the algorithm and its analysis emerged through extensive interactions with GPT-5.6 Sol. The initial goal was to improve the dimensional dependence beyond $d^{1/3}$. In response, GPT proposed considering the smoothed potential $\widehat{V}_\eta := \mathbb{E}_{G \sim \mathcal{N}(0, I)} V(\cdot + \sqrt{\eta} G)$, which led to a sampler with dimensional dependence $d^{1/4}$. We subsequently suggested the more natural smoothing adopted in this paper and, through further refinements of the analysis and extensive interactions between the authors and GPT, obtained the improved dimensional dependence and the algorithms presented here. The authors independently checked the arguments, prepared the manuscript, and take full responsibility for its contents.

2 Preliminaries

Except in the generic warm start result [Theorem 6.1](#) and its consequence [Theorem 7.1](#), throughout the remainder of the paper $V \in C^2(\mathbb{R}^d)$ and [\(1.1\)](#) holds pointwise. For most of the paper, we normalize $\beta = 1$, so that $\kappa^{-1}I \preceq \nabla^2 V \preceq I$; this can be achieved by rescaling.

For simplicity of exposition, we initially allow ourselves access to proximal queries to V , in addition to queries to ∇V . For step size $a > 0$, one proximal query returns

$$\text{prox}_{aV}(y) := \operatorname{argmin}_{x \in \mathbb{R}^d} \left\{ V(x) + \frac{\|x - y\|^2}{2a} \right\}. \quad (2.1)$$

In this paper, we only ever need to call the proximal oracle for a β -smooth potential with step size at most $1/(2\beta)$. In this regime, the computation of prox_{aV} is a well-conditioned optimization problem and can be easily implemented via standard routines. In [Section D](#), we remove the use of proximal queries altogether.

The sampling analogue of the proximal operator is the restricted Gaussian oracle (RGO), which is the distribution

$$R_{a,y}(\mathrm{d}x) \propto \exp \left\{ -V(x) - \frac{\|x - y\|^2}{2a} \right\} \mathrm{d}x.$$

In [Section A](#), we show how to approximately implement it in $\widetilde{O}(1 + (a^2 d)^{1/3})$ queries, based on the exact diffusion sampler of [CCRZ26](#).

We assume that an admissible reference point is supplied:

$$\|\nabla V(x_{\text{ref}})\| \leq \sqrt{\alpha d}. \quad (2.2)$$

This reference point could be obtained via an optimization routine.

The total variation distance is $\text{TV}(P, Q) := \sup_A |P(A) - Q(A)|$. We also use

$$\text{KL}(P \parallel Q) := \int \log\left(\frac{\mathrm{d}P}{\mathrm{d}Q}\right) \mathrm{d}P, \quad W_2^2(P, Q) := \inf_{(X, Y)} \mathbb{E}\|X - Y\|^2,$$

with the usual value $+\infty$ for KL without absolute continuity; the infimum in the definition of W_2 is over all couplings of P and Q . For $q > 1$, define Rényi divergence by

$$R_q(P \parallel Q) := \frac{1}{q-1} \log \int \left(\frac{dP}{dQ} \right)^q dQ,$$

with value $+\infty$ when $P \not\ll Q$; this is the standard convention of [vEH14].

For matrices and tensors, $\|\cdot\|_{\text{op}}$ and $\|\cdot\|_{\text{HS}}$ denote the operator and Hilbert–Schmidt (or Frobenius) norms, respectively. For an r -tensor A on $\mathbb{R}^d$, this means

$$\|A\|_{\text{HS}}^2 = \sum_{j_1, \dots, j_r \in [d]} |A[e_{j_1}, \dots, e_{j_r}]|^2,$$

where $(e_j)_{j \in [d]}$ is any orthonormal basis of $\mathbb{R}^d$.

The notation $\tilde{O}(B)$ means B times a fixed universal power of a logarithm of the dimension, condition number, inverse accuracy, and any explicit confidence parameter. Throughout, $C, c > 0$ denote universal constants that may change from line to line.

3 Technical overview

Recall our convention that $\beta = 1$, so that $\kappa^{-1}I \preceq \nabla^2 V \preceq I$.

3.1 The smoothing trick

For a smoothing variance $\eta > 0$, write

$$\pi_\eta := \pi * \mathbf{N}(0, \eta I), \quad \Pi_\eta := \pi_\eta \otimes \mathbf{N}(0, I).$$

The **key algorithmic idea** of this paper is to pick a smoothing parameter $\eta > 0$ and to sample from the smoothed distribution π_η instead of the original target π . If $\text{Law}(Y) \approx \pi_\eta$, then we can recover a sample from π via the RGO: for $X \sim R_{\eta, Y}$, we have $\text{Law}(X) \approx \pi$.

The smoothed potential V_η , defined by $e^{-V_\eta} := e^{-V} * \mathbf{N}(0, \eta I)$, is given by

$$\nabla V_\eta(y) = \mathbb{E}[\nabla V(X) \mid X + \sqrt{\eta} G = y], \quad (3.1)$$

where $G \sim \mathbf{N}(0, I)$ and $X \sim \pi$ are independent. The smoothed potential admits quantitative higher-order derivative bounds, allowing us to apply higher-order integrators. The catch is that (3.1) is given as a conditional expectation, and is therefore not explicitly computable. Write $g_\eta := \nabla V_\eta$.

Therefore, we replace evaluations of ∇V_η with

$$\hat{g}_\eta(y; G) := \nabla V(\text{prox}_{\eta V}(y) + \sqrt{\eta} G), \quad G \sim \mathbf{N}(0, I). \quad (3.2)$$

The intuition is that $\nabla V_\eta(y) = \int \nabla V dR_{\eta, y}$, and $\text{Law}(\text{prox}_{\eta V}(y) + \sqrt{\eta} G) \approx R_{\eta, y}$, so that $\hat{g}_\eta(y; G)$ is almost an unbiased estimator of $\nabla V_\eta(y)$.

3.2 The higher-order integrator and the W_2 guarantee

Picard iteration. The numerical scheme we consider below is a discretization of an unadjusted generalized Hamiltonian sampler with partial Ornstein–Uhlenbeck (OU) momentum refreshment,

following the terminology of [GLBMM25]. Between momentum refreshments, we follow the Hamiltonian dynamics

$$\dot{X}_t = P_t, \quad \dot{P}_t = -\nabla V_\eta(X_t).$$

The dynamics can be written in integral form as

$$X_t = X_0 + tP_0 - \int_0^t (t-s) g_\eta(X_s) ds, \quad P_t = P_0 - \int_0^t g_\eta(X_s) ds. \quad (3.3)$$

Since the integral cannot generally be computed explicitly, we simulate it with Picard iteration. Let $(X_t^{[0]}, P_t^{[0]})_{t \geq 0}$ be a computable initial guess for the trajectory. The ideal Picard iteration maps this to an updated trajectory $(X_t^{[1]}, P_t^{[1]})_{t \geq 0}$ by applying (3.3) but with the non-linear term g_η evaluated at the preceding trajectory. That is,

$$X_t^{[1]} := X_0 + tP_0 - \int_0^t (t-s) g_\eta(X_s^{[0]}) ds, \quad P_t^{[1]} := P_0 - \int_0^t g_\eta(X_s^{[0]}) ds. \quad (3.4)$$

A particularly simple choice for the initial trajectory is $X_t^{[0]} := X_0 + tP_0$.

This describes one Picard iteration, and in principle one can iterate further. For the main results of this paper, it suffices to iterate up to depth 2. Higher depth iteration is treated in [Section E](#).

The ideal Picard iteration requires exact integration, which is not directly implementable. To develop an algorithm, we apply numerical integration, or quadrature.

Chebyshev–Lobatto quadrature. Fix an integer $J \geq 2$. The Chebyshev–Lobatto nodes on the interval $[0, h]$ are defined to be

$$t_j := \frac{h}{2} \left\{ 1 - \cos\left(\frac{j-1}{J-1} \pi\right) \right\}, \quad j = 1, \dots, J.$$

Associated with the nodes are certain interpolating polynomials ℓ_j , $j = 1, \dots, J$, characterized by $\ell_j(t_i) = \mathbf{1}_{\{i=j\}}$. Then, for every vector-valued function f defined at the nodes, let

$$(\mathcal{I}_{J,h}f)(t) := \sum_{j=1}^J f(t_j) \ell_j(t) \quad (3.5)$$

be its Lagrange interpolant; this is an approximation to $f(t)$ itself. Background material on Chebyshev–Lobatto quadrature is collected in [Section B](#).

To obtain a numerical scheme, we replace g_η in (3.4) with its interpolant. This produces the iterations

$$\begin{aligned} X_t^{[1]} &= X_0 + tP_0 - \int_0^t (t-s) (\mathcal{I}_{J,h}[g_\eta \circ X^{[0]}])(s) ds, & P_t^{[1]} &= P_0 - \int_0^t \mathcal{I}_{J,h}[g_\eta \circ X^{[0]}](s) ds, \\ X_t^{[2]} &= X_0 + tP_0 - \int_0^t (t-s) (\mathcal{I}_{J,h}[g_\eta \circ X^{[1]}])(s) ds, & P_t^{[2]} &= P_0 - \int_0^t \mathcal{I}_{J,h}[g_\eta \circ X^{[1]}](s) ds. \end{aligned}$$

Note that we only ever need to evaluate these paths at the nodes t_j . We can write the iterations in terms of the precomputed quantities

$$\omega_j := \int_0^h \ell_j(t) dt, \quad \omega_{i,j} := \int_0^{t_i} (t_i - t) \ell_j(t) dt, \quad 1 \leq i, j \leq J. \quad (3.6)$$

Finally, we replace g_η by the stochastic-gradient approximation $\hat{g}_\eta$ from (3.2). Thus, throughout this subsection, the algorithm has exact access to $\text{prox}_{\eta V}$. We replace these proximal queries by gradient queries in Section D.

This leads to the following algorithm.

Algorithm 3.1 (Smoothed Picard HMC). Initialize $X_{\text{init}} := x_{\text{ref}}$ and $P_{\text{init}} \sim \mathcal{N}(0, I)$. Given $(X_{\text{init}}, P_{\text{init}})$, one phase draws independent standard Gaussians $\zeta_0, \zeta_1, (G_j^{[0]}, G_j^{[1]})_{j=1}^J$, and performs the following steps.

1. Partially refresh the momentum and create the starting trajectory.

$$\text{Set } P_0 := e^{-h/2} P_{\text{init}} + \sqrt{1 - e^{-h}} \zeta_0 \text{ and } X_{t_j}^{[0]} := X_{\text{init}} + t_j P_0.$$

2. Perform the first Picard update:

$$X_{t_i}^{[1]} := X_{\text{init}} + t_i P_0 - \sum_{j=1}^J \omega_{i,j} \hat{g}_\eta(X_{t_j}^{[0]}; G_j^{[0]}).$$

3. Perform the second Picard update:

$$X_h^{[2]} := X_{\text{init}} + h P_0 - \sum_{j=1}^J \omega_{J,j} \hat{g}_\eta(X_{t_j}^{[1]}; G_j^{[1]}), \quad P_h^{[2]} := P_0 - \sum_{j=1}^J \omega_j \hat{g}_\eta(X_{t_j}^{[1]}; G_j^{[1]}).$$

4. Partially refresh the momentum and initialize the next phase.

$$\text{Set } X_{\text{init}} := X_h^{[2]} \text{ and } P_{\text{init}} := e^{-h/2} P_h^{[2]} + \sqrt{1 - e^{-h}} \zeta_1.$$

After the prescribed number of phases, return $Y := X_{\text{init}}$.

Finally, recall from Section 3.1 that once we have an approximate sample from the smoothed distribution π_η , we can recover an approximate sample from π by sampling from the RGO. We implement this last step via FORS, leading to the full algorithm.

Algorithm 3.2 (Smoothed Picard HMC with a terminal FORS step). Given the final position Y of Algorithm 3.1, independently run the terminal FORS routine of Theorem A.4 with target $R_{\eta, Y}$. Return its output X_{out} .

The exact guarantee is Theorem 1.1, and it is proven in Section 4. We next describe some of the ideas of the analysis at a high level in order to explain the source of the $d^{1/6}$ scaling. For the sake of exposition, we ignore the dependence on κ and ε .

Analysis. The analysis is based on the local error framework [MT21], which produces multi-step W_2 bounds via computations of “local” or one-step errors. The key is to decompose the one-step error into “deterministic” and “stochastic” components. Over the course of $\asymp h^{-1}$ iterations, the former is expected to accrue with a prefactor h^{-1} , whereas the latter only accrues with a prefactor $h^{-1/2}$ due to cancellations. In our analysis, the “deterministic” component consists of two terms: the *bias* of our stochastic gradient estimator, since $\mathbb{E} \hat{g}_\eta(\cdot; G) \neq g_\eta$, and the *discretization error* of the Picard integrator. The “stochastic” component consists of the fluctuations, or *variance*, produced by the stochastic gradient estimator. We handle these terms individually.

- **Discretization error.** Here, the error further consists of two parts: the quadrature error, and the error of the Picard iteration itself. They are controlled in Proposition 4.15.

- *Quadrature error.* It is classical that the error of approximating an integral via numerical quadrature depends strongly on the smoothness of the integrand. In our setting, this amounts to the smoothness of V_η . By studying Gaussian cumulants, we present a refined estimate showing that the size of $\|D^k V_\eta\|_{\text{HS}}$ is of order at most $\sqrt{d} \eta^{-(k-2)/2}$, for $k \geq 2$ (Lemma 4.3). From this bound, we are able to show that the quadrature error is bounded by a term involving $(h/\sqrt{\eta})^J$, where J is the number of Chebyshev–Lobatto nodes. By taking polylogarithmic J , this error is made negligible, provided $h < \sqrt{\eta}$. This shows that the natural scale is to take $h \asymp \sqrt{\eta}$, and we henceforth impose this choice in order to facilitate the interpretation of the subsequent bounds.
- *Picard error.* The error of Picard iteration decreases exponentially with the depth. We show that already for the depth-2 scheme, this error is of order $\sqrt{d} h^5$, which is negligible compared to the other error terms.
- **Bias.** Deterministically, we have $\|\mathbb{E} \hat{g}_\eta(\cdot; G) - g_\eta\| \leq \eta^{3/2} \sqrt{d}$ (Lemma 4.2). Propagating this through the dynamics, it is straightforward to show that the bias error is of order $\sqrt{d} (h\eta^{3/2} + \eta^{1/2} h^3) \asymp \sqrt{d} h^4$ (Proposition 4.9).
- **Variance.** As shown in [PW26], sharp analyses of stochastic gradient samplers should leverage the fact that *centered* perturbations of a standard Gaussian are closer in W_2 to the Gaussian than a naïve coupling argument would predict; in fact, in this case, the W_2 distance scales with the square of the size of the perturbation. We refer to this type of result as a second-order W_2 estimate. We extend the convolution bound of [PW26] to a more general setting in Lemma 4.11, based on a novel heat flow interpolation argument. We use this to prove a bound on the variance error term of order $\sqrt{d} (\eta h^{3/2} + \eta^{1/2} h^3) \lesssim \sqrt{d} h^{7/2}$ (Proposition 4.13).

After accumulating these errors through the local error framework, we find that the multi-step error is of order $\sqrt{d} h^3$, leading to the scaling $h \asymp d^{-1/6}$.

Finally, since smoothed Picard HMC step targets the smoothed distribution π_η , we need a terminal step in which we sample from the RGO R_η . As shown in Section A, when implemented with the exact diffusion sampler of [CCRZ26], the cost of this step is $\tilde{O}(1 + (\eta^2 d)^{1/3})$. At our scale $\eta \asymp d^{-1/3}$, this terminal step only costs $\tilde{O}(d^{1/9})$ queries, so it is not dominant.

3.3 Higher moment control

For our warm start construction, we need a stronger W_q guarantee, $q \geq 2$, for smoothed Picard HMC. We establish the following result.

Informal theorem (see Theorem 5.1 for precise statement). *Suppose that $\kappa^{-1}I \preceq \nabla^2 V \preceq I$ and we are given a reference point x_{ref} satisfying $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{d/\kappa}$. For any $q \geq 2$, $0 < \varepsilon \leq 1$, and fixed $c_0 > 0$, gradient-only smoothed Picard HMC can choose a smoothing variance $0 < \eta \leq c_0$ and return a law $\hat{\pi}_\eta$ satisfying $W_q(\hat{\pi}_\eta, \pi_\eta) \leq \varepsilon$, using*

$$O\left(\left\{\kappa^2 + \frac{\kappa^{4/3} (d+q)^{1/6}}{\varepsilon^{1/3}}\right\} \left\{q + \log \frac{e\kappa d}{\varepsilon}\right\}^6\right)$$

gradient queries in expectation, with an implicit constant depending only on c_0 .

This is taken up in Section 5 and follows the same outline as the analysis in Section 4, except that we need several L^q extensions of the ingredients from Section 4 which can be somewhat subtle. Note that this theorem only guarantees closeness to the smoothed distribution π_η , but this is all that is required for our warm start generator described in Section 3.4.

We highlight in particular that [Lemma 5.3](#) establishes an L^q extension of the classical local error framework, which could be useful for future works.

Note that when we upgrade the W_2 guarantee in [Theorem 1.1](#) to a W_q guarantee, we pay a price which is polynomial in q . However, for our warm start application, we only need to take q to be logarithmic in the problem parameters, so this turns out to be harmless.

3.4 Recursive warm start generation

Next, we develop a powerful recursive framework for warm start generation. Here, we suppose we have access to a sampling subroutine which we call a *smoothed sampler*. Given a strongly convex and smooth potential V , normalized to satisfy $\kappa^{-1}I \preceq \nabla^2 V \preceq I$, as well as an admissible reference point x_{ref} with $\|\nabla V(x_{\text{ref}})\| \leq \sqrt{d/\kappa}$, the smoothed sampler returns a smoothing level $0 \leq \eta \ll 1$ and a sample which is ε -close in law to $\pi_V * \mathcal{N}(0, \eta I)$, where $\pi_V \propto \exp(-V)$.

We remark that if V satisfies $\alpha I \preceq \nabla^2 V \preceq \beta I$, then we can apply the smoothed sampler to $V(\cdot/\sqrt{\beta})$, and then rescale the output by $\beta^{-1/2}$. After this rescaling, the reported smoothing level η corresponds to the law of the rescaled sample being close to $\pi_V * \mathcal{N}(0, \beta^{-1}\eta I)$.

We consider three types of guarantees for the smoothed sampler: the output law is close to the smoothed distribution in W_2 , in W_p (for $p \geq 2$), or in the sub-Gaussian Orlicz–Wasserstein metric W_{ψ_2} . In these three cases, by taking advantage of regularization along the heat flow, we develop a recursive sampler that upgrades the guarantee to KL, to mixed TV and R_q (as in the guarantee of [Theorem 1.2](#)), and to R_q , respectively.

For $V \in C^2(\mathbb{R}^d)$, $A > 0$, and $u \in \mathbb{R}^d$, consider the RGO potential and distribution

$$V_{A,u} := V + \frac{\|\cdot - u\|^2}{2A}, \quad R_{A,u}^V \propto \exp(-V_{A,u}).$$

The following recursive sampler aims to draw an approximate sample from the RGO distribution $R_{A,u}^V$. Note that when $A = \infty$, this just corresponds to $\pi \propto \exp(-V)$. The sampling task is easier for smaller A as $V_{A,u}$ then has a smaller condition number. The trick of the following algorithm is to recursively reduce the RGO sampling task to one with a smaller A , until the terminal level is reached.

Algorithm 3.3 (Recursive RGO sampler). Fix a terminal level $\underline{A} > 0$.

- If $A \leq \underline{A}$, then simply apply the localized FORS routine in [Theorem A.4](#) directly to $R_{A,u}^V$.
- Otherwise, proceed via the following steps.
 - Apply the smoothed sampler to $V_{A,u}$, and let $\eta > 0$ be the reported smoothing level, and let Z be the rescaled output. Thus, $\text{Law}(Z) \approx R_{A,u}^V * \mathcal{N}(0, \beta_A^{-1}\eta I)$, where β_A is the smoothness of $V_{A,u}$.
 - Pick $\tau > 0$, draw an independent $G \sim \mathcal{N}(0, I)$, and form

$$Y := Z + \sqrt{\frac{\tau}{\beta_A}} G, \quad a := \frac{\eta + \tau}{\beta_A}.$$

Thus, $\text{Law}(Y) \approx R_{A,u}^V * \mathcal{N}(0, aI)$.

- We wish to obtain an approximate sample from $R_{A,u}^V$. It suffices to sample from $R_{a,Y}^{V_{A,u}}$. Indeed, by the definition of the RGO, for any potential V , we have the identity $X \sim \pi_V$ if $Y \sim \pi_V * \mathcal{N}(0, aI)$ and $X \sim R_{a,Y}^V$.

In [Lemma 6.4](#), we justify the following closure property of the RGO: $R_{a,Y}^{V_{A,u}} = R_{A^+,u^+}^V$, where the updated parameters are

$$\frac{1}{A^+} := \frac{1}{A} + \frac{1}{a}, \quad u^+ := A^+ \left(\frac{u}{A} + \frac{Y}{a} \right).$$

Thus, we call the recursive RGO sampler on R_{A^+,u^+}^V , and return its output.

At this point, it may be unclear why we add an additional Gaussian to the output of the smoothed sampler. In fact, this is the key step which enables the use of regularization along the heat flow—specifically, in the form of reverse transport inequalities—to upgrade the smoothed sampler guarantee from Wasserstein to stronger distances. The guarantee for [Algorithm 3.3](#) is given in [Theorem 6.5](#).

We can apply [Algorithm 3.3](#) to sample from π by taking $A = \infty$. However, if the iteration complexity of the smoothed sampler has poor dependence on the condition number, then it is better to wrap the entire recursive scheme within the proximal sampler [[LST21](#), [CCSW22](#)].

Algorithm 3.4 (Proximal sampler wrapper). Given an initial sample X_0 , an outer variance $a_{\text{out}} > 0$, and a number N of outer steps, iterate as follows. At step $n \in \{0, \dots, N-1\}$, draw $Y_n = X_n + \mathbf{N}(0, a_{\text{out}}I)$, and set X_{n+1} to be the output of [Algorithm 3.3](#) with target R_{a_{out},Y_n}^V . Return the final iterate X_N .

This wrapper ensures that we only ever call the smoothed sampler on well-conditioned targets, leading to an overall improved condition number dependence. This aligns with the original use of the proximal sampler in [[LST21](#)]. The full guarantee for the recursive warm start generator is given in [Theorem 6.1](#). We emphasize that the guarantee is fully generic and applies to any smoothed sampler implementing our interface.

Instantiating the recursive warm start generator with smoothed Picard HMC as a subroutine then establishes our warm start guarantee in [Theorem 1.2](#).

4 Proof of the W_2 guarantee

In this section, our goal is to prove the W_2 guarantee ([Theorem 1.1](#)). However, since we will later need to obtain stronger moment estimates for [Theorem 1.2](#), we will first develop the part of the analysis that is common to both results.

4.1 Setup and smoothing estimates

For a positive-definite matrix M , write

$$W_{q,M}(P, Q) := \inf_{X \sim P, Y \sim Q} (\mathbb{E} \|X - Y\|_M^q)^{1/q}, \quad \|x\|_M := (x^\top M x)^{1/2}.$$

Recall that

$$\pi_\eta := \pi * \mathbf{N}(0, \eta I), \quad \Pi_\eta(dx dp) := \pi_\eta(dx) \mathbf{N}(0, I)(dp).$$

Let $X \mid Y = y$ be drawn from $R_{\eta,y}$. Recalling the definition (2.1) of the proximal operator, throughout this section we write

$$x_y^+ := \text{prox}_{\eta V}(y).$$

It is easy to check the identities

$$g_\eta(y) = \nabla V_\eta(y) = \eta^{-1} (y - \mathbb{E}[X \mid Y = y]), \quad \nabla^2 V_\eta(y) = \eta^{-1} I - \eta^{-2} \text{Cov}(X \mid Y = y). \quad (4.1)$$

Lemma 4.1 (Hessian bounds for the smoothed potential). *For $0 < \eta \leq 1$,*

$$\frac{1}{\kappa + \eta} I \preceq \nabla^2 V_\eta \preceq \frac{1}{1 + \eta} I \preceq I.$$

In particular g_η is 1-Lipschitz.

Proof. The RGO potential has Hessian between $(\kappa^{-1} + \eta^{-1}) I$ and $(1 + \eta^{-1}) I$. The Brascamp–Lieb covariance inequality [BL76] gives

$$\text{Cov}(X \mid Y = y) \preceq (\kappa^{-1} + \eta^{-1})^{-1} I.$$

For the reverse bound, the Cramér–Rao inequality (see, e.g., [Che26, Theorem 3.5.8]) gives

$$\text{Cov}(X \mid Y = y) \succeq (1 + \eta^{-1})^{-1} I.$$

Substitute both bounds in (4.1). □

The following is the key estimate that decomposes the stochastic gradient error into a bias term and a stochastic error term.

Lemma 4.2 (Stochastic gradient estimates). *Write*

$$\widehat{g}_\eta(y; G) := g_\eta(y) + b_\eta(y) + \xi_\eta(y; G), \quad \mathbb{E} \xi_\eta(y; G) = 0.$$

Then, for all u, y, y', G, G' ,

$$\|b_\eta(y)\| \leq \eta^{3/2} \sqrt{d}, \tag{4.2}$$

$$\mathbb{E} e^{\langle u, \xi_\eta(y; G) \rangle} \leq e^{\eta \|u\|^2 / 2}, \tag{4.3}$$

$$\|\widehat{g}_\eta(y; G) - \widehat{g}_\eta(y'; G)\| \leq \|y - y'\|, \tag{4.4}$$

$$\|\widehat{g}_\eta(y; G) - \widehat{g}_\eta(y; G')\| \leq \sqrt{\eta} \|G - G'\|. \tag{4.5}$$

Proof. Proximal optimality gives $\nabla V(x_y^+) = (y - x_y^+)/\eta$. Define

$$\rho_y(u) := V(x_y^+ + \sqrt{\eta}u) - V(x_y^+) - \sqrt{\eta} \langle \nabla V(x_y^+), u \rangle.$$

Then $0 \leq \nabla^2 \rho_y \preceq \eta I$, and under $R_{\eta, y}$, $(X - x_y^+)/\sqrt{\eta}$ has density proportional to $u \mapsto e^{-\|u\|^2/2 - \rho_y(u)}$. Call this law r_y . We have

$$W_2(r_y, \mathbf{N}(0, I)) \leq \sqrt{\mathbb{E}_{r_y}[\|\nabla \rho_y(U)\|^2]} \leq \eta \sqrt{\mathbb{E}_{r_y}[\|U\|^2]} \leq \eta \sqrt{d}. \tag{4.6}$$

Indeed, the first inequality combines Talagrand’s inequality for the standard Gaussian with the Gaussian log-Sobolev inequality; the second uses that $\nabla \rho_y(0) = 0$ and $\nabla \rho_y$ is η -Lipschitz; and the last follows by integration by parts since r_y is 1-strongly log-concave with its mode at 0.

Moreover,

$$g_\eta(y) = \nabla V(x_y^+) + \eta^{-1/2} \mathbb{E}_{r_y} \nabla \rho_y(U), \quad \widehat{g}_\eta(y; G) = \nabla V(x_y^+) + \eta^{-1/2} \nabla \rho_y(G).$$

The gradient of ρ_y is η -Lipschitz, so (4.6) gives (4.2). Since $\widehat{g}_\eta(y; \cdot)$ is $\sqrt{\eta}$ -Lipschitz, the standard Gaussian concentration inequality gives (4.3), and the same observation gives (4.5). Finally, $y \mapsto x_y^+$ is non-expansive and ∇V is 1-Lipschitz, proving (4.4). □

We will also need high-order regularity of the smoothed potential. The key point is a near-Gaussian cancellation, which is stronger than the generic moment bound for a strongly log-concave law. The Gaussian cumulant estimates used in the proof are collected in [Section C](#).

Lemma 4.3 (Derivatives of the smoothed potential). *For every integer $k \geq 2$,*

$$\sup_{y \in \mathbb{R}^d} \|D^k V_\eta(y)\|_{\text{HS}} \leq (k-1)! \sqrt{d} \eta^{-(k-2)/2}, \quad (4.7)$$

$$\sup_{y \in \mathbb{R}^d, \|v\|=1} \|D^k V_\eta(y)[v, \cdot^{k-1}]\|_{\text{HS}} \leq (k-1)! \eta^{-(k-2)/2}. \quad (4.8)$$

Consequently,

$$\|\nabla^2 V_\eta(y) - \nabla^2 V_\eta(y')\|_{\text{HS}} \leq 2\eta^{-1/2} \|y - y'\|. \quad (4.9)$$

Proof. Let r_y be the normalized RGO from the proof of [Lemma 4.2](#). Its potential has Hessian between I and $(1+\eta)I$. Let T_y be the optimal transport from the standard Gaussian to r_y . Caffarelli's contraction theorem [[Caf00](#)], applied in both directions, gives

$$(1+\eta)^{-1/2}I \preceq DT_y \preceq I, \quad \|DT_y - I\|_{\text{op}} \leq \eta/2, \quad (4.10)$$

almost everywhere.

By [\(C.7\)](#), for $k \geq 3$,

$$D^k V_\eta(y) = -\eta^{-k} \text{Cum}_k(X \mid Y = y).$$

Under the RGO, $X = x_y^+ + \sqrt{\eta} T_y(G)$, $G \sim \mathcal{N}(0, I)$. Translation invariance and homogeneity of cumulants, followed by [\(4.10\)](#) and [Lemma C.4](#) with $\varepsilon_T \leq \eta/2$, prove [\(4.7\)](#)–[\(4.8\)](#). The case $k = 2$ is [Lemma 4.1](#); integrating the $k = 3$ estimate [\(4.8\)](#) along a segment gives [\(4.9\)](#). $\square$

4.2 Local error analysis

The W_2 proof will be based on the local error framework.

Lemma 4.4 (Local error framework). *Let $P, \hat{P}$ be Markov kernels on $\mathbb{R}^m$. Suppose that, for every $x, y \in \mathbb{R}^m$, there is a coupling $X \sim \delta_x P$, $Y \sim \delta_y P$ such that*

$$\|X - Y\|_{L^2} \leq \rho \|x - y\|, \quad \|(X - x) - (Y - y)\|_{L^2} \leq \chi \|x - y\|. \quad (4.11)$$

Suppose also that, writing $\hat{X} \sim \delta_x \hat{P}$,

$$W_2(\delta_x \hat{P}, \delta_x P) \leq \mathcal{E}_{\text{strong}}, \quad \|\mathbb{E} \hat{X} - \mathbb{E} X\| \leq \mathcal{E}_{\text{weak}}.$$

Assume $0 < \rho < 1$. Then, for all probability laws μ, ν on $\mathbb{R}^m$,

$$W_2^2(\mu \hat{P}, \nu P) \leq \rho W_2^2(\mu, \nu) + \mathcal{E}_{\text{strong}}^2 + \frac{(\mathcal{E}_{\text{weak}} + \chi \mathcal{E}_{\text{strong}})^2}{\rho(1-\rho)}.$$

Proof. This is the standard local error theorem [[Che26](#), Lemma 5.1.2]. $\square$

We introduce three kernels and associated mappings.

- Let $\hat{K}$ denote the one-phase kernel defined by [Algorithm 3.1](#) with stochastic gradients $\hat{g}_\eta$. We can express the output of the phase as $\hat{\Phi}_z(\zeta, G)$ for Gaussian variables ζ, G .

- Define

$$\bar{\Phi}_z(\zeta) := \mathbb{E}[\widehat{\Phi}_z(\zeta, G) \mid \zeta], \quad \bar{K}(z, \cdot) := (\bar{\Phi}_z)_\# \mathbf{N}(0, I_{2d}).$$

- Finally, let K be the exact gradient numerical kernel obtained by replacing every stochastic gradient $\widehat{g}_\eta$ by g_η . Write Φ_z for the corresponding map, so that

$$K(z, \cdot) := (\Phi_z)_\# \mathbf{N}(0, I_{2d}).$$

It is standard that contraction for HMC is only witnessed after a change of coordinates. Set

$$M_\kappa := \begin{bmatrix} \frac{1}{2\kappa} + \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \otimes I_d.$$

Since $\kappa \geq 1$, this defines a metric which is uniformly equivalent to the Euclidean norm; such twisted metrics are standard in the analysis of kinetic dynamics; see, e.g., [CCBJ18, GLBMM25]. We apply the lemma after the linear change of coordinates $z \mapsto M_\kappa^{1/2} z$, with $P := K$ and $\widehat{P} := \widehat{K}$. The structure of the proof is given by the following key theorem.

Theorem 4.5 (Local error analysis in W_2). *Let $J \geq 2$, and assume that $h \leq c/\kappa$. Then, for every phase-space law $\widehat{\mu}$,*

$$W_{2, M_\kappa}^2(\widehat{\mu} \widehat{K}, \Pi_\eta) \leq (1 - ch/\kappa) W_{2, M_\kappa}^2(\widehat{\mu}, \Pi_\eta) + C \mathcal{E}_{\text{var}}^2 + \frac{C\kappa}{h} (\mathcal{E}_{\text{bias}}^2 + \mathcal{E}_{\text{disc}}^2),$$

where we define

$$\mathcal{E}_{\text{bias}} := \sup_{z, \zeta \in \mathbb{R}^{2d}} \|\bar{\Phi}_z(\zeta) - \Phi_z(\zeta)\|, \quad \mathcal{E}_{\text{var}} := \sup_{z \in \mathbb{R}^{2d}} W_2(\delta_z \widehat{K}, \delta_z \bar{K}), \quad \mathcal{E}_{\text{disc}} := W_2(\Pi_\eta K, \Pi_\eta).$$

Consequently,

$$W_{2, M_\kappa}^2(\widehat{\mu} \widehat{K}^N, \Pi_\eta) \leq e^{-cNh/\kappa} W_{2, M_\kappa}^2(\widehat{\mu}, \Pi_\eta) + \frac{C\kappa}{h} \mathcal{E}_{\text{var}}^2 + \frac{C\kappa^2}{h^2} (\mathcal{E}_{\text{bias}}^2 + \mathcal{E}_{\text{disc}}^2). \quad (4.12)$$

Proof. Fix $z, z' \in \mathbb{R}^{2d}$ and set $Z := \Phi_z(\zeta)$, $Z' := \Phi_{z'}(\zeta)$. Since $J \geq 2$ and $h \leq c/\kappa$, the assumption on h implies the hypotheses of Proposition 4.7 and Lemma 4.8 below. Those results verify (4.11) with (Z, Z') in place of (X, Y) . We can bound

$$\mathcal{E}_{\text{weak}} := \sup_{z \in \mathbb{R}^{2d}} \|\mathbb{E} \widehat{\Phi}_z(\zeta, G) - \mathbb{E} \Phi_z(\zeta)\|_{M_\kappa} \leq \sup_{z \in \mathbb{R}^{2d}} \mathbb{E} \|\bar{\Phi}_z(\zeta) - \Phi_z(\zeta)\|_{M_\kappa} \leq C \mathcal{E}_{\text{bias}},$$

and

$$\begin{aligned} \mathcal{E}_{\text{strong}} &:= \sup_{z \in \mathbb{R}^{2d}} W_{2, M_\kappa}(\delta_z \widehat{K}, \delta_z K) \leq \sup_{z \in \mathbb{R}^{2d}} \{W_{2, M_\kappa}(\delta_z \widehat{K}, \delta_z \bar{K}) + W_{2, M_\kappa}(\delta_z \bar{K}, \delta_z K)\} \\ &\leq C (\mathcal{E}_{\text{var}} + \mathcal{E}_{\text{bias}}), \end{aligned}$$

since the M_κ norm is equivalent to the standard Euclidean norm. Thus, the hypotheses of Lemma 4.4 hold with

$$\rho := 1 - \frac{ch}{\kappa}, \quad \chi := Ch, \quad \mathcal{E}_{\text{strong}} \leq C (\mathcal{E}_{\text{var}} + \mathcal{E}_{\text{bias}}), \quad \mathcal{E}_{\text{weak}} \leq C \mathcal{E}_{\text{bias}}.$$

We obtain

$$W_{2,M_\kappa}^2(\widehat{\mu}\widehat{K}, \Pi_\eta K) \leq \left(1 - \frac{ch}{\kappa}\right) W_{2,M_\kappa}^2(\widehat{\mu}, \Pi_\eta) + C\mathcal{E}_{\text{var}}^2 + \frac{C\kappa}{h} \mathcal{E}_{\text{bias}}^2.$$

By Young's inequality, for $\lambda > 0$,

$$W_{2,M_\kappa}^2(\widehat{\mu}\widehat{K}, \Pi_\eta) \leq (1 + \lambda) W_{2,M_\kappa}^2(\widehat{\mu}\widehat{K}, \Pi_\eta K) + (1 + \lambda^{-1}) W_{2,M_\kappa}^2(\Pi_\eta K, \Pi_\eta).$$

Uniform norm equivalence gives $W_{2,M_\kappa}(\Pi_\eta K, \Pi_\eta) \leq C\mathcal{E}_{\text{disc}}$. Choose $\lambda \asymp h/\kappa$. After reducing the universal constant in the contraction factor ch/κ , this proves the one-step bound; summing the recurrence proves the global bound. $\square$

The three errors $\mathcal{E}_{\text{bias}}$, $\mathcal{E}_{\text{var}}$, and $\mathcal{E}_{\text{disc}}$ correspond, respectively, to the bias and centered fluctuation of the stochastic gradient phase and to the discretization error of the numerical kernel. Thus, the main task is to bound these errors.

4.3 Properties of the exact-gradient kernel

Write $a := e^{-h/2}$, and for a symmetric matrix H , let

$$A_H := \begin{bmatrix} 0 & I \\ -H & -I \end{bmatrix}.$$

We use the integrated weight bounds in [Proposition B.1](#).

Lemma 4.6 (First-order difference expansion). *Couple draws $Z_1 \sim \delta_z K$ and $Z'_1 \sim \delta_{z'} K$ from phase-space inputs $z := (x_{\text{init}}, p_{\text{init}})$ and $z' := (x'_{\text{init}}, p'_{\text{init}})$ using the same Gaussian variables. Set*

$$\Delta_0 := z - z', \quad \Delta_1 := Z_1 - Z'_1.$$

If $h^2 \leq 1$, then

$$\Delta_1 = (I + hA_H)\Delta_0 + \mathcal{R}_h\Delta_0, \quad \|\mathcal{R}_h\|_{\text{op}} \leq Ch^2,$$

for a symmetric H with $\frac{1}{2\kappa} I \preceq H \preceq I$.

Proof. To verify this, write $\Delta x := x_{\text{init}} - x'_{\text{init}}$ and $\Delta p := p_{\text{init}} - p'_{\text{init}}$. The difference of gradients can be written as

$$g_\eta(x) - g_\eta(x') = H(x, x')(x - x'), \quad H(x, x') := \int_0^1 \nabla^2 V_\eta(x' + t(x - x')) dt.$$

By [Lemma 4.1](#), $\frac{1}{2\kappa} I \preceq H \preceq I$.

The first OU half-step changes Δp to $a\Delta p$. Thus,

$$\Delta X_{t_j}^{[0]} = \Delta x + at_j \Delta p.$$

Writing $H_j^{[0]} := H(X_{t_j}^{[0]}, X_{t_j}^{\prime[0]})$, the first Picard update gives

$$\Delta X_{t_i}^{[1]} = \Delta x + at_i \Delta p - \underbrace{\sum_{j \in [J]} \omega_{i,j} H_j^{[0]} (\Delta x + at_j \Delta p)}_{=: E_i}.$$

The bounds $\|H_j^{[0]}\|_{\text{op}} \leq 1$, $a \leq 1$, $t_j \leq h$, and $\sum_{j \in [J]} |\omega_{i,j}| \leq h^2$ (Proposition B.1) imply

$$\|E_i\| \leq \sum_{j \in [J]} |\omega_{i,j}| (\|\Delta x\| + at_j \|\Delta p\|) \leq h^2 (\|\Delta x\| + h \|\Delta p\|).$$

Then, set

$$H_j := H(X_{t_j}^{[1]}, X_{t_j}'^{[1]}), \quad H := h^{-1} \sum_{j \in [J]} \omega_j H_j.$$

Positivity and $\sum_{j \in [J]} \omega_j = h$ imply $\frac{1}{2\kappa} I \preceq H \preceq I$. The exact endpoint differences are

$$\Delta X_{\text{init}}^+ = \Delta x + ah \Delta p - \sum_{j \in [J]} \omega_{j,j} H_j \Delta X_{t_j}^{[1]}, \quad \Delta P_{\text{init}}^+ = a^2 \Delta p - a \sum_{j \in [J]} \omega_j H_j \Delta X_{t_j}^{[1]}.$$

Subtract $\Delta x + h \Delta p$ from the first line and $\Delta p - hH \Delta x - h \Delta p$ from the second. Use $a = 1 - h/2 + O(h^2)$, $a^2 = 1 - h + O(h^2)$, (B.2), and the preceding bound on E_i . Every remaining term is bounded by $Ch^2 \|(\Delta x, \Delta p)\|$, which proves the claimed expansion. $\square$

Proposition 4.7 (Contraction). *If $h \leq c/\kappa$, then, for all $z, z', \zeta \in \mathbb{R}^{2d}$,*

$$\|\Phi_z(\zeta) - \Phi_{z'}(\zeta)\|_{M_\kappa} \leq \left(1 - \frac{ch}{\kappa}\right) \|z - z'\|_{M_\kappa}. \quad (4.13)$$

Proof. Let q be a unit eigenvector of H with eigenvalue $\lambda \in [\frac{1}{2\kappa}, 1]$. The two-dimensional phase-space subspace $\{(uq, vq) : u, v \in \mathbb{R}\}$ is invariant under A_H . In the coordinates (u, v) , the restrictions of A_H and M_κ to this subspace are represented by

$$A_H(uq, vq) = (vq, -(\lambda u + v)q), \quad A_\lambda := \begin{bmatrix} 0 & 1 \\ -\lambda & -1 \end{bmatrix}, \quad M_\kappa^{(2)} := \begin{bmatrix} \frac{1}{2\kappa} + \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}.$$

A direct calculation gives

$$A_\lambda^\top M_\kappa^{(2)} + M_\kappa^{(2)} A_\lambda = - \begin{bmatrix} \lambda & \lambda - \frac{1}{2\kappa} \\ \lambda - \frac{1}{2\kappa} & 1 \end{bmatrix} \preceq -\frac{1}{8\kappa} M_\kappa^{(2)}. \quad (4.14)$$

Indeed, the positive matrix after the minus sign has trace at most 2 and determinant

$$\lambda - \left(\lambda - \frac{1}{2\kappa}\right)^2 \geq \frac{1}{2\kappa}.$$

Its smallest eigenvalue is therefore at least $1/(4\kappa)$. Since $M_\kappa^{(2)} \preceq 2I$, this proves (4.14). Decomposing along an orthonormal eigenbasis of H yields the full phase-space inequality

$$A_H^\top M_\kappa + M_\kappa A_H \preceq -\frac{1}{8\kappa} M_\kappa. \quad (4.15)$$

Apply Lemma 4.6. Uniform equivalence of the Euclidean and M_κ norms gives

$$\|\mathcal{R}_h \Delta_0\|_{M_\kappa} \leq Ch^2 \|\Delta_0\|_{M_\kappa}.$$

Expanding the square and applying (4.15) gives

$$\|\Delta_1\|_{M_\kappa}^2 \leq \{1 - ch/\kappa + Ch^2\} \|\Delta_0\|_{M_\kappa}^2.$$

The step size restriction absorbs the last term. Taking square roots and changing the universal constant gives (4.13). $\square$

Lemma 4.8 (Synchronous increment coupling). *If $h \leq c/\kappa$, then, for all $z, z', \zeta \in \mathbb{R}^{2d}$,*

$$\|\{\Phi_z(\zeta) - z\} - \{\Phi_{z'}(\zeta) - z'\}\|_{M_\kappa} \leq Ch \|z - z'\|_{M_\kappa}. \quad (4.16)$$

Proof. Subtract Δ_0 from the expansion in Lemma 4.6. Uniform equivalence of the Euclidean and M_κ norms, together with $\|H\|_{\text{op}} \leq 1$, gives $\|A_H \Delta_0\|_{M_\kappa} \leq C \|\Delta_0\|_{M_\kappa}$. Moreover, $h^2 \leq h$ under the step size assumption. Hence $\|\Delta_1 - \Delta_0\|_{M_\kappa} \leq Ch \|\Delta_0\|_{M_\kappa}$, which is (4.16). $\square$

4.4 Local error bounds

4.4.1 Bias bound

We first control the displacement between the conditional mean and exact gradient kernels. Put

$$\bar{g}_\eta(y) := \mathbb{E}\hat{g}_\eta(y; G) = g_\eta(y) + b_\eta(y).$$

Proposition 4.9 (Bias bound). *If $h \leq 1$, then*

$$\mathcal{E}_{\text{bias}} \leq C\sqrt{d} (h\eta^{3/2} + \eta^{1/2}h^3/\sqrt{J}). \quad (4.17)$$

Proof. Fix $z, \zeta \in \mathbb{R}^{2d}$ and write

$$B := \sup_{y \in \mathbb{R}^d} \|b_\eta(y)\| \leq \eta^{3/2}\sqrt{d},$$

by (4.2). To distinguish the two first Picard updates, let $X_{t_i}^{[1]}$ denote the iterate obtained with the exact gradient g_η , and let $\hat{X}_{t_i}^{[1]}$ denote the iterate obtained with $\hat{g}_\eta$. Taking the expectation over the first stochastic gradient block,

$$\bar{X}_{t_i}^{[1]} - X_{t_i}^{[1]} := \mathbb{E}[\hat{X}_{t_i}^{[1]} | \zeta] - X_{t_i}^{[1]} = - \sum_{j \in [J]} \omega_{i,j} b_\eta(X_{t_j}^{[0]}).$$

Consequently, (B.2) gives the estimate

$$\max_{j \in [J]} \|\bar{X}_{t_j}^{[1]} - X_{t_j}^{[1]}\| \leq h^2 B. \quad (4.18)$$

We also need to control the fluctuations:

$$\hat{X}_{t_i}^{[1]} - \bar{X}_{t_i}^{[1]} = - \sum_{j \in [J]} \omega_{i,j} \xi_\eta(X_{t_j}^{[0]}; G_j^{[0]}).$$

The variables in the sum are independent and centered, and (4.3) implies $\mathbb{E}\|\xi_\eta(y; G)\|^2 \leq \eta d$. Therefore,

$$\mathbb{E}[\|\hat{X}_{t_i}^{[1]} - \bar{X}_{t_i}^{[1]}\|^2 | \zeta] = \sum_{j \in [J]} \omega_{i,j}^2 \mathbb{E}[\|\xi_\eta(X_{t_j}^{[0]}; G_j^{[0]})\|^2 | \zeta] \leq \frac{5\pi^2 h^4}{128(J-1)} \eta d \leq \frac{Ch^4 \eta d}{J}. \quad (4.19)$$

It remains to propagate these errors through the endpoint update. The endpoint stochastic gradient block is independent of the first block, so for each j ,

$$\mathbb{E}[\hat{g}_\eta(\hat{X}_{t_j}^{[1]}; G_j^{[1]} | \zeta] = \mathbb{E}[\bar{g}_\eta(\hat{X}_{t_j}^{[1]} | \zeta].$$

Since $\widehat{g}_\eta(\cdot; G)$ is 1-Lipschitz for every G , so is $\bar{g}_\eta$. Using $\bar{g}_\eta = g_\eta + b_\eta$, followed by (4.18), (4.19), and Jensen's inequality, we obtain

$$\begin{aligned} \|\mathbb{E}[\bar{g}_\eta(\widehat{X}_{t_j}^{[1]} \mid \zeta) - g_\eta(X_{t_j}^{[1]})] &\leq \mathbb{E}[\|\widehat{X}_{t_j}^{[1]} - \bar{X}_{t_j}^{[1]}\| \mid \zeta] + \|\bar{X}_{t_j}^{[1]} - X_{t_j}^{[1]}\| + \|b_\eta(X_{t_j}^{[1]})\| \\ &\leq Ch^2\sqrt{\eta d/J} + (1 + h^2)B. \end{aligned}$$

The position and momentum components of $\bar{\Phi}_z(\zeta) - \Phi_z(\zeta)$ are obtained by weighting these gradient differences by $\omega_{J,j}$ and $a\omega_j$, respectively. Since

$$\sum_{j \in [J]} |\omega_{J,j}| \leq h^2, \quad a \sum_{j \in [J]} \omega_j = ah \leq h,$$

we conclude that

$$\begin{aligned} \|\bar{\Phi}_z(\zeta) - \Phi_z(\zeta)\| &\leq (h + h^2) \{Ch^2\sqrt{\eta d/J} + (1 + h^2)B\} \\ &\leq Ch(B + Ch^2\sqrt{\eta d/J}) \\ &\leq C\sqrt{d}(\eta^{3/2}h + \eta^{1/2}h^3/\sqrt{J}). \end{aligned}$$

The bound is uniform in z and ζ , proving (4.17). $\square$

4.4.2 Variance bound

Fix a phase-space input $z := (x_{\text{init}}, p_{\text{init}})$, and put $\sigma := (1 - e^{-h})^{1/2}$. Collect the OU variables as $\zeta := (\zeta_0, \zeta_1) \sim \mathcal{N}(0, I_{2d})$ and all Gaussian variables used by the stochastic gradients as G_{all} . Define

$$\mathcal{G} := h^{-1} \sum_{j \in [J]} \omega_j \widehat{g}_\eta(X_{t_j}^{[1]}; G_j^{[1]}), \quad \mathcal{G}' := h^{-2} \sum_{j \in [J]} \omega_{J,j} \widehat{g}_\eta(X_{t_j}^{[1]}; G_j^{[1]}).$$

Direct substitution in Algorithm 3.1 gives

$$\begin{bmatrix} X_{\text{init}}^+ \\ P_{\text{init}}^+ \end{bmatrix} = u + B_h \{\zeta + F(\zeta, G_{\text{all}})\},$$

where

$$u := \begin{bmatrix} x_{\text{init}} + ah p_{\text{init}} \\ a^2 p_{\text{init}} \end{bmatrix}, \quad B_h := \sigma \begin{bmatrix} hI & 0 \\ aI & I \end{bmatrix}, \quad F := \begin{bmatrix} -h\mathcal{G}'/\sigma \\ ah(\mathcal{G}' - \mathcal{G})/\sigma \end{bmatrix}.$$

Moreover, $h/\sigma \asymp \sqrt{h}$ and $\|M_\kappa^{1/2} B_h\|_{\text{op}} \leq C\sqrt{h}$. The purpose of this decomposition is to apply the refined second-order W_2 estimate in Lemma 4.11 below. First, we record some Lipschitz estimates.

Lemma 4.10 (Lipschitz bounds). *If $h^2 \leq 1$, then, uniformly in the input,*

$$\begin{aligned} \|F(\zeta, G) - F(\zeta', G)\| &\leq Ch^2 \|\zeta - \zeta'\|, \\ \|F(\zeta, G) - F(\zeta, G')\| &\leq C\sqrt{\eta h/J} \|G - G'\|. \end{aligned} \tag{4.20}$$

Proof. Consider first two OU inputs $\zeta := (\zeta_0, \zeta_1)$ and $\zeta' := (\zeta'_0, \zeta'_1)$, with the stochastic gradient variables fixed. The map F does not depend on ζ_1 . Moreover,

$$\|X_{t_j}^{[0]}(\zeta_0) - X_{t_j}^{[0]}(\zeta'_0)\| = \sigma t_j \|\zeta_0 - \zeta'_0\| \leq \sigma h \|\zeta_0 - \zeta'_0\|.$$

Using (4.4) and then (B.2) gives

$$\max_{j \in [J]} \|X_{t_j}^{[1]}(\zeta_0, G^{[0]}) - X_{t_j}^{[1]}(\zeta'_0, G^{[0]})\| \leq \sigma h \|\zeta_0 - \zeta'_0\| + h^2 \sigma h \|\zeta_0 - \zeta'_0\| \leq 2\sigma h \|\zeta_0 - \zeta'_0\|.$$

Since $\omega_j \geq 0$, $\sum_{j \in [J]} \omega_j = h$, and $\sum_{j \in [J]} |\omega_{J,j}| \leq h^2$, it follows that

$$\|\Delta \mathcal{G}\| \leq C\sigma h \|\Delta \zeta_0\|, \quad \|\Delta \mathcal{G}'\| \leq C\sigma h \|\Delta \zeta_0\|.$$

Multiplying by the factor h/σ in the definition of F proves (4.20).

Now fix ζ and compare two collections $G = (G^{[0]}, G^{[1]})$ and $G' = (G'^{[0]}, G'^{[1]})$. Cauchy–Schwarz, (4.5), and Lemma B.2 give

$$\max_{j \in [J]} \|\Delta X_{t_j}^{[1]}\| \leq Ch^2 \sqrt{\eta/J} \|\Delta G^{[0]}\|.$$

Use the Chebyshev–Lobatto weight bounds. Since the normalizations of $\mathcal{G}$ and $\mathcal{G}'$ are h^{-1} and h^{-2} , respectively, both satisfy

$$\|\Delta \mathcal{G}\| \vee \|\Delta \mathcal{G}'\| \leq C\sqrt{\eta/J} \{h^2 \|\Delta G^{[0]}\| + \|\Delta G^{[1]}\|\}.$$

Multiplying by $h/\sigma \leq C\sqrt{h}$ proves the second bound. $\square$

The key to our analysis is the following refined bound.

Lemma 4.11 (Second-order W_2 estimate). *Let $\mathcal{H}$ be a Hilbert space, let $Z \sim \mathbf{N}(0, I_m)$ and $G \sim \mathbf{N}(0, I_n)$ be independent, and let $f : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfy*

$$\|f(z, g) - f(z', g)\| \leq A \|z - z'\|, \quad \|f(z, g) - f(z, g')\| \leq B \|g - g'\|.$$

Assume that $f(z, g) - f(z, g')$ belongs to a fixed subspace $E \subseteq \mathbb{R}^m$, and let Π_E denote the orthogonal projection onto E . If $L : \mathbb{R}^m \rightarrow \mathcal{H}$ is linear and $f(Z) := \mathbb{E}[f(Z, G) \mid Z]$, then

$$W_2(\text{Law}\{L(Z + f(Z, G))\}, \text{Law}\{L(Z + \bar{f}(Z))\}) \leq \sqrt{m} \left\{ \frac{1}{2} \|L\Pi_E\|_{\text{op}} B^2 + 5 \|L\|_{\text{op}} AB \right\}. \quad (4.21)$$

It is instructive to compare this to a more trivial bound. The obvious coupling, together with the Gaussian Poincaré inequality applied conditionally on Z and over the subspace E , yields

$$\begin{aligned} \mathbb{E}[\|L(Z + f(Z, G)) - L(Z + \bar{f}(Z))\|^2] &= \mathbb{E}[\|L\Pi_E(f(Z, G) - \bar{f}(Z))\|^2] \\ &\leq \|L\Pi_E\|_{\text{op}}^2 \mathbb{E}[\|f(Z, G) - \bar{f}(Z)\|^2] \leq m \|L\Pi_E\|_{\text{op}}^2 B^2. \end{aligned}$$

This yields

$$W_2(\text{Law}\{L(Z + f(Z, G))\}, \text{Law}\{L(Z + \bar{f}(Z))\}) \leq \sqrt{m} \|L\Pi_E\|_{\text{op}} B.$$

This argument does not use any assumption on the distribution of Z . In contrast, Lemma 4.11 uses the fact that Z is Gaussian, together with the fact that the difference $f(Z, G) - \bar{f}(Z)$ is conditionally centered, to obtain a second-order bound which depends quadratically on A and B . The proof, given in Section C.3, is based on heat flow interpolation and Gaussian integration by parts. We remark that for the purpose of this paper, we only need the case $E = \mathbb{R}^m$.

Remark 4.12. Related second-order estimates are known for centered perturbations independent of the Gaussian variable being perturbed: see [SN26, Lemma 1] for bounded perturbations on a one-dimensional subspace, [PW26, Theorems 2.1 and 2.7] for more general Gaussian convolution inequalities, and the Ingster–Suslina second moment method [IS03] for divergence bounds. A complementary heat flow perspective is given by [CNW22, Theorem 2.1]: the first unmatched moment determines the asymptotic decay of W_2 under Gaussian smoothing. These independent perturbation results do not apply directly when $f(Z, G) - \bar{f}(Z)$ depends on the Gaussian Z providing the smoothing; the AB term in Lemma 4.11 accounts for this dependence.

With this in hand, we can now bound the variance term.

Proposition 4.13 (Variance bound). *Under $h^2 \leq 1$,*

$$\mathcal{E}_{\text{var}} \leq C\sqrt{d} (\eta h^{3/2}/J + \eta^{1/2} h^3/\sqrt{J}).$$

Proof. Apply [Lemma 4.11](#) with

$$m = 2d, \quad Z = \zeta, \quad G = G_{\text{all}}, \quad f = F, \quad L = B_h, \quad E = \mathbb{R}^{2d}.$$

By [Lemma 4.10](#), we may take

$$A = Ch^2 \quad \text{and} \quad B = C\sqrt{\eta h/J}.$$

Since $\|B_h\|_{\text{op}} \leq C\sqrt{h}$, [Eq. \(4.21\)](#) yields

$$\mathcal{E}_{\text{var}} \leq C\sqrt{d} \|B_h\|_{\text{op}} (B^2 + AB) \leq C\sqrt{d} (\eta h^{3/2}/J + \eta^{1/2} h^3/\sqrt{J}). \quad \square$$

4.4.3 Discretization error bound

Consider the Hamiltonian generator.

$$\mathcal{L}_{\text{H}} := \langle p, \nabla_x \rangle - \langle g_\eta(x), \nabla_p \rangle$$

For a position vector field $W = (W_i)_{i \in [d]}$, put $\delta_{\pi_\eta} W := \langle g_\eta, W \rangle - \text{div}_x W$. Integration by parts under e^{-V_η} gives the identity

$$\|\delta_{\pi_\eta} W\|_{L^2(\pi_\eta)}^2 \leq \|W\|_{L^2(\pi_\eta; \text{HS})}^2 + \|D_x W\|_{L^2(\pi_\eta; \text{HS})}^2. \quad (4.22)$$

Indeed, expanding the square and integrating the two cross terms produces a contraction of the derivative arrays and the quadratic form generated by $\nabla^2 V_\eta$; Cauchy-Schwarz and $0 \preceq \nabla^2 V_\eta \preceq I$ give the displayed bound.

The following lemma relies on Gaussian polynomial bounds collected in [Lemma C.5](#). In particular, if a Hilbert-valued momentum polynomial $F(p)$ and a polynomial vector field $W(p)$ have degree at most r , then

$$\|D_p F\|_{L^2(\gamma; \text{HS})} \leq \sqrt{r} \|F\|_{L^2(\gamma)}, \quad \|\delta_\gamma W\|_{L^2(\gamma)} \leq \sqrt{r+1} \|W\|_{L^2(\gamma; \text{HS})}. \quad (4.23)$$

Lemma 4.14 (Stationary iterated derivatives in L^2). *For every integer $k \geq 0$,*

$$\|\mathcal{L}_{\text{H}}^k g_\eta\|_{L^2(\Pi_\eta)} \leq C^{k+1} (k!)^{3/2} \sqrt{d} \eta^{-k/2}.$$

Proof. For a tensor-valued function F , define the weighted Sobolev norm

$$\|F\|_{\mathbb{W}^{N,2}} := \sum_{j=0}^N \frac{\eta^{j/2}}{j!} \|D_x^j F\|_{L^2(\Pi_\eta; \text{HS})}.$$

The weight ratio gives $\|D_x F\|_{\mathbb{W}^{N-1,2}} \leq N\eta^{-1/2} \|F\|_{\mathbb{W}^{N,2}}$. We next establish the corresponding bound for δ_{π_η} . For $0 \leq j \leq N-1$, the notation

$$\text{contr}(D_x^\ell g_\eta, D_x^{j-\ell} W) := \text{Sym} \left(\sum_{a \in [d]} D_x^\ell (g_\eta)_a \otimes D_x^{j-\ell} W_a \right)$$

denotes contraction over the position index a , followed by normalized symmetrization over the j derivative indices. The higher-order Leibniz rule [Nua06, Exercise 1.2.13] gives

$$D_x^j(\delta_{\pi_\eta} W) = \delta_{\pi_\eta}(D_x^j W) + \sum_{\ell=1}^j \binom{j}{\ell} \text{contr}(D_x^\ell g_\eta, D_x^{j-\ell} W).$$

By (4.22), the weighted sum of the first terms is bounded by

$$\begin{aligned} \sum_{j=0}^{N-1} \frac{\eta^{j/2}}{j!} \|\delta_{\pi_\eta}(D_x^j W)\|_{L^2(\Pi_\eta)} &\leq \sum_{j=0}^{N-1} \frac{\eta^{j/2}}{j!} (\|D_x^j W\|_{L^2(\Pi_\eta; \text{HS})} + \|D_x^{j+1} W\|_{L^2(\Pi_\eta; \text{HS})}) \\ &\leq \frac{2N}{\sqrt{\eta}} \|W\|_{\mathbf{W}^{N,2}}. \end{aligned}$$

For the remaining terms, (4.8) implies

$$\|\text{contr}(D_x^\ell g_\eta, D_x^{j-\ell} W)\|_{\text{HS}} \leq \ell! \eta^{-(\ell-1)/2} \|D_x^{j-\ell} W\|_{\text{HS}}.$$

Relative to the weight of $D_x^{j-\ell} W$, the coefficient of the term with ℓ derivatives on g_η is

$$\frac{\eta^{j/2}}{j!} \binom{j}{\ell} \ell! \eta^{-(\ell-1)/2} \Big/ \frac{\eta^{(j-\ell)/2}}{(j-\ell)!} = \sqrt{\eta}.$$

After setting $s = j - \ell$, each fixed position derivative $D_x^s W$ appears for at most $N - 1 - s \leq N$ choices of ℓ . Thus, the commutator contribution is at most $N\sqrt{\eta} \|W\|_{\mathbf{W}^{N,2}}$. Combining the two contributions, using $0 < \eta \leq 1$ and $N \geq 1$, yields

$$\|\delta_{\pi_\eta} W\|_{\mathbf{W}^{N-1,2}} \leq \frac{CN}{\sqrt{\eta}} \|W\|_{\mathbf{W}^{N,2}}.$$

It remains to control the Hamiltonian generator. One can check that

$$\mathcal{L}_H F = \delta_\gamma(D_x F) - \delta_{\pi_\eta}(D_p F). \quad (4.24)$$

Suppose that F has momentum degree at most r . Since D_x commutes with δ_γ , the Gaussian divergence estimate in (4.23) gives

$$\|\delta_\gamma(D_x F)\|_{\mathbf{W}^{N-1,2}} \leq \sqrt{r+1} \|D_x F\|_{\mathbf{W}^{N-1,2}} \leq \frac{N\sqrt{r+1}}{\sqrt{\eta}} \|F\|_{\mathbf{W}^{N,2}}.$$

For the second term, the Sobolev divergence estimate above and the Gaussian gradient estimate in (4.23) yield

$$\|\delta_{\pi_\eta}(D_p F)\|_{\mathbf{W}^{N-1,2}} \leq \frac{CN}{\sqrt{\eta}} \|D_p F\|_{\mathbf{W}^{N,2}} \leq \frac{CN\sqrt{r}}{\sqrt{\eta}} \|F\|_{\mathbf{W}^{N,2}}.$$

Combining the last two displays proves

$$\|\mathcal{L}_H F\|_{\mathbf{W}^{N-1,2}} \leq \frac{CN\sqrt{r+1}}{\sqrt{\eta}} \|F\|_{\mathbf{W}^{N,2}}. \quad (4.25)$$

Finally, integration by parts gives $\|g_\eta\|_{L^2(\pi_\eta)}^2 = \mathbb{E}_{\pi_\eta} \Delta V_\eta \leq d$, while (4.7) gives

$$\|g_\eta\|_{\mathbf{W}^{N,2}} \leq \sqrt{d} + N\sqrt{\eta d} \leq (N+1)\sqrt{d}.$$

Since $\mathcal{L}_H^j g_\eta$ has momentum degree at most j , iterating the bound (4.25) for $j = 0, \dots, k-1$ produces $k!\sqrt{k!}$. The extra factor $k+1$ from the initialization is absorbed into C^{k+1} , proving the claim. $\square$

Let $(X_t, P_t)_{t \geq 0}$ be the exact Hamiltonian flow and put $g(t) := g_\eta(X_t)$ when the initial state has law Π_η . Invariance and [Lemma 4.14](#) give

$$\sup_{0 \leq t \leq h} \|g^{(k)}(t)\|_{L^2} \leq C^{k+1} (k!)^{3/2} \sqrt{d} \eta^{-k/2}. \quad (4.26)$$

Proposition 4.15 (Discretization error bound). *If $h^2 \leq 1$, then*

$$\mathcal{E}_{\text{disc}} \leq C\sqrt{d} \left[B_{J,2} h \left(\frac{h}{\sqrt{\eta}} \right)^J + h^5 \right]. \quad (4.27)$$

Here, one may take $B_{J,2} = C_0^{J+1} \sqrt{J!}$, where C_0 is universal.

Proof. For an $L^2(\Pi_\eta; \mathbb{R}^d)$ -valued curve f on $[0, h]$, write

$$\|f\|_{\infty,2} := \sup_{0 \leq t \leq h} \|f(t)\|_{L^2}.$$

In particular, we can identify the position coordinates X , $X^{[0]}$, $X^{[1]}$ of the Hamiltonian flow and Picard iterations, respectively, as well as g , as $L^2(\Pi_\eta; \mathbb{R}^d)$ -valued curves.

Apply the Banach-valued interpolation remainder in [Lemma B.4](#) to the curve $g : [0, h] \rightarrow L^2(\Pi_\eta; \mathbb{R}^d)$. Together with (4.26), it gives

$$\begin{aligned} \mathcal{E}_{J,2} &:= \sup_{0 \leq t \leq h} \|g(t) - \mathcal{I}_{J,h} g(t)\|_{L^2} \leq \frac{h^J}{J!} \sup_{0 \leq t \leq h} \|g^{(J)}(t)\|_{L^2} \\ &\leq C^{J+1} \sqrt{J!} \sqrt{d} \left(\frac{h}{\sqrt{\eta}} \right)^J \leq B_{J,2} \sqrt{d} \left(\frac{h}{\sqrt{\eta}} \right)^J. \end{aligned}$$

Since the exact flow is stationary under Π_η , integration by parts gives

$$\|g_\eta \circ X\|_{\infty,2}^2 = \mathbb{E}_{\pi_\eta} \|g_\eta\|^2 = \mathbb{E}_{\pi_\eta} \Delta V_\eta \leq d.$$

Therefore, the integral equation for X shows that

$$\|X^{[0]} - X\|_{\infty,2} \leq \sqrt{d} \frac{h^2}{2}. \quad (4.28)$$

To control the first Picard iteration, insert the interpolant of the force along the exact trajectory:

$$\begin{aligned} X_t^{[1]} - X_t &= \int_0^t (t-s) \{g_\eta(X_s) - \mathcal{I}_{J,h}[g_\eta \circ X^{[0]}](s)\} ds \\ &= \int_0^t (t-s) \{g_\eta(X_s) - \mathcal{I}_{J,h}[g_\eta \circ X](s)\} ds + \int_0^t (t-s) \mathcal{I}_{J,h}[g_\eta \circ X - g_\eta \circ X^{[0]}](s) ds. \end{aligned}$$

The proof of [Lemma B.2](#) applies to $k = (t - \cdot)_+$ for every $t \in [0, h]$, so its consequence is uniform in t . Since g_η is 1-Lipschitz, (4.28) gives

$$\|X^{[1]} - X\|_{\infty,2} \leq \frac{h^2}{2} \mathcal{E}_{J,2} + h^2 \|X^{[0]} - X\|_{\infty,2} \leq \frac{h^2}{2} \mathcal{E}_{J,2} + \frac{\sqrt{d} h^4}{2}.$$

Set $(\hat{X}_h, \hat{P}_h) = (X_h^{[2]}, P_h^{[2]})$. At the momentum endpoint, positivity and the sum of the weights give

$$\|\hat{P}_h - P_h\|_{L^2} \leq h \mathcal{E}_{J,2} + h \|X^{[1]} - X\|_{\infty,2}$$

$$\leq (1 + h^2/2) h \mathcal{E}_{J,2} + \sqrt{d} h^5/2 \leq C \mathcal{E}_{J,2} h + C \sqrt{d} h^5.$$

Similarly,

$$\begin{aligned} \|\widehat{X}_h - X_h\|_{L^2} &\leq \frac{h^2}{2} \mathcal{E}_{J,2} + h^2 \|X^{[1]} - X\|_{\infty,2} \\ &\leq \frac{h^2}{2} (1 + h^2) \mathcal{E}_{J,2} + \sqrt{d} h^6/2 \leq C \mathcal{E}_{J,2} h^2 + C \sqrt{d} h^6. \end{aligned}$$

In the last lines, we used $h^2 \leq 1$.

Finally, start both the numerical phase and an exact Hamiltonian phase from Π_η , and use the same Gaussian variables in the two OU refreshes. The first refresh preserves Π_η , so the preceding stationary bounds apply. The exact Hamiltonian flow and the final refresh also preserve Π_η . In the final refresh the common Gaussian increment cancels, while the momentum difference is multiplied by $e^{-h/2}$. Consequently, using also $h \leq 1$, which follows from $h^2 \leq 1$ and $h > 0$,

$$\mathcal{E}_{\text{disc}} \leq \|\widehat{X}_h - X_h\|_{L^2} + \|\widehat{P}_h - P_h\|_{L^2} \leq C \mathcal{E}_{J,2} h + C \sqrt{d} h^5 \leq C \sqrt{d} \left[B_{J,2} h \left(\frac{h}{\sqrt{\eta}} \right)^J + h^5 \right].$$

This proves (4.27). $\square$

4.5 Proof of Theorem 1.1

We record the following standard initialization lemma.

Lemma 4.16 (Initialization). *Let $\widehat{\mu}_0 := \delta_{x_{\text{ref}}} \otimes \mathbf{N}(0, I)$. For every $q \geq 2$,*

$$W_{q, M_\kappa}(\widehat{\mu}_0, \Pi_\eta) \leq C \sqrt{\kappa(d+q)}.$$

Proof. Let $x_\star$ minimize V . Strong monotonicity and the assumption on $\|\nabla V(x_{\text{ref}})\|$ give

$$\|x_{\text{ref}} - x_\star\| \leq \kappa \|\nabla V(x_{\text{ref}})\| \leq \sqrt{\kappa d}.$$

If $X \sim \pi$, concentration from strong log-concavity [BGL14, Chapter 5] and integration by parts give $\|X - x_\star\|_{L^q} \leq C \sqrt{\kappa(d+q)}$. A draw from π_η is given by $X + \sqrt{\eta} G$, and $\sqrt{\eta} \|G\|_{L^q} \leq C \sqrt{\kappa(d+q)}$ because $\eta \leq 1 \leq \kappa$. Coupling the initial and stationary momenta identically, the claim follows from the triangle inequality and $M_\kappa \leq CI$. $\square$

The following lemma handles the terminal step.

Lemma 4.17 (Terminal RGO step in W_2). *Suppose $\kappa^{-1}I \preceq \nabla^2 V \preceq I$. For every $\eta > 0$, write $\pi_\eta := \pi * \mathbf{N}(0, \eta I)$ and $R_\eta(y, dx) := R_{\eta, y}(dx)$. Then, for every $y, y' \in \mathbb{R}^d$,*

$$W_2(R_{\eta, y}, R_{\eta, y'}) \leq \frac{\kappa}{\kappa + \eta} \|y - y'\| \leq \|y - y'\|.$$

Moreover, $\pi_\eta R_\eta = \pi$. Consequently, for any center laws μ, μ' ,

$$W_2(\mu R_\eta, \mu' R_\eta) \leq W_2(\mu, \mu').$$

Proof. These are the standard invariance and contraction properties of the restricted Gaussian oracle; see [LST21, CCSW22]. $\square$

The proof of [Theorem 1.1](#) is completed by plugging the bias, variance, and discretization error bounds from [Section 4.4](#) into (4.12) and performing some bookkeeping.

Proof of Theorem 1.1. Throughout, we assume access to exact proximal queries; this assumption can be removed via the arguments in [Section D](#). After the change of variables $z = \sqrt{\beta}x$, it suffices to work under $\beta = 1$ and prove $W_2 \leq \sqrt{\kappa}\varepsilon$. Choose

$$\mathfrak{L} := \log \frac{C\kappa d}{\varepsilon}, \quad J := \lceil C\mathfrak{L}^2 \rceil, \quad h := \frac{c}{\mathfrak{L}} \min \left\{ \frac{1}{\kappa}, \frac{\varepsilon^{1/3}}{\kappa^{1/6}d^{1/6}} \right\}, \quad N := \left\lceil \frac{C\kappa\mathfrak{L}}{h} \right\rceil,$$

where $c, C > 0$ are universal constants; the following assertions hold after adjusting these constants appropriately. Here, $\mathfrak{L}$ is introduced to streamline the bookkeeping of logarithmic terms.

Let $\theta > 0$, to be chosen below, and choose $\eta > 0$ so that $h^2/\eta = \theta$. The explicit choice of $B_{J,2}$ in [Proposition 4.15](#) gives

$$B_{J,2}\theta^{J/2} = C_0^{J+1}\sqrt{J!}\theta^{J/2} \leq C_0^{J+1}(J\theta)^{J/2} \leq e^{-c_1J} \leq h^4, \quad (4.29)$$

provided that $\theta \leq c\mathfrak{L}^{-2}$. We henceforth choose $\theta := c/\mathfrak{L}^2$. Then, $J \geq 2$ and $h \leq c/\kappa$, so the restrictions in [Propositions 4.7](#) and [4.15](#) hold, and moreover $\eta \leq 1$.

Take square roots in (4.12) and substitute the local bounds from [Propositions 4.9](#), [4.13](#) and [4.15](#). Together with [Lemma 4.16](#), this gives

$$\begin{aligned} W_{2,M_\kappa}(\hat{\mu}_N^{\text{ph}}, \Pi_\eta) &\leq C\sqrt{\kappa d}e^{-cNh/\kappa} + C\sqrt{\kappa d}(\eta h/J + \eta^{1/2}h^{5/2}/\sqrt{J}) \\ &\quad + C\kappa\sqrt{d}(\eta^{3/2} + \eta^{1/2}h^2/\sqrt{J} + h^4), \end{aligned}$$

where $\hat{\mu}_N^{\text{ph}}$ is the final phase-space law. The Picard iteration error term is included in the last monomial by (4.29). Since $\eta = h^2/\theta$, the preceding bounds on h, η show that the sum of the last five terms is at most $\sqrt{\kappa}\varepsilon/4$ for c appropriately small. This bookkeeping is uniform over $0 < \varepsilon \leq 1$. The definition of N makes the initial term at most $\sqrt{\kappa}\varepsilon/4$.

By [Lemma 4.17](#),

$$W_2(\hat{\mu}_N R_\eta, \pi) = W_2(\hat{\mu}_N R_\eta, \pi_\eta R_\eta) \leq W_2(\hat{\mu}_N, \pi_\eta) \leq C W_{2,M_\kappa}(\hat{\mu}_N^{\text{ph}}, \Pi_\eta) \leq \frac{\sqrt{\kappa}\varepsilon}{2},$$

after adjusting universal constants. Let $\hat{R}_{\eta,y}$ be the conditional output law of the terminal FORS routine in [Algorithm 3.2](#), and set

$$\delta_{\text{term}} := \min \left\{ \frac{1}{2}, \frac{\kappa\varepsilon^2}{8\eta} \right\}.$$

By [Theorem A.4](#) with $q = 2$ and $\varepsilon^2 := \min\{1/4, \delta_{\text{term}}\}$, uniformly in y , we can ensure that

$$\text{KL}(\hat{R}_{\eta,y} \parallel R_{\eta,y}) \leq R_2(\hat{R}_{\eta,y} \parallel R_{\eta,y}) \leq \delta_{\text{term}}.$$

The RGO potential is $(\kappa^{-1} + \eta^{-1})$ -strongly convex. Talagrand's inequality therefore gives

$$W_2^2(\hat{R}_{\eta,y}, R_{\eta,y}) \leq \frac{2\delta_{\text{term}}}{\kappa^{-1} + \eta^{-1}} \leq 2\eta\delta_{\text{term}} \leq \frac{\kappa\varepsilon^2}{4}.$$

The triangle inequality then yields $W_2(\hat{\mu}_N \hat{R}_\eta, \pi) \leq \sqrt{\kappa}\varepsilon$.

Each phase uses $2J$ gradients and $2J$ proximal calls. Substitution of N gives

$$2JN \leq C \left\{ \kappa^2 + \frac{\kappa^{7/6}d^{1/6}}{\varepsilon^{1/3}} \right\} \mathfrak{L}^4$$

calls of each type. By [Theorem A.4](#), the terminal routine has expected cost at most $C\mathfrak{L}^{5/3}(1 + \varepsilon^{4/9}d^{1/9}/\kappa^{2/9})$. Any fixed power of $\mathfrak{L}$ is dominated by a positive power of $\kappa d/\varepsilon$. Hence this cost is dominated by the Picard HMC bound for $\kappa, d \geq 1$ and $0 < \varepsilon \leq 1$. This proves the theorem. $\square$

5 W_q analysis

For our warm start application, we need to upgrade the W_2 guarantee to a W_q guarantee, for $q \geq 2$. We organize the proof in parallel with [Section 4](#), and along the way, we describe the new ingredients needed for this part of the analysis.

5.1 W_q guarantee

Theorem 5.1 (W_q guarantee). *Fix any $c_0 > 0$. There is a constant $C = C(c_0) > 0$ with the following property. Suppose $\kappa \geq 1$, $V \in C^2(\mathbb{R}^d)$, and*

$$\kappa^{-1}I \preceq \nabla^2 V \preceq I, \quad \|\nabla V(x_{\text{ref}})\| \leq \sqrt{d/\kappa}.$$

For every $q \geq 2$ and $0 < \varepsilon \leq 1$, put

$$\mathfrak{L}_q := q + \log \frac{e\kappa d}{\varepsilon}.$$

Then, there is a smoothing variance $0 < \eta \leq c_0$ for which a gradient-only implementation of Picard HMC returns a sample whose output law $\hat{\pi}_\eta$ satisfies

$$W_q(\hat{\pi}_\eta, \pi_\eta) \leq \varepsilon.$$

Its expected number of gradient queries is at most

$$C \left\{ \kappa^2 + \frac{\kappa^{4/3} (d+q)^{1/6}}{\varepsilon^{1/3}} \right\} \mathfrak{L}_q^{9/2}.$$

Note that unlike [Theorem 1.1](#), this guarantee targets the smoothed distribution π_η ; that is, we omit the terminal FORS step. This is because the terminal FORS guarantee does not immediately hold in W_q ; moreover, the guarantee for sampling from π_η will be enough for our eventual warm start application. Indeed, the guarantee is intended to implement the smooth sampler [A2](#) in [Section 6](#).

5.2 W_q local error analysis

In the W_2 proof, the local error framework ([Lemma 4.4](#)) separates a weak error from a strong error by expanding a square. For $q > 2$, we replace it with the following smoothness inequality.

Lemma 5.2 (L^q smoothness). *Let $q \geq 2$. If Z is a centered random vector in a Hilbert space and a is deterministic, then*

$$(\mathbb{E}\|a + Z\|^q)^{2/q} \leq \|a\|^2 + (q-1) (\mathbb{E}\|Z\|^q)^{2/q}. \quad (5.1)$$

Proof. This follows from the smoothness of L^q ; see [\[Pin94\]](#). It also follows directly by differentiating $t \mapsto (\mathbb{E}\|a + tZ\|^q)^{2/q}$: centeredness gives zero derivative at the origin and Hölder's inequality bounds the second derivative by $2(q-1) \|Z\|_{L^q}^2$. $\square$

Lemma 5.3 (Local error framework in W_q). *Let $P, \hat{P}$ be Markov kernels on $\mathbb{R}^m$, and let $q \geq 2$. Suppose that, for every $x, y \in \mathbb{R}^m$, there is a coupling $X \sim \delta_x P$, $Y \sim \delta_y P$ satisfying*

$$\|\mathbb{E}(X - Y)\| \leq \rho \|x - y\|, \quad \|(X - Y) - \mathbb{E}(X - Y)\|_{L^q} \leq \chi_q \|x - y\|.$$

Suppose also that, writing $\hat{X} \sim \delta_x \hat{P}$ and $X \sim \delta_x P$,

$$W_q(\delta_x \hat{P}, \delta_x P) \leq \mathcal{E}_{\text{strong}}, \quad \|\mathbb{E}\hat{X} - \mathbb{E}X\| \leq \mathcal{E}_{\text{weak}}.$$

Assume $0 < \rho < 1$ and $4(q-1)\chi_q^2 \leq \rho(1-\rho)$. Then, for all probability laws μ, ν on $\mathbb{R}^m$,

$$W_q^2(\mu \hat{P}, \nu P) \leq \rho W_q^2(\mu, \nu) + \frac{16(q-1)}{1+\rho} \mathcal{E}_{\text{strong}}^2 + \frac{2}{1-\rho} \mathcal{E}_{\text{weak}}^2.$$

Proof. Let (Z, Z') be an optimal W_q coupling of μ and ν . Given (Z, Z') , let $\hat{X} \sim \hat{P}(Z, \cdot)$ and $X \sim P(Z, \cdot)$ be optimally coupled in W_q , and let $Y \sim P(Z', \cdot)$ be coupled to X via the coupling in the theorem statement. Write $\Delta := \hat{X} - X$. Conditionally on (Z, Z') , Jensen's inequality gives

$$\|\mathbb{E}[\Delta \mid Z, Z']\| \leq \mathcal{E}_{\text{weak}}, \quad \|\Delta - \mathbb{E}[\Delta \mid Z, Z']\|_{L^q|Z, Z'} \leq 2\mathcal{E}_{\text{strong}}.$$

By Minkowski's inequality and Young's inequality, for $\lambda > 0$,

$$\|\hat{X} - Y\|_{L^q|Z, Z'}^2 \leq (1 + \lambda) \|\Delta - \mathbb{E}[\Delta \mid Z, Z'] + X - Y\|_{L^q|Z, Z'}^2 + (1 + \lambda^{-1}) \|\mathbb{E}[\Delta \mid Z, Z']\|_{L^q|Z, Z'}^2.$$

For the first term, by (5.1),

$$\begin{aligned} & \|\Delta - \mathbb{E}[\Delta \mid Z, Z'] + X - Y\|_{L^q|Z, Z'}^2 \\ & \leq \|\mathbb{E}[X - Y \mid Z, Z']\|^2 + (q - 1) \|\Delta - \mathbb{E}[\Delta \mid Z, Z'] + X - Y - \mathbb{E}[X - Y \mid Z, Z']\|_{L^q|Z, Z'}^2 \\ & \leq \rho^2 \|Z - Z'\|^2 + (q - 1) (2\mathcal{E}_{\text{strong}} + \chi_q \|Z - Z'\|)^2. \end{aligned}$$

Using $(u + v)^2 \leq 2u^2 + 2v^2$ and the assumption on χ_q , the right-hand side is at most

$$\frac{\rho(1 + \rho)}{2} \|Z - Z'\|^2 + 8(q - 1) \mathcal{E}_{\text{strong}}^2.$$

Take the $L^{q/2}$ norm over (Z, Z') . Then, choosing $\lambda = \frac{1 - \rho}{1 + \rho}$,

$$\begin{aligned} \|\hat{X} - Y\|_{L^q}^2 &= \|\|\hat{X} - Y\|_{L^q|Z, Z'}\|_{L^{q/2}}^2 \\ &\leq \left\| (1 + \lambda) \left(\frac{\rho(1 + \rho)}{2} \|Z - Z'\|^2 + 8(q - 1) \mathcal{E}_{\text{strong}}^2 \right) + (1 + \lambda^{-1}) \mathcal{E}_{\text{weak}}^2 \right\|_{L^{q/2}} \\ &= \rho \|Z - Z'\|_{L^q}^2 + \frac{16(q - 1)}{1 + \rho} \mathcal{E}_{\text{strong}}^2 + \frac{2}{1 - \rho} \mathcal{E}_{\text{weak}}^2. \quad \square \end{aligned}$$

Recall the kernels $\hat{K}$, $\bar{K}$, and K from Section 4.1. For $q \geq 2$, define the centered variance and bias errors

$$\mathcal{E}_{\text{var}, q} := \sup_{z \in \mathbb{R}^{2d}} W_q(\delta_z \hat{K}, \delta_z \bar{K}), \quad \mathcal{E}_{\text{bias}} := \sup_{z, \zeta \in \mathbb{R}^{2d}} \|\bar{\Phi}_z(\zeta) - \Phi_z(\zeta)\|.$$

Take $\hat{P} := \hat{K}$ and $P := K$. By the definition of $\bar{\Phi}_z$, the laws $\delta_z \hat{K}$ and $\delta_z \bar{K}$ have the same mean. The triangle inequality yields

$$\mathcal{E}_{\text{strong}} \leq \mathcal{E}_{\text{var}, q} + \mathcal{E}_{\text{bias}}, \quad \mathcal{E}_{\text{weak}} \leq \mathcal{E}_{\text{bias}}.$$

Also, define

$$\mathcal{E}_{\text{disc}, q} := W_q(\Pi_\eta K, \Pi_\eta).$$

The analogue of Theorem 4.5 is the following theorem.

Theorem 5.4 (Local error analysis in W_q). *Let $J \geq 2$, and assume $h \leq c/(\kappa q)$. Then, for every phase-space law $\hat{\mu}$,*

$$W_{q, M_\kappa}^2(\hat{\mu} \hat{K}, \Pi_\eta) \leq (1 - ch/\kappa) W_{q, M_\kappa}^2(\hat{\mu}, \Pi_\eta) + Cq \mathcal{E}_{\text{var}, q}^2 + \frac{C\kappa}{h} (\mathcal{E}_{\text{bias}}^2 + \mathcal{E}_{\text{disc}, q}^2). \quad (5.2)$$

Consequently,

$$W_{q, M_\kappa}^2(\hat{\mu} \hat{K}^N, \Pi_\eta) \leq e^{-cNh/\kappa} W_{q, M_\kappa}^2(\hat{\mu}, \Pi_\eta) + \frac{C\kappa q}{h} \mathcal{E}_{\text{var}, q}^2 + \frac{C\kappa^2}{h^2} (\mathcal{E}_{\text{bias}}^2 + \mathcal{E}_{\text{disc}, q}^2). \quad (5.3)$$

Proof. Repeat the proof of Theorem 4.5, using Lemma 5.3 in place of the W_2 local error framework; Proposition 4.7 and Lemma 4.8 still apply and give $\rho = 1 - ch/\kappa$ and $\chi_q \leq 2Ch$. $\square$

5.3 W_q local error bounds

The bias term $\mathcal{E}_{\text{bias}}$ is the same as before, and is controlled by [Proposition 4.9](#). Thus, we focus on controlling the other two terms.

5.3.1 Variance bound

For the variance bound, we must extend the second-order W_2 estimate in [Lemma 4.11](#) to W_q .

Lemma 5.5 (Second-order W_q estimate). *In the setting of [Lemma 4.11](#), for all $q \geq 2$, it holds that*

$$W_q(\text{Law}\{L(Z + f(Z, G))\}, \text{Law}\{L(Z + \bar{f}(Z))\}) \leq C\sqrt{m}(\sqrt{q}\|L\Pi_E\|_{\text{op}}B^2 + q\|L\|_{\text{op}}AB). \quad (5.4)$$

The proof, which is given in [Section C.3](#), relies on the same strategy as [Lemma 4.11](#), but relies on a more involved Gaussian divergence inequality ([Lemma C.1](#)). Using this, we can control the variance term in W_q .

Proposition 5.6 (Variance bound in W_q). *If $h \leq 1$, then, for every $q \geq 2$,*

$$\mathcal{E}_{\text{var},q} \leq C\sqrt{d}(\eta h^{3/2}q^{1/2}/J + \eta^{1/2}h^3q/\sqrt{J}).$$

Proof. Apply [Lemma 5.5](#) with the constants in [Lemma 4.10](#). □

5.3.2 Discretization error bound

Next, we need to replace the L^2 bound ([4.22](#)) with an L^q version. The following lemma reduces this to an L^q bound for the Gaussian divergence operator, which in turn is handled in [Lemma C.1](#).

Lemma 5.7 (Divergence bound). *Let $X \sim \pi$, $G \sim \mathbf{N}(0, I)$, and $P \sim \mathbf{N}(0, I)$ be independent, and set $Y := X + \sqrt{\eta}G$. For a smooth tensor-valued position vector field $W = (W_i)_{i \in [d]}$, we have the representation*

$$\delta_{\pi_\eta} W(Y, P) = \mathbb{E}[\eta^{-1/2} \delta_\gamma \{W(X + \sqrt{\eta} \cdot, P)\}(G) \mid Y, P]. \quad (5.5)$$

Consequently, for every $q \geq 2$,

$$\|\delta_{\pi_\eta} W\|_{L^q(\Pi_\eta)} \leq C\sqrt{q/\eta}\|W\|_{L^q(\Pi_\eta; \text{HS})} + Cq\|DW\|_{L^q(\Pi_\eta; \text{HS})}. \quad (5.6)$$

The constant is independent of all tensor dimensions.

Proof. The identity in ([4.1](#)) is equivalent to $g_\eta(Y) = \eta^{-1/2} \mathbb{E}[G \mid Y]$. The Gaussian divergence gives

$$\delta_\gamma \{W(X + \sqrt{\eta} \cdot, P)\}(G) = \langle G, W(Y, P) \rangle - \eta^{1/2} \text{div}_x W(Y, P).$$

Here $\text{div}_x W := \sum_{i \in [d]} \partial_{x_i} W_i$ is the divergence in the position argument; the $\eta^{1/2}$ factor comes from the chain rule. Conditioning on (Y, P) proves ([5.5](#)). Conditional expectation is an L^q contraction. Apply Jensen's inequality and ([C.4](#)) to obtain ([5.6](#)). □

Lemma 5.8 (Stationary iterated derivatives in L^q). *For $q \geq 2$ and $k \geq 0$,*

$$\|\mathcal{L}_H^k g_\eta\|_{L^q(\Pi_\eta)} \leq C^{k+1} q^k (k!)^{3/2} \sqrt{d + q} \eta^{-k/2}. \quad (5.7)$$

Proof. For a tensor-valued F , define the weighted Sobolev norm

$$\|F\|_{\mathbf{W}^{N,q}} := \sum_{j=0}^N \frac{\eta^{j/2}}{j!} \|D_x^j F\|_{L^q(\Pi_\eta; \text{HS})}. \quad (5.8)$$

Tracing through the proof of [Lemma 4.14](#), but using (5.6) in place of (4.22), we obtain

$$\|\delta_{\pi_\eta} W\|_{\mathbf{W}^{N-1,q}} \leq \frac{CqN}{\sqrt{\eta}} \|W\|_{\mathbf{W}^{N,q}}. \quad (5.9)$$

Recall the factorization $\mathcal{L}_H F = \delta_\gamma(D_x F) - \delta_{\pi_\eta}(D_p F)$ from (4.24). If F is polynomial of degree at most r in p , then [Lemma C.6](#), (5.8), and (5.9) yield the dimension-free bound

$$\|\mathcal{L}_H F\|_{\mathbf{W}^{N-1,q}} \leq \frac{CN(\sqrt{q} + q\sqrt{r})}{\sqrt{\eta}} \|F\|_{\mathbf{W}^{N,q}}. \quad (5.10)$$

It remains to initialize the induction. From [[Che26](#), Lemma 6.2.7], we have $\|g_\eta\|_{L^q(\pi_\eta)} \leq C\sqrt{d+q}$. For $j \geq 1$, the uniform bound (4.7) and the weights in (5.8) give

$$\|g_\eta\|_{\mathbf{W}^{N,q}} \leq C(N+1)\sqrt{d+q}.$$

Finally, $\mathcal{L}_H^j g_\eta$ is polynomial of degree at most j in momentum. Apply (5.10) k times, starting at $N = k$ and using $r = j$ at step j . Since $\sqrt{q} + q\sqrt{j} \leq 2q\sqrt{j+1}$ for $q \geq 2$, the product of the one-step factors is bounded by

$$\prod_{j=0}^{k-1} [(k-j)(\sqrt{q} + q\sqrt{j})] \leq (2q)^k (k!)^{3/2}.$$

For $k \geq 1$, the initialization factor satisfies $k+1 \leq 2^k$ and can therefore be absorbed into C^{k+1} ; the case $k = 0$ follows directly from the initialization bound. This proves (5.7). $\square$

Proposition 5.9 (Discretization error bound). *If $h^2 \leq 1$, then, for every $q \geq 2$,*

$$\mathcal{E}_{\text{disc},q} \leq C\sqrt{d+q} \left[B_{J,q} h \left(\frac{h}{\sqrt{\eta}} \right)^J + h^5 \right].$$

Here, one may take $B_{J,q} = C_0^{J+1} q^J \sqrt{J!}$, where C_0 is universal.

Proof. Repeat the proof of [Proposition 4.15](#) with the uniform-in-time L^q norm in place of the uniform-in-time L^2 norm. The interpolation estimate is replaced by

$$\mathcal{E}_{J,q} := \sup_{0 \leq t \leq h} \|g(t) - \mathcal{I}_{J,h} g(t)\|_{L^q(\Pi_\eta)} \leq \frac{h^J}{J!} \sup_{0 \leq t \leq h} \|g^{(J)}(t)\|_{L^q(\Pi_\eta)} \leq C^{J+1} q^J \sqrt{J!} \sqrt{d+q} \left(\frac{h}{\sqrt{\eta}} \right)^J. \quad (5.11)$$

Also, stationarity and [[Che26](#), Lemma 6.2.7] give

$$\sup_{0 \leq t \leq h} \|g_\eta(X_t)\|_{L^q(\Pi_\eta)} = \|g_\eta\|_{L^q(\pi_\eta)} \leq C\sqrt{d+q}$$

in place of the corresponding L^2 bound by $\sqrt{d}$. $\square$

5.4 Proof of Theorem 5.1

Proof of Theorem 5.1. We first assume exact proximal queries. The modifications needed to remove proximal access are deferred to Section D.

Fix $q \geq 2$ and $0 < \varepsilon \leq 1$, and let $\mathfrak{L}_q$ be as in the theorem. Choose

$$h := c\mathfrak{L}_q^{-3/2} \min\left\{\frac{1}{\kappa}, \left(\frac{\varepsilon}{\kappa\sqrt{d+q}}\right)^{1/3}\right\}, \quad \eta := C_\eta \mathfrak{L}_q^3 h^2,$$

where $C_\eta \geq 1$ is universal and $c > 0$ will be chosen sufficiently small. Set

$$J := \lceil C\mathfrak{L}_q \rceil, \quad N := \lceil C\kappa\mathfrak{L}_q/h \rceil. \quad (5.12)$$

With appropriately large C and small c , the definitions give

$$e^{-C\mathfrak{L}_q} \leq h \leq \frac{c}{\kappa\mathfrak{L}_q^{3/2}}, \quad h\mathfrak{L}_q \leq \frac{c}{\kappa\mathfrak{L}_q^{1/2}} \leq \frac{c}{\kappa}, \quad \eta \leq \frac{C_\eta c^2}{\kappa^2} \leq \min\{c_0, 1/2\},$$

and $\log(h^{-1}) \leq C\mathfrak{L}_q$. Since $B_{J,q} = C_0^{J+1} q^J \sqrt{J!}$, we have

$$B_{J,q} C_\eta^{-J/2} \mathfrak{L}_q^{-3J/2} \leq e^{-cJ}.$$

A sufficiently large C therefore gives

$$B_{J,q} \left(\frac{h}{\sqrt{\eta}}\right)^J \leq h^4.$$

Tedious bookkeeping shows that

$$W_{q,M_\kappa}(\hat{\mu}_N, \Pi_\eta) \leq C\mathfrak{L}_q^{9/2} \kappa \sqrt{d+q} h^3, \quad (5.13)$$

Each phase makes $2J$ gradient queries and $2J$ exact proximal queries. Since $J \leq C\mathfrak{L}_q$ and $N \leq C\kappa\mathfrak{L}_q/h + 1 \leq C'\kappa\mathfrak{L}_q/h$, the numbers of direct gradient queries and exact proximal queries are each at most $C\kappa\mathfrak{L}_q^2/h$. Taking position marginals in (5.13) and the definition of h gives

$$W_q(\hat{\pi}_\eta, \pi_\eta) \leq C\mathfrak{L}_q^{9/2} \kappa \sqrt{d+q} h^3 \leq \varepsilon.$$

The modifications in Section D remove the proximal queries altogether with expected gradient cost at most $C\kappa\mathfrak{L}_q^3/h$. Substitution of h gives the query bound in the theorem. $\square$

6 Recursive warm start generator

In this section, we describe the recursive warm start generator and its guarantee. See Section 3.4 for the algorithm and an overview.

6.1 Main result

For probability laws P, Q on $\mathbb{R}^d$, define the sub-Gaussian Orlicz–Wasserstein distance

$$W_{\psi_2}(P, Q) := \inf_{(X,Y)} \inf\left\{\varepsilon > 0 : \mathbb{E} \exp\left(\frac{\|X - Y\|^2}{\varepsilon^2}\right) \leq 2\right\},$$

where the outer infimum ranges over all couplings of P and Q .

The recursive warm start generator assumes access to a smoothed sampler as follows.

Assumption (Smoothed sampler). There is a sampler $\mathfrak{A}$ which, provided any normalized potential U satisfying $\kappa_U^{-1}I \preceq \nabla^2 U \preceq I$, and an admissible reference point x_{ref} with $\|\nabla U(x_{\text{ref}})\| \leq \sqrt{d/\kappa_U}$, implements one of the interfaces below. Here, $\pi_{U,\eta} := \pi_U * \mathbf{N}(0, \eta I)$ denotes the smoothed distribution.

A1 Given $\varepsilon > 0$, the sampler returns $0 < \eta \leq c_0$ and a sample with law $\hat{\pi}_{U,\eta}$ such that

$$\kappa_U^{-1/2} W_2(\hat{\pi}_{U,\eta}, \pi_{U,\eta}) \leq \varepsilon$$

using at most $C_2(\kappa_U, d, \varepsilon)$ expected gradient queries.

A2 Given $p \geq 2$ and $\varepsilon > 0$, it returns $0 < \eta \leq c_0$ and a sample with law $\hat{\pi}_{U,\eta}$ such that

$$\kappa_U^{-1/2} W_p(\hat{\pi}_{U,\eta}, \pi_{U,\eta}) \leq \varepsilon$$

using at most $C_p(\kappa_U, d, \varepsilon)$ expected gradient queries.

A3 Given $\varepsilon > 0$, it returns $0 < \eta \leq c_0$ and a sample with law $\hat{\pi}_{U,\eta}$ such that

$$\kappa_U^{-1/2} W_{\psi_2}(\hat{\pi}_{U,\eta}, \pi_{U,\eta}) \leq \varepsilon$$

using at most $C_{\psi_2}(\kappa_U, d, \varepsilon)$ expected gradient queries.

Assume that each complexity function is non-decreasing in its condition number argument and non-increasing in its accuracy argument, and that $p \mapsto C_p$ is non-decreasing.

We will apply this interface directly to potentials whose smoothness is not normalized to 1. Concretely, if $\alpha_U I \preceq \nabla^2 U \preceq \beta_U I$, apply $\mathfrak{A}$ to the normalized potential $U(\cdot/\sqrt{\beta_U})$, and then divide its output by $\sqrt{\beta_U}$. If η is the normalized smoothing level, then the law of the rescaled output approximates $\pi_U * \mathbf{N}(0, \beta_U^{-1} \eta I)$. The corresponding scale-invariant guarantee is $\sqrt{\alpha_U} W_\bullet \leq \varepsilon$, and the query cost is unchanged.

In [Sections 4](#) and [5](#), we have shown that Picard HMC implements the first two interfaces above. We have not shown that it implements the third, but we include this case for completeness because, as shown below, [A3](#) leads to a genuine Rényi warm start.

We now present the main guarantee.

Theorem 6.1 (Recursive warm start generator). *Suppose that $\pi \propto \exp(-V)$ is β -log-smooth, in the sense that ∇V is β -Lipschitz, and satisfies the logarithmic Sobolev inequality with constant α^{-1} . Write $\kappa := \beta/\alpha$, and suppose that we are given a point x_{ref} satisfying $V(x_{\text{ref}}) - \inf V \leq d$.*

Let the smoothed sampler $\mathfrak{A}$ satisfy one of [A1](#), [A2](#), and [A3](#). Fix $0 < \delta, \Delta \leq 1/2$. For $q \geq 2$, write

$$\mathfrak{L}_q := q + \log \frac{d\kappa q}{\Delta}.$$

Then the proximal sampler wrapper in [Algorithm 3.4](#), with each inner RGO sampled by [Algorithm 3.3](#), has the following guarantees.

(i) Under [A1](#), the generator returns a sample with law $\hat{\pi}$ satisfying

$$\text{KL}(\hat{\pi} \parallel \pi) \leq \Delta^2.$$

The expected total number of gradient queries is at most

$$C\kappa\mathfrak{L}_2^2 C_2\left(6, d, \frac{c\Delta}{\sqrt{\kappa\mathfrak{L}_2}}\right) + C\kappa\mathfrak{L}_2^2.$$

- (ii) Fix $2 \leq q < \infty$. Under [A2](#), the generator returns a sample with law $\hat{\pi}$ for which there exists a law $\hat{\pi}^\dagger$ satisfying

$$\text{TV}(\hat{\pi}, \hat{\pi}^\dagger) \leq \delta, \quad \mathbf{R}_q(\hat{\pi}^\dagger \parallel \pi) \leq \Delta^2.$$

The expected total number of gradient queries is at most

$$C\kappa \mathfrak{L}_q^2 \mathbf{C}_p \left(6, d, \frac{c\Delta}{\sqrt{q\kappa \log(eq)} \mathfrak{L}_q} \right) + C\kappa \mathfrak{L}_q \left(\mathfrak{L}_q + \log \frac{C\kappa \mathfrak{L}_q}{\delta} \right),$$

where

$$p := 2 \vee \left\lceil \log \frac{C\kappa \mathfrak{L}_q^2}{\delta} \right\rceil.$$

- (iii) Fix $2 \leq q < \infty$. Under [A3](#), the generator returns a sample with law $\hat{\pi}$ satisfying

$$\mathbf{R}_q(\hat{\pi} \parallel \pi) \leq \Delta^2.$$

The expected total number of gradient queries is at most

$$C\kappa \mathfrak{L}_q^2 \mathbf{C}_{\psi_2} \left(6, d, c \min \left\{ \frac{1}{q}, \frac{\Delta}{\sqrt{q\kappa \log(eq)} \mathfrak{L}_q} \right\} \right) + C\kappa \mathfrak{L}_q^2.$$

Although the statement of [Theorem 6.1](#) is cumbersome, it is quite powerful: it states that once we have a smoothed sampler with Wasserstein guarantees, we can immediately obtain a stronger guarantee with a negligible overhead. Indeed, the cost of the warm start generation is essentially the cost of the original smoothed sampler, called with a slightly smaller accuracy $\Delta/\sqrt{q\kappa}$ up to logarithmic factors. Crucially, in case (ii), we only need the smoothed sampler guarantee in W_p for an order p which is logarithmic in $1/\delta$.

Due to the use of the outer proximal sampler wrapper, note that the smoothed sampler is only ever called on distributions with condition number bounded by a universal constant; above we take it to be 6. This will ultimately improve the condition number dependence of our combined sampler.

6.2 Technical tools

6.2.1 Regularization and reverse transport inequalities

The key to the recursive warm start is the regularizing property of the heat flow, which enables proving reverse transport inequalities. We collect together the key inequalities here.

We begin with a simple lemma on truncation.

Lemma 6.2 (Truncation). *Let P, Q be probability laws satisfying $W_p(P, Q) \leq \varepsilon_p$ for some $p \geq 2$. For every $0 < \delta < 1$, there is a probability law $P^\dagger$ such that*

$$\text{TV}(P, P^\dagger) \leq \delta, \quad W_\infty(P^\dagger, Q) \leq \varepsilon_p \delta^{-1/p}.$$

Proof. Choose an optimal W_p coupling $(Y, Y^\star)$ of P and Q . Define

$$Y^\dagger := \begin{cases} Y, & \|Y - Y^\star\| \leq \varepsilon_p \delta^{-1/p}, \\ Y^\star, & \|Y - Y^\star\| > \varepsilon_p \delta^{-1/p}. \end{cases}$$

Markov's inequality shows that $Y \neq Y^\dagger$ with probability at most δ , whereas $\|Y^\dagger - Y^\star\| \leq \varepsilon_p \delta^{-1/p}$ almost surely. Taking $P^\dagger := \text{Law}(Y^\dagger)$ proves the result. $\square$

For $\tau > 0$, let $\gamma_\tau := \mathbf{N}(0, \tau I)$. The following lemma shows that after applying the heat flow, we can bound stronger information-theoretic divergences in terms of transport distances.

Lemma 6.3 (Reverse transport along the heat flow). *Let P, Q be probability laws on $\mathbb{R}^d$.*

(i) *If $P, Q \in \mathcal{P}_2(\mathbb{R}^d)$, then*

$$\text{KL}(P * \gamma_\tau \| Q * \gamma_\tau) \leq \frac{W_2^2(P, Q)}{2\tau}.$$

(ii) *If $P, Q \in \mathcal{P}_p(\mathbb{R}^d)$, then for every $0 < \delta < 1$, there is a probability law $P^\dagger$ such that*

$$\text{TV}(P, P^\dagger) \leq \delta, \quad \mathbf{R}_q(P^\dagger * \gamma_\tau \| Q * \gamma_\tau) \leq \frac{q W_p^2(P, Q)}{2\delta^{2/p_\tau}}, \quad \text{for all } q > 1.$$

(iii) *Let $q \geq 2$ and suppose that $W_{\psi_2}(P, Q) \leq \varepsilon_{\psi_2}$. Then,*

$$\frac{q(q-1)\varepsilon_{\psi_2}^2}{2\tau} \leq 1 \implies \mathbf{R}_q(P * \gamma_\tau \| Q * \gamma_\tau) \leq \frac{q \log 2}{2\tau} \varepsilon_{\psi_2}^2.$$

Proof. Statement (i) follows from the joint convexity of the KL divergence.

For (ii), choose $P^\dagger$ as in [Lemma 6.2](#) and use the fact that

$$\mathbf{R}_q(P^\dagger * \gamma_\tau \| Q * \gamma_\tau) \leq \frac{q}{2\tau} W_\infty^2(P^\dagger, Q).$$

Part (iii) is [\[AC24, Remark 3.8\]](#), with the regularization parameter there specialized to zero, with zero initial shift, the source displacement parameter equal to $W_{\psi_2}(P, Q)$, and $\sigma^2 = t$. $\square$

6.2.2 RGO calculus

For $U \in C^2(\mathbb{R}^d)$, $A \in (0, \infty]$, and $u \in \mathbb{R}^d$, define the RGO potential

$$U_{A,u} := U + \frac{\|\cdot - u\|^2}{2A},$$

and let $R_{A,u}^U \propto \exp(-U_{A,u})$ be the corresponding RGO.

In the lemma below, we note that the RGO of an RGO is itself an RGO for the original distribution, and we record how the parameters change. The proof is an elementary calculation and is omitted.

Lemma 6.4 (RGO calculus). *Suppose that U satisfies $\kappa^{-1}I \preceq \nabla^2 U \preceq I$. Then, $U_{A,u}$ is α_A -strongly convex and β_A -smooth, where*

$$\alpha_A := \kappa^{-1} + A^{-1}, \quad \beta_A := 1 + A^{-1}, \quad \kappa_A := \frac{\beta_A}{\alpha_A}.$$

Moreover,

$$R_{a,y}^{U_{A,u}} = R_{A^+, u^+}^U,$$

where

$$\frac{1}{A^+} := \frac{1}{A} + \frac{1}{a}, \quad u^+ := A^+ \left(\frac{u}{A} + \frac{y}{a} \right), \quad \kappa_{A^+} = \frac{a\beta_A + 1}{a\beta_A + \kappa_A} \kappa_A. \quad (6.1)$$

6.3 Recursive RGO sampler

The following result gives the formal guarantee for [Algorithm 3.3](#). This is essentially the same as [Theorem 6.1](#), but without the outer proximal sampler wrapper.

Theorem 6.5 (Recursive RGO sampler). *Assume (1.1), and suppose that the sampler $\mathfrak{A}$ satisfies one of [A1](#), [A2](#), or [A3](#). Fix $A_0 \in (0, \infty]$, $u_0 \in \mathbb{R}^d$, and correspondingly a target R_{A_0, u_0}^V for which an admissible reference point x_{ref} is supplied, let $\kappa_0 := \kappa_{A_0}$, and fix $0 < \delta_0, \Delta_0 \leq 1/2$. For $q \geq 2$, write*

$$\mathfrak{L}_q := q + \log \frac{\kappa_0 dq}{\Delta_0}.$$

Then [Algorithm 3.3](#) has the following guarantees.

- (i) Under [A1](#), it returns a sample with law $\widehat{R}_{A_0, u_0}^V$ satisfying

$$\text{KL}(\widehat{R}_{A_0, u_0}^V \parallel R_{A_0, u_0}^V) \leq \Delta_0^2.$$

The expected total number of gradient queries is at most

$$C \mathfrak{L}_2 C_2 \left(2\kappa_0, d, \frac{c\Delta_0}{\sqrt{\mathfrak{L}_2}} \right) + C\kappa_0 \mathfrak{L}_2.$$

- (ii) Fix $q \geq 2$. Under [A2](#), it returns a sample with law $\widehat{R}_{A_0, u_0}^V$ for which there exists $(\widehat{R}_{A_0, u_0}^V)^\dagger$ satisfying

$$\text{TV}(\widehat{R}_{A_0, u_0}^V, (\widehat{R}_{A_0, u_0}^V)^\dagger) \leq \delta_0, \quad \mathbf{R}_q((\widehat{R}_{A_0, u_0}^V)^\dagger \parallel R_{A_0, u_0}^V) \leq \Delta_0^2.$$

The expected total number of gradient queries is at most

$$C \mathfrak{L}_q C_p \left(2\kappa_0, d, \frac{c\Delta_0}{\sqrt{q\mathfrak{L}_q}} \right) + C\kappa_0 \mathfrak{L}_q,$$

where $p := 2 \vee \lceil \log(C\mathfrak{L}_q/\delta_0) \rceil$.

- (iii) Fix $q \geq 2$. Under [A3](#), it returns a sample with law $\widehat{R}_{A_0, u_0}^V$ satisfying

$$\mathbf{R}_q(\widehat{R}_{A_0, u_0}^V \parallel R_{A_0, u_0}^V) \leq \Delta_0^2.$$

The expected total number of gradient queries is at most

$$C \mathfrak{L}_q C_{\psi_2} \left(2\kappa_0, d, c \min \left\{ \frac{1}{q}, \frac{\Delta_0}{\sqrt{q\mathfrak{L}_q}} \right\} \right) + C\kappa_0 \mathfrak{L}_q.$$

Although [Algorithm 3.3](#) is written recursively, it is convenient to analyze it iteratively. Thus, we break down the algorithm into stages, so that the target at stage j is V_{A_j, u_j} . Let η_j be the reported smoothing level at stage j , and denote the rescaled output by X_j ; its law approximates $R_{A_j, u_j}^V * \mathbf{N}(0, \beta_{A_j}^{-1} \eta_j I)$ in a Wasserstein metric. Draw an independent $G_j \sim \mathbf{N}(0, I)$ and set

$$Y_j := X_j + \sqrt{\frac{\tau_j}{\beta_{A_j}}} G_j, \quad a_j := \frac{\eta_j + \tau_j}{\beta_{A_j}}.$$

Thus, $\text{Law}(Y_j)$ is close to $R_{A_j, u_j}^V * \mathbf{N}(0, a_j I)$. The algorithm then updates (A_{j+1}, u_{j+1}) by (6.1) with RGO variance a_j and center Y_j , as in [Algorithm 3.3](#). At the terminal level J , it samples the remaining target R_{A_J, u_J}^V with the FORS subroutine.

At stage j , write $\kappa_j := \kappa_{A_j}$ and choose the heat flow duration

$$\tau_j := \begin{cases} \kappa_j, & \kappa_j \geq 2, \\ c_0, & \kappa_j < 2. \end{cases} \quad (6.2)$$

Lemma 6.6 (Geometric progress of the recursive stages). *Suppose that $c_0 \in (0, 1/4)$ and $0 < \eta_j \leq c_0$ at every stage.*

(i) *If $\kappa_j \geq 2$, then*

$$\frac{1}{2} \kappa_j \leq \kappa_{j+1} \leq \frac{4}{5} \kappa_j. \quad (6.3)$$

(ii) *If $\kappa_j < 2$, then $A_{j+1} \leq 2c_0$, and all later well-conditioned stages obey*

$$A_{j+1} \leq \rho_0 A_j, \quad \rho_0 := \frac{2c_0}{1 + 2c_0} < 1.$$

Proof. In the first case, by (6.2), $\kappa_j = \tau_j \leq \eta_j + \tau_j \leq \kappa_j + c_0$. Substitution in (6.1) gives (6.3) after decreasing c_0 if necessary.

In the well-conditioned regime, $c_0 \leq \eta_j + \tau_j \leq 2c_0$. By (6.1), the definition of a_j , and $A_j \beta_{A_j} = A_j + 1$,

$$A_{j+1} = \frac{A_j a_j}{A_j + a_j} = \frac{A_j (\eta_j + \tau_j)}{A_j \beta_{A_j} + \eta_j + \tau_j} = \frac{A_j (\eta_j + \tau_j)}{A_j + 1 + \eta_j + \tau_j}.$$

This yields

$$A_{j+1} \leq \tau_j + \eta_j \leq 2c_0, \quad \frac{A_{j+1}}{A_j} = \frac{\eta_j + \tau_j}{A_j + \eta_j + \tau_j + 1} \leq \frac{2c_0}{1 + 2c_0}. \quad \square$$

Proof of Theorem 6.5. Recursive error propagation. Here and below, take $q = 2$ under A1. For sufficiently small and large universal constants $c, C > 0$, respectively, set

$$\underline{A} := \frac{c}{\sqrt{d} \mathfrak{L}_q + \mathfrak{L}_q}, \quad J := \left\lceil C \log \frac{e \kappa_0}{\underline{A}} \right\rceil \leq C \mathfrak{L}_q. \quad (6.4)$$

At every stage $j \in \{0, \dots, J-1\}$, call the smoothed sampler with accuracy

$$\varepsilon_{2,j}^2 := \frac{2\tau_j \Delta_0^2}{\kappa_j (J+1)} \quad \text{under A1}, \quad (6.5)$$

$$\varepsilon_{p,j}^2 := \frac{2e^{-2}\tau_j \Delta_0^2}{q\kappa_j (J+1)}, \quad p := 2 \vee \left\lceil \log \frac{C \mathfrak{L}_q}{\delta_0} \right\rceil \quad \text{under A2}, \quad (6.6)$$

$$\varepsilon_{\psi_2,j}^2 := \frac{c\tau_j}{\kappa_j} \min \left\{ \frac{1}{q(q-1)}, \frac{\Delta_0^2}{q(J+1)} \right\} \quad \text{under A3}. \quad (6.7)$$

Fix a stage $j < J$ and condition on the complete history up to that stage. Write $\hat{P}_j$ for the conditional law of Y_j , and write $P_j := R_{A_j, u_j}^V * \mathbf{N}(0, a_j I)$ for its ideal law. Also, let $K_j(y, \cdot) := R_{a_j, y}^{V_{A_j, u_j}}$, and let $\hat{K}_j(y, \cdot)$ be the conditional law returned by the remaining recursive call after the center y is selected. Thus, $R_j := R_{A_j, u_j}^V = P_j K_j$, and the output law from level j is $\hat{R}_j = \hat{P}_j \hat{K}_j$. We prove by backward induction that, conditional on every history,

$$\text{KL}(\hat{R}_j \parallel R_j) \leq \frac{J-j+1}{J+1} \Delta_0^2 \quad \text{under A1},$$

$$\text{R}_q(\hat{R}_j^\dagger \parallel R_j) \leq \frac{J-j+1}{J+1} \Delta_0^2, \quad \text{TV}(\hat{R}_j, \hat{R}_j^\dagger) \leq \frac{J-j}{J} \delta_0 \quad \text{under A2},$$

$$\text{R}_q(\hat{R}_j \parallel R_j) \leq \frac{J-j+1}{J+1} \Delta_0^2 \quad \text{under A3}.$$

The terminal construction below establishes the base case $j = J$, with $\widehat{R}_J^\dagger = \widehat{R}_J$ under [A2](#).

Suppose that these inequalities are valid at stage $j + 1$. Under [A1](#), after rescaling the output, the smoothed sampler provides the guarantee $W_2^2 \leq \kappa_j \varepsilon_{2,j}^2 / \beta_{A_j}$. After convolving with $\mathcal{N}(0, \beta_{A_j}^{-1} \tau_j I)$, part (i) of [Lemma 6.3](#) and (6.5) yield

$$\text{KL}(\widehat{P}_j \parallel P_j) \leq \frac{\kappa_j \varepsilon_{2,j}^2 / \beta_{A_j}}{2\tau_j / \beta_{A_j}} = \frac{\kappa_j \varepsilon_{2,j}^2}{2\tau_j} = \frac{\Delta_0^2}{J+1}.$$

Under [A2](#), the corresponding guarantee is $W_p^2 \leq \kappa_j \varepsilon_{p,j}^2 / \beta_{A_j}$. Apply part (ii) of the same lemma with TV budget δ_0 / J . Since $p \geq \log(J / \delta_0)$, we have $(J / \delta_0)^{2/p} \leq e^2$. Hence there exists $\widehat{P}_j^\dagger$ with

$$\text{TV}(\widehat{P}_j, \widehat{P}_j^\dagger) \leq \frac{\delta_0}{J}, \quad \text{R}_q(\widehat{P}_j^\dagger \parallel P_j) \leq \frac{qe^2 \kappa_j \varepsilon_{p,j}^2 / \beta_{A_j}}{2\tau_j / \beta_{A_j}} = \frac{qe^2 \kappa_j \varepsilon_{p,j}^2}{2\tau_j} = \frac{\Delta_0^2}{J+1}.$$

Under [A3](#), the corresponding guarantee is $W_{\psi_2}^2 \leq \kappa_j \varepsilon_{\psi_2,j}^2 / \beta_{A_j}$. For a sufficiently small universal c , (6.7) verifies both

$$\frac{q(q-1)\kappa_j \varepsilon_{\psi_2,j}^2 / \beta_{A_j}}{2\tau_j / \beta_{A_j}} = \frac{q(q-1)\kappa_j \varepsilon_{\psi_2,j}^2}{2\tau_j} \leq 1 \quad \text{and} \quad \text{R}_q(\widehat{P}_j \parallel P_j) \leq \frac{\Delta_0^2}{J+1}.$$

We now verify the induction step. Under [A1](#), the KL chain rule and data processing give

$$\begin{aligned} \text{KL}(\widehat{R}_j \parallel R_j) &\leq \text{KL}(\widehat{P}_j \parallel P_j) + \int \text{KL}(\widehat{K}_j(y, \cdot) \parallel K_j(y, \cdot)) \widehat{P}_j(dy) \\ &\leq \frac{\Delta_0^2}{J+1} + \frac{J-j}{J+1} \Delta_0^2 = \frac{J-j+1}{J+1} \Delta_0^2. \end{aligned}$$

Under [A3](#), the same argument applies except that we replace the KL chain rule with the Rényi composition rule. Finally, under [A2](#), we take $\widehat{R}_j^\dagger := \widehat{P}_j^\dagger \widehat{K}_j^\dagger$, and we obtain the bound on $\text{R}_q(\widehat{R}_j^\dagger \parallel R_j)$ as before. Also, we maximally couple $\widehat{P}_j$ and $\widehat{P}_j^\dagger$, and on the event that their draws agree, we use the inductive coupling of $\widehat{K}_j^\dagger$ and $\widehat{K}_j$. This gives

$$\text{TV}(\widehat{R}_j, \widehat{R}_j^\dagger) \leq \text{TV}(\widehat{P}_j, \widehat{P}_j^\dagger) + \sup_{y \in \mathbb{R}^d} \text{TV}(\widehat{K}_j(y, \cdot), \widehat{K}_j^\dagger(y, \cdot)) \leq \frac{\delta_0}{J} + \frac{J-j-1}{J} \delta_0 = \frac{J-j}{J} \delta_0.$$

This completes the induction.

Reference point construction. We verify that every smoothed sampler call is supplied with an admissible reference point and account for the work needed to construct these points. Let U be the current normalized potential and Y the output of the smoothed sampler, and write τ for the current heat flow duration. Under [A1](#) and [A2](#), use $W_2 \leq W_p$. Under [A3](#), Jensen's inequality in the definition of W_{ψ_2} gives $W_2 \leq \sqrt{\log 2} W_{\psi_2}$. Thus, in every case, we can couple Y to the ideal $Y^* = X + \sqrt{\eta} G'$, where $X \sim \pi_U$. This yields, by 1-smoothness of U ,

$$\mathbb{E} \|\nabla U(Y)\|^2 \leq 2 (\mathbb{E} \|\nabla U(Y^*)\|^2 + \kappa_U \varepsilon_j^2) \leq C (d + \eta d + \kappa_U \varepsilon_j^2),$$

where ε_j denotes the accuracy at stage j .

In normalized coordinates, the next RGO potential is

$$U^+ := U + \frac{\|\cdot - (Y + \sqrt{\tau} G)\|^2}{2(\eta + \tau)}.$$

Since $\kappa_U^{-1}I \preceq \nabla^2 U \preceq I$, the potential U^+ is α^+ -strongly convex and β^+ -smooth, where

$$\alpha^+ := \kappa_U^{-1} + (\eta + \tau)^{-1}, \quad \beta^+ := 1 + (\eta + \tau)^{-1}.$$

Its gradient at Y adds only $-\sqrt{\tau}G/(\eta + \tau)$, and hence

$$\mathbb{E}\|\nabla U^+(Y)\|^2 \leq C \left(d + \kappa_U \varepsilon_J^2 + \frac{\tau d}{(\eta + \tau)^2} \right).$$

Gradient descent with step size $1/\beta^+$ produces iterates $\{x_k\}_{k \geq 0}$ with

$$\|\nabla U^+(x_k)\|^2 \leq \kappa^+ \left(1 - \frac{1}{\kappa^+} \right)^k \|\nabla U^+(Y)\|^2, \quad \kappa^+ := \frac{\beta^+}{\alpha^+}.$$

Stopping when $\|\nabla U^+(x_k)\| \leq \sqrt{\alpha^+ d}$ gives the admissible reference point for the next stage. The requested radii make the initial squared gradient polynomial in the stage parameters, so Jensen's inequality bounds the expected work at this stage by $\tilde{O}(\kappa^+)$. The condition numbers in the large conditioning regime form a decreasing geometric sequence, and all later stages have universally bounded condition number. Summing over the stages in (6.4) bounds the expected cost of all inner reference points by $C\kappa_0 \mathfrak{L}_q$.

Terminal stage. By Lemma 6.6(i), the regime $\kappa_j \geq 2$ lasts for at most $C \log(e\kappa_0)$ stages. Thereafter, part (ii) of the same lemma contracts A_j by the fixed factor $\rho_0 < 1$. Thus (6.4) ensures that $A_J \leq \underline{A}$. Set $\varepsilon_J := \Delta_0/\sqrt{J+1}$; this is the accuracy requested from the terminal FORS sampler.

Starting from the retained reference point, gradient descent on the terminal potential produces x_+ satisfying $\|\nabla V_{A_J, u_J}(x_+)\| \leq \sqrt{d/A_J}$. The same descent estimate used for the inner reference points shows that this terminal construction costs at most $C\kappa_0 \mathfrak{L}_q$ expected gradient queries. Moreover,

$$\|u_J - A_J \nabla V(x_+) - x_+\| = A_J \|\nabla V_{A_J, u_J}(x_+)\| \leq \sqrt{dA_J},$$

which is exactly the condition in Theorem A.4 with target R_{A_J, u_J}^V .

Under A1, let $\hat{R}_J$ be the output of FORS at order 2. Under A2, let $\hat{R}_J$ be the output of FORS at order q and set $\hat{R}_J^\dagger = \hat{R}_J$. Under A3, let $\hat{R}_J$ be the output of FORS at order q . The choice of ε_J verifies the localized-scale condition in Theorem A.4. Hence every terminal call contributes at most $\Delta_0^2/(J+1)$ in the relevant divergence and has constant expected gradient query cost, establishing the base cases of the backward induction. $\square$

6.4 Outer proximal sampler

From Theorems 1.1 and 5.1, the smoothed Picard HMC sampler requires at least κ^2 queries, and if we use it as our smoothed sampler then this cost is inherited by the recursive RGO sampler. Instead, the full warm start generator runs an outer proximal chain with step size $(2\beta)^{-1}$ and applies the recursive RGO sampler only to its uniformly well-conditioned RGO targets. This is the construction described in Algorithm 3.4.

Initialize $X_0 \sim \hat{\mu}_0 := \mathbf{N}(x_{\text{ref}}, (2\beta)^{-1}I)$. An outer step draws $Y_n = X_n + \mathbf{N}(0, (2\beta)^{-1}I)$, and then applies an approximate RGO kernel at Y_n .

Proof of Theorem 6.1. Under A1, choose

$$\Delta_0^2 := \frac{c\Delta^2}{\kappa}, \quad N := \left\lceil C\kappa \log \frac{ed\kappa}{\Delta} \right\rceil.$$

Under [A2](#) and [A3](#), keep the prescribed order $q \geq 2$, and choose

$$\Delta_0^2 := \frac{c\Delta^2}{\kappa \log(eq)}, \quad N_0 := \left\lceil C\kappa \log \frac{ed\kappa}{\Delta} \right\rceil, \quad N_1 := \left\lceil \frac{\log(q-1)}{\log(1+(2\kappa)^{-1})} \right\rceil, \quad N := N_0 + N_1.$$

Initialization. Write $V_\star := \inf V$ and $g = \nabla V(x_{\text{ref}})$. The logarithmic Sobolev inequality implies the Poincaré inequality, and hence $\text{Cov}_\pi \preceq \alpha^{-1}I$. The Gaussian maximum entropy bound, together with $-\int \log \pi \, d\pi \geq V_\star + \log \int e^{-V}$, therefore gives

$$\log \int_{\mathbb{R}^d} e^{-V} + V_\star \leq \frac{d}{2} \log \frac{2\pi e}{\alpha}.$$

On the other hand, β -smoothness yields

$$V(x) \leq V(x_{\text{ref}}) + \langle g, x - x_{\text{ref}} \rangle + \frac{\beta}{2} \|x - x_{\text{ref}}\|^2$$

and by the descent lemma,

$$\frac{\|g\|^2}{2\beta} \leq V(x_{\text{ref}}) - V_\star \leq d.$$

Comparing the density of $\hat{\mu}_0 = \mathbf{N}(x_{\text{ref}}, (2\beta)^{-1}I)$ with that of π , and maximizing the resulting quadratic upper bound, gives the pointwise estimate

$$\log \frac{d\hat{\mu}_0}{d\pi} \leq \frac{d}{2} \log(2e\kappa) + 2d \leq Cd \log(e\kappa).$$

Consequently, $R_q(\hat{\mu}_0 \parallel \pi) \leq Cd \log(e\kappa)$ for every $q \geq 1$, with the convention $R_1 = \text{KL}$.

Approximate RGO kernels. For every outer RGO target, apply [Theorem 6.5](#) with the local accuracy Δ_0 chosen above. Under [A2](#), use $\delta_0 = \delta/(2N)$. Every outer RGO has curvature between βI and $3\beta I$. Apply [Theorem 6.5](#) to sample from the RGO, taking $A_0 = \infty$. Thus, its three explicit bounds apply with $\kappa_0 \leq 3$, uniformly over the outer center and complete history.

Outer contraction and error propagation. Let $\hat{\mu}_n$ denote the law of the n -th iterate X_n of the proximal sampler, and, under [A2](#), let $\mu_n^\dagger$ denote the comparison law. Write $\mu_n^\# = \hat{\mu}_n$ under [A1](#) and [A3](#), and $\mu_n^\# = \mu_n^\dagger$ under [A2](#).

Under [A1](#), the forward half of the proximal sampler contraction in [[CCSW22](#), Theorem 3 and Appendix A.4] and the KL chain rule give

$$\text{KL}(\hat{\mu}_{n+1} \parallel \pi) \leq (1 + (2\kappa)^{-1})^{-1} \text{KL}(\hat{\mu}_n \parallel \pi) + \Delta_0^2.$$

Consequently,

$$\text{KL}(\hat{\mu}_N \parallel \pi) \leq (1 + (2\kappa)^{-1})^{-N} Cd \log(e\kappa) + C\kappa \Delta_0^2 \leq \Delta^2.$$

It remains to treat [A2](#) and [A3](#). During the first N_0 steps, use the Rényi contraction from [[CCSW22](#), Theorem 3 and Appendix A.4] and Rényi composition. This yields

$$R_2(\mu_{N_0}^\# \parallel \pi) \leq (1 + (2\kappa)^{-1})^{-N_0/2} Cd \log(e\kappa) + C\kappa \Delta_0^2 \leq \frac{\Delta^2}{2}.$$

For the remaining N_1 steps, we leverage hypercontractivity to obtain a sharper dependence on q . Define $q_{N_0} := 2$ and

$$q_{n+1} - 1 := \min\{q - 1, (1 + (2\kappa)^{-1})(q_n - 1)\}, \quad n = N_0, \dots, N - 1.$$

Thus, $q_N = q$. Since π has log-Sobolev constant at most α^{-1} , hypercontractivity of simultaneous heat flow [KV26, Theorem 1.2 and Corollary 1.3] gives

$$R_{q_{n+1}}(\mu_n^\sharp * \gamma_{(2\beta)^{-1}} \parallel \pi * \gamma_{(2\beta)^{-1}}) \leq R_{q_n}(\mu_n^\sharp \parallel \pi).$$

Data processing through the exact backward RGO, followed by Rényi composition with the approximate RGO kernel, therefore yields

$$R_{q_{n+1}}(\mu_{n+1}^\sharp \parallel \pi) \leq R_{q_n}(\mu_n^\sharp \parallel \pi) + \Delta_0^2.$$

Iterating and using $N_1 \leq C\kappa \log(eq)$, we obtain

$$R_q(\mu_N^\sharp \parallel \pi) \leq R_2(\mu_{N_0}^\sharp \parallel \pi) + N_1 \Delta_0^2 \leq \Delta^2.$$

Under A2, the TV errors also sum to at most $\delta/2$.

Outer reference points. Under A1 and A3, the preceding bounds on the KL divergence, followed by Talagrand's inequality and smoothness, bounds the second moment of $\nabla V(Y_n)$. If we run gradient descent starting from Y_n , Jensen's inequality implies that we can find an admissible reference point to apply Theorem 6.5 using logarithmic expected work per outer step.

Under A2, the implementable chain need not have controlled moments of all orders, so cap the outer reference gradient descent after

$$N_{\text{cap}} := \left\lceil \log_2 \frac{C\kappa N}{\delta} \right\rceil$$

steps. Markov's inequality shows that the sum of the probabilities that we hit N_{cap} iterations is at most $\delta/2$. Together with TV bounds from the preceding argument, this gives $\text{TV}(\hat{\mu}_N, \mu_N^\dagger) \leq \delta$. Set $\hat{\pi} := \hat{\mu}_N$, and, under A2, set $\hat{\pi}^\dagger := \mu_N^\dagger$.

Query cost. The choice above satisfies $N \leq C\kappa \mathfrak{L}_2$ under A1, and $N \leq C\kappa \mathfrak{L}_q$ under A2 and A3. Since the capped outer reference construction uses at most

$$C\kappa \mathfrak{L}_q \left(\mathfrak{L}_q + \log \frac{C\kappa \mathfrak{L}_q}{\delta} \right)$$

queries, and in the other two cases the cost is at most the final displayed term, Theorem 6.5 and careful bookkeeping proves the result. $\square$

7 Picard HMC warm starts and the high-accuracy sampler

7.1 Picard HMC warm starts

We first state a strengthened form of Theorem 1.2. In particular, strong log-concavity is replaced by log-smoothness and a logarithmic Sobolev inequality, and we also state KL divergence guarantee.

Theorem 7.1 (Picard HMC warm starts). *Suppose that $\pi \propto \exp(-V)$ is β -log-smooth and satisfies the logarithmic Sobolev inequality with constant α^{-1} . Write $\kappa := \beta/\alpha$, and suppose that we are given a point x_{ref} satisfying $V(x_{\text{ref}}) - \inf V \leq d$. For every $0 < \Delta, \delta \leq 1/2$, Picard HMC within the recursive warm start generator satisfies the following guarantees.*

(i) It returns a sample with law $\hat{\pi}$ satisfying

$$\text{KL}(\hat{\pi} \parallel \pi) \leq \Delta^2.$$

The expected number of gradient queries is at most

$$C \left(\kappa^{7/6} d^{1/6} \Delta^{-1/3} + \kappa \right) \log^7 \frac{e\kappa d}{\Delta}.$$

(ii) It returns a sample with law $\hat{\pi}_\delta$ for which there exists $\hat{\pi}_\delta^\dagger$ satisfying

$$\text{TV}(\hat{\pi}_\delta, \hat{\pi}_\delta^\dagger) \leq \delta, \quad \text{R}_2(\hat{\pi}_\delta^\dagger \parallel \pi) \leq \Delta^2.$$

The expected number of gradient queries is at most

$$C \left(\kappa^{7/6} d^{1/6} \Delta^{-1/3} + \kappa \right) \log^7 \frac{e\kappa d}{\Delta \delta}.$$

Proof. We verify that Picard HMC implements [A1](#) and [A2](#), and then invoke [Theorem 6.1](#). Fix a normalized potential U with condition number κ_U , let $p \geq 2$, and put $\bar{\varepsilon} := \min\{\varepsilon, \kappa_U^{-1/2}\}$. Apply [Theorem 5.1](#) with accuracy $\sqrt{\kappa_U} \bar{\varepsilon}$. It returns $0 < \eta \leq c_0$ and a law $\hat{\pi}_{U,\eta}$ satisfying

$$\kappa_U^{-1/2} W_p(\hat{\pi}_{U,\eta}, \pi_{U,\eta}) \leq \bar{\varepsilon} \leq \varepsilon.$$

Thus Picard HMC implements [A2](#) with complexity

$$C_p(\kappa_U, d, \varepsilon) \leq C \left(\kappa_U^2 + \kappa_U^{7/6} (d+p)^{1/6} \bar{\varepsilon}^{-1/3} \right) \left(p + \log \frac{e\sqrt{\kappa_U} d}{\bar{\varepsilon}} \right)^{9/2}.$$

Taking $p = 2$ also implements [A1](#). Careful bookkeeping shows that the costs are as displayed. $\square$

Under (1.1), the reference assumption of [Theorem 1.2](#) implies

$$V(x_{\text{ref}}) - \inf V \leq \frac{\|\nabla V(x_{\text{ref}})\|^2}{2\alpha} \leq \frac{d}{2}.$$

Thus [Theorem 1.2](#) follows from [Theorem 7.1\(ii\)](#) by taking $\Delta = 1/2$. Part (i) also gives a genuine KL warm start under the more general assumptions of [Theorem 7.1](#).

7.2 From the proxy warm start to high accuracy

We now record the guarantee for proximal BPS.

Theorem 7.2 (Proximal BPS). *Assume $V \in C^2$. Let $\Delta \geq 1$ and $0 < \tau < 1/4$, and set*

$$\mathfrak{L}_{\text{BPS}} := \Delta^2 + \log \frac{Cd\kappa}{\tau}.$$

Then, the proximal BPS Markov kernel $K_{\tau,\Delta}$ satisfies the following guarantee: for every λ_0 with $\text{R}_2(\lambda_0 \parallel \pi) \leq \Delta^2$, it holds that $\text{TV}(\lambda_0 K_{\tau,\Delta}, \pi) \leq \tau$. Moreover, if $N_{\nabla V}$ is the total number of gradient queries made by this kernel, then

$$\mathbb{E} N_{\nabla V} \leq C\sqrt{\kappa} \left(d\mathfrak{L}_{\text{BPS}} + \mathfrak{L}_{\text{BPS}}^2 \right)^{1/4} \mathfrak{L}_{\text{BPS}}^{3/2} \log(\kappa \mathfrak{L}_{\text{BPS}}). \quad (7.1)$$

Proof. This is Algorithm 4.2, Theorem 4.3, and Corollary 4.4 of [CCLZ26a]. $\square$

Proof of Theorem 1.3. Apply Theorem 7.1(ii) with $\Delta = 1/2$ and $\delta = \varepsilon/2$. This produces an implementable law $\hat{\pi}_{\varepsilon/2}$ and a comparison law $\hat{\pi}_{\varepsilon/2}^\dagger$ such that

$$\text{TV}(\hat{\pi}_{\varepsilon/2}, \hat{\pi}_{\varepsilon/2}^\dagger) \leq \frac{\varepsilon}{2}, \quad \text{R}_2(\hat{\pi}_{\varepsilon/2}^\dagger \parallel \pi) \leq \frac{1}{4}.$$

Run the proximal BPS kernel $K_{\varepsilon/2,1}$ from the implementable law. By data processing,

$$\text{TV}(\hat{\pi}_{\varepsilon/2} K_{\varepsilon/2,1}, \hat{\pi}_{\varepsilon/2}^\dagger K_{\varepsilon/2,1}) \leq \frac{\varepsilon}{2}.$$

Since $\hat{\pi}_{\varepsilon/2}^\dagger$ satisfies the hypothesis of Theorem 7.2 with $\Delta = 1$, its image under this kernel is within $\varepsilon/2$ of π in total variation. The triangle inequality therefore gives the desired error ε .

Put $\mathfrak{L} := \log(e\kappa d/\varepsilon)$. The warm start cost is at most $C\kappa^{7/6}d^{1/6}\mathfrak{L}^7$, since $d, \kappa \geq 1$. For the subsequent proximal BPS call, $\Delta = 1$ and $\tau = \varepsilon/2$, so $\mathfrak{L}_{\text{BPS}} \leq C\mathfrak{L}$. Thus (7.1) gives the cost bound

$$C\sqrt{\kappa}(d^{1/4}\mathfrak{L}^{7/4} + \mathfrak{L}^2) \log(\kappa\mathfrak{L}) \leq C\sqrt{\kappa}(d^{1/4}\mathfrak{L}^{11/4} + \mathfrak{L}^3).$$

Here we used $\log(\kappa\mathfrak{L}) \leq C\mathfrak{L}$. The remaining term $\sqrt{\kappa}\mathfrak{L}^3$ is absorbed by $\kappa^{7/6}d^{1/6}\mathfrak{L}^7$, proving the stated query bound. $\square$

A Terminal RGO implementation by FORS

This appendix records the FORS subroutine that we use in the paper. After reviewing the generic FORS mechanism, we describe a variant that is tailored to sampling from RGO distributions. Since the main arguments here were already introduced in prior works, we will be more terse and only emphasize the differences from the analyses of [CCDR26, CCRZ26].

A.1 Review of first-order rejection sampling (FORS)

For $q \geq 1$, write

$$\bar{D}_q(P, Q) := \max \left\{ \int \left(\frac{dP}{dQ} \right)^q dQ - 1, \int \left(\frac{dQ}{dP} \right)^q dP - 1 \right\}.$$

Either term is understood to be infinite when the corresponding absolute continuity condition fails.

Let $\mathbf{Q}$ be a proposal law for a path $\mathbf{Z}$, and suppose the desired path law satisfies

$$\frac{d\mathbf{P}_\star}{d\mathbf{Q}}(\mathbf{Z}) \propto \exp\{w_\star(\mathbf{Z})\}.$$

Given an auxiliary variable $\xi \sim \Xi$, FORS uses an estimator $W(\xi; \mathbf{Z})$ with $w(\mathbf{Z}) := \mathbb{E}_{\xi \sim \Xi} W(\xi; \mathbf{Z})$. At clipping level $B > 0$, its implemented path law $\hat{\mathbf{P}}$ satisfies

$$\frac{d\hat{\mathbf{P}}}{d\mathbf{Q}}(\mathbf{Z}) \propto \exp\left\{ \mathbb{E}_{\xi \sim \Xi} \text{clip}_{[-B, B]}(W(\xi; \mathbf{Z})) \right\}.$$

Lemma A.1 (FORS likelihood comparison). *Let $P_\star$ and $\hat{P}$ be the laws of the terminal endpoint of $\mathbf{Z}$ under $\mathbf{P}_\star$ and $\hat{\mathbf{P}}$, respectively. For every $q \geq 2$,*

$$2 \log\{1 + \bar{D}_q(\hat{P}, P_\star)\} \leq \log \mathbb{E}_{\mathbf{P}_\star} \exp\{4q |w_\star(\mathbf{Z}) - w(\mathbf{Z})|\} + \log \mathbb{E}_{\substack{\mathbf{Z} \sim \mathbf{P}_\star \\ \xi \sim \Xi}} \exp\{4q (|W(\xi; \mathbf{Z})| - B)_+\}.$$

Proof. This is [CCRZ26, Lemma 2.4]. □

To draw a sample from the law $\hat{\mathbf{P}}$, we use the following mechanism.

Lemma A.2 (Poisson exponentiation). *Let $X \sim Q$. Suppose that, conditionally on $X = x$, one can simulate a random variable $W_x \in [-B, B]$ with $w(x) := \mathbb{E}[W_x \mid X = x]$. Draw $N \sim \text{Poisson}(2B)$ and, conditionally on $X = x$, independent copies $W_x^{(1)}, \dots, W_x^{(N)}$. Accept x with probability*

$$\prod_{k=1}^N \frac{B + W_x^{(k)}}{2B}.$$

Then the accepted proposal has law proportional to $\exp(w)Q$. The acceptance probability of one attempt is at least $\exp(-2B)$.

Proof. See [CCRZ26, Algorithm 1 and Theorem 2.3]. □

A.2 FORS for RGO distributions

Given a convex potential V , a variance $a > 0$, and a center y , let

$$R_{a,y} \propto \exp\left\{-V(\cdot) - \frac{\|\cdot - y\|^2}{2a}\right\}.$$

Here, we describe a variant of FORS diffusion simulation, specialized to distributions of this form. Compared to [CCRZ26], we exactly integrate the linear part of the drift in the underdamped Langevin proposal. Namely, from (x, p) , the proposal is the path measure of

$$\begin{aligned} dX_t &= P_t dt, \\ dP_t &= -\left\{\frac{X_t - y}{a} + \nabla V(x) + \gamma P_t\right\} dt + \sqrt{2\gamma} dB_t, \quad c_0 a^{-1/2} \leq \gamma \leq C_0 a^{-1/2}. \end{aligned} \quad (\text{A.1})$$

The full algorithm is described below.

Algorithm A.3 (FORS for RGO). Fix a block length $T > 0$, a number N of blocks, a mesh width h_{mesh} dividing T , a clipping level $B_{\text{clip}} > 0$, and a stopping set $\mathcal{E} \subseteq \mathbb{R}^d \times \mathbb{R}^d$. Compute $x^+ := \text{prox}_{aV}(y)$. On each attempt, initialize

$$X^0 \sim \mathcal{N}\left(x^+, \frac{a}{1+a\beta}I\right) \quad \text{and} \quad P^0 \sim \mathcal{N}(0, I)$$

independently. Restart the attempt if $(X^0, P^0) \notin \mathcal{E}$.

For $n = 0, \dots, N-1$, repeat the following proposal until it is accepted. Starting from $x = X^n$ and $p = P^n$, draw $M \sim \text{Poisson}(2B_{\text{clip}})$ and independent $t_j \sim \text{Unif}[0, T]$, $j \in [M]$. Jointly sample the linear proposal (A.1), its driving Brownian motion, and all values needed below. Let $r_j \in [T/h_{\text{mesh}}]$ be the unique index such that $(r_j - 1)h_{\text{mesh}} \leq t_j < r_j h_{\text{mesh}}$, and set

$$\begin{aligned} \mu_t &:= \frac{\nabla V(X_t) - \nabla V(x)}{\sqrt{2\gamma}}, \\ -\mathcal{W}(t_j) &:= \frac{T}{\sqrt{2\gamma} h_{\text{mesh}}} \langle B_T - B_{r_j h_{\text{mesh}}}, \nabla V(X_{r_j h_{\text{mesh}}}) - \nabla V(X_{(r_j-1)h_{\text{mesh}}}) \rangle + \frac{T}{2} \|\mu_{t_j}\|^2. \end{aligned}$$

Accept the proposed endpoint (X_T, P_T) with probability

$$\prod_{j \in [M]} \frac{B_{\text{clip}} + \text{clip}_{[-B_{\text{clip}}, B_{\text{clip}}]}(\mathcal{W}(t_j))}{2B_{\text{clip}}}.$$

Upon acceptance, set $(X^{n+1}, P^{n+1}) := (X_T, P_T)$. If the accepted state lies outside $\mathcal{E}$, restart from the initialization step. After N accepted blocks, return X^N .

Theorem A.4 (FORS for RGO distributions). *Let $V : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex and β -smooth, and let $R_{a,y}$ be the law above. Fix $q \geq 2$ and $0 < \varepsilon, p \leq 1/2$.*

1. *If $0 < a\beta \leq 1$, there is a randomized routine using evaluations of ∇V and one evaluation of $\text{prox}_{aV}(y)$ whose output law $\hat{R}_{a,y}$ satisfies*

$$\sup_{y \in \mathbb{R}^d} R_q(\hat{R}_{a,y} \parallel R_{a,y}) \leq \varepsilon^2. \quad (\text{A.2})$$

Writing

$$\mathfrak{L} := q + \log \frac{Cd}{a\beta\varepsilon p},$$

its expected number of queries and, with probability at least $1 - p$, its realized number of queries are at most

$$C\mathfrak{L}^{5/3} \{1 + ((a\beta)^2 d)^{1/3}\}. \quad (\text{A.3})$$

2. Suppose that a point x_+ is supplied such that

$$\|y - a\nabla V(x_+) - x_+\| \leq \sqrt{da}.$$

If

$$a^{-1} \geq C\beta [\sqrt{d(q + \log(1/\varepsilon))} + q + \log(1/\varepsilon)],$$

then a gradient-only routine returns a law satisfying (A.2). It uses $O(1)$ expected evaluations of ∇V , and at most $C \log(2/p)$ evaluations with probability at least $1 - p$.

Proof. The second statement is the smooth case of [CCDR26, Theorem D.1]. We therefore focus on the first statement, which is based on Algorithm A.3.

By rescaling, it suffices to consider $\beta = 1$. Let $x^+ := \text{prox}_{aV}(y)$, set $x = x^+ + \sqrt{a}z$, and define

$$F(z) := V(x^+ + \sqrt{a}z) - V(x^+) - \sqrt{a} \langle \nabla V(x^+), z \rangle.$$

After a change of variables, we consider the density proportional to

$$z \mapsto \exp\left\{-\frac{1}{2}\|z\|^2 - F(z)\right\}, \quad F(0) = 0, \quad \nabla F(0) = 0, \quad 0 \preceq \nabla^2 F \preceq aI.$$

Writing $g := \nabla F$ and $\bar{\gamma} := \sqrt{a}\gamma$, the corresponding space-time rescaling transforms (A.1), conditionally on the blockwise initial state $(Z_0, P_0) = (z, p)$, into

$$\begin{aligned} dZ_t &= P_t dt, \\ dP_t &= -\{Z_t + g(z) + \bar{\gamma}P_t\} dt + \sqrt{2\bar{\gamma}} dB_t, \quad c_0 \leq \bar{\gamma} \leq C_0. \end{aligned} \quad (\text{A.4})$$

We explain the only changes needed in the FORS proof of [CCRZ26]. Let S solve

$$S'' + \bar{\gamma}S' + S = 0, \quad S(0) = 0, \quad S'(0) = 1, \quad b_t := \int_0^t S_r dr.$$

For bounded $\bar{\gamma}$ and sufficiently small t , $|S_t| \leq Ct$, $|b_t| \leq Ct^2$, and $|S'_t| + |S''_t| \leq C$. Under (A.4),

$$Z_t - z = S_t p - b_t \{z + g(z)\} + \sqrt{2\bar{\gamma}} \int_0^t S_{t-s} dB_s. \quad (\text{A.5})$$

Consequently the Girsanov control contains only the residual force,

$$\mu_t := \frac{g(Z_t) - g(z)}{\sqrt{2\bar{\gamma}}}.$$

This is the main improvement due to exact integration of the linear term.

Now put

$$\mathcal{R}(z, p) := \|p\|^2 + \|z\|^2 + a^{-1} \|g(z)\|^2 + d.$$

The calculations in [CCRZ26, Lemmas B.1 and B.3], applied to (A.5), gives the same estimate as in that paper with its full smoothness replaced by the residual smoothness a . In particular, the random likelihood factor has sub-Gaussian scale

$$\log \mathbb{E} \exp\{\lambda |\mathcal{W}_t|\} \leq \log 2 + C\lambda^2 \frac{a^2 T^3}{\bar{\gamma}} \mathcal{R}(z, p), \quad 1 \leq \lambda \leq \frac{c}{aT^2}. \quad (\text{A.6})$$

If the control is frozen on a mesh of width h_{mesh} , its discretization error $\mathcal{E}_{\text{mesh}}$ satisfies

$$\log \mathbb{E} \exp\{\lambda |\mathcal{E}_{\text{mesh}}|\} \leq C\lambda \frac{ah_{\text{mesh}}}{\bar{\gamma}} (d^{-1/2} + aT^2) \mathcal{R}(z, p). \quad (\text{A.7})$$

The proofs are the same as [CCRZ26, Lemma C.1 and the proof of Proposition 3.1]; the bounds on S_t above replace the exponential Euler bounds there.

Applying Lemma A.1 using Eqs. (A.6) and (A.7), and then choosing the clipping level and mesh exactly as in that proof, gives the required error on one block. The stopped composition and query tail arguments are unchanged from [CCRZ26, Theorem B.11]. One $\mathcal{W}_t$ factor uses g at no more than three times. Since the linear process (Z, P, B) can be sampled by Gaussian regression at only those times and the two adjacent mesh points, refining the mesh makes no additional gradient queries.

Initialize from $\mathcal{N}(0, (1+a)^{-1}I) \otimes \mathcal{N}(0, I)$. The bounds

$$\frac{1}{2} \|z\|^2 \leq \frac{1}{2} \|z\|^2 + F(z) \leq \frac{1+a}{2} \|z\|^2$$

imply that the initial $\mathcal{R}_{2q}$ divergence is at most $ad/2$. For the choices below, the logarithmic factor in the stopped composition argument satisfies $2q + \log \frac{CNq}{\epsilon^2} \leq C\mathfrak{L}$. Take

$$T := c\mathfrak{L}^{-2/3} \min\{1, (a^2d)^{-1/3}\} \quad \text{and} \quad N := \lceil C\mathfrak{L}/T \rceil.$$

Indeed, the cubic restriction from Eq. (A.6) is $T^3 \leq c/(a^2d\mathfrak{L}^2)$, while the quadratic restriction is $T^2 \leq c/(a\mathfrak{L})$; the displayed choice satisfies both. The mesh is then refined until Eq. (A.7) gives the required path accuracy, without additional gradient queries. Since the target is 1-strongly log-concave, a total time $NT \geq C\mathfrak{L}$ suffices in the LSI mixing and Rényi composition argument of [CCRZ26, Theorem 3.2(ii)], run at Rényi order $2q$. This yields (A.2). Moreover,

$$N \leq C\mathfrak{L}^{5/3} \{1 + (a^2d)^{1/3}\}.$$

The stopped implementation uses $O(N)$ expected queries and $O(N + \log(1/p))$ queries with probability at least $1 - p$. Since $\log(1/p) \leq \mathfrak{L}$, this proves (A.3). $\square$

B Chebyshev–Lobatto interpolation and quadrature

This appendix collects the interpolation facts used by the Picard integrator.

B.1 Nodes, interpolating polynomials, and integrated weights

For an integer $J \geq 2$ indicating the total number of nodes, define the increasing Chebyshev–Lobatto nodes on $[0, h]$ by

$$t_j := \frac{h}{2} \left\{ 1 - \cos\left(\frac{j-1}{J-1}\pi\right) \right\}, \quad 1 \leq j \leq J.$$

Their interpolating polynomials are

$$\ell_j(t) := \prod_{\substack{1 \leq k \leq J \\ k \neq j}} \frac{t - t_k}{t_j - t_k}, \quad \ell_j(t_k) = \mathbf{1}_{\{j=k\}}.$$

Thus, the operator $\mathcal{I}_{J,h}$ in (3.5) is the unique polynomial interpolant of degree at most $J - 1$. Its Lebesgue constant is defined as

$$\Lambda_J := \sup_{0 \leq t \leq h} \sum_{j=1}^J |\ell_j(t)|. \quad (\text{B.1})$$

For a normed space E and $\{f(t_j)\}_{j=1}^J \subseteq E$, the triangle inequality immediately gives

$$\sup_{0 \leq t \leq h} \|(\mathcal{I}_{J,h}f)(t)\|_E \leq \Lambda_J \max_{j \in [J]} \|f(t_j)\|_E.$$

Recall the quadrature coefficients $\omega_{i,j}$ and ω_j from (3.6). By their definition, for every polynomial f of degree at most $J-1$,

$$\int_0^{t_i} (t_i - t) f(t) dt = \sum_{j=1}^J \omega_{i,j} f(t_j), \quad \int_0^h f(t) dt = \sum_{j=1}^J \omega_j f(t_j).$$

Proposition B.1 (Chebyshev–Lobatto coefficient bounds). *For every $J \geq 2$,*

$$\max_{i \in [J]} \sum_{j \in [J]} |\omega_{i,j}| \leq h^2, \quad \omega_j \geq 0, \quad \sum_{j \in [J]} \omega_j = h, \quad \Lambda_J \leq \frac{2}{\pi} \log J + 1. \quad (\text{B.2})$$

Proof. The classical estimate for Λ_J can be found in [Tre19, Theorem 15.2]. The first estimate follows from Lemma B.2 and Remark B.3 below. The ω_j 's are the Clenshaw–Curtis weights, which are non-negative. Since the rule integrates constants exactly on $[0, h]$, they sum to h . $\square$

Lemma B.2 (Square estimates). *For every $J \geq 2$,*

$$\max_{i \in [J]} \sum_{j \in [J]} \omega_{i,j}^2 \leq \frac{5\pi^2 h^4}{128(J-1)}, \quad \sum_{j \in [J]} \omega_j^2 \leq \frac{\pi^2 h^2}{8(J-1)}.$$

Proof. For $z = (z_j)_{j \in [J]} \in \mathbb{R}^J$, let $f := \mathcal{I}_{J,h}z = \sum_{j \in [J]} z_j \ell_j$. The discrete norm comparison in [ST06, Lemma 1.3.1], with the Chebyshev–Lobatto weights in [ST06, (1.3.11)], gives

$$\int_0^h \frac{|f(s)|^2}{\sqrt{s(h-s)}} ds \leq \frac{\pi}{J-1} \left\{ \frac{|z_1|^2 + |z_J|^2}{2} + \sum_{j=2}^{J-1} |z_j|^2 \right\} \leq \frac{\pi}{J-1} \|z\|^2.$$

Indeed, the change of variables $x = 1 - 2s/h$ sends t_j to $\cos(\frac{j-1}{J-1}\pi)$ and the measure $ds/\sqrt{s(h-s)}$ to $dx/\sqrt{1-x^2}$, with the limits reversed. The two endpoint weights are $\frac{\pi}{2(J-1)}$ and every interior weight is $\frac{\pi}{J-1}$.

For a continuous function $k : [0, h] \rightarrow \mathbb{R}$, put $a_j(k) := \int_0^h k(s) \ell_j(s) ds$, $j \in [J]$. Cauchy–Schwarz with the reciprocal weights $1/\sqrt{s(h-s)}$ and $\sqrt{s(h-s)}$ yields

$$\begin{aligned} \left| \sum_{j \in [J]} a_j(k) z_j \right|^2 &= \left| \int_0^h f(s) k(s) ds \right|^2 \leq \left(\int_0^h \frac{|f(s)|^2}{\sqrt{s(h-s)}} ds \right) \left(\int_0^h |k(s)|^2 \sqrt{s(h-s)} ds \right) \\ &\leq \frac{\pi \|z\|^2}{J-1} \int_0^h |k(s)|^2 \sqrt{s(h-s)} ds. \end{aligned}$$

Taking the supremum over $\|z\| = 1$ gives

$$\sum_{j \in [J]} a_j(k)^2 \leq \frac{\pi}{J-1} \int_0^h |k(s)|^2 \sqrt{s(h-s)} ds.$$

For $k = (t_i - \cdot)_+$, these coefficients are $a_j(k) = \omega_{i,j}$. Since $0 \leq (t_i - s)_+ \leq h - s$ for $s \in [0, h]$,

$$\int_0^h (t_i - s)_+^2 \sqrt{s(h-s)} \, ds \leq \int_0^h (h-s)^2 \sqrt{s(h-s)} \, ds = \frac{5\pi h^4}{128}.$$

This proves the first estimate, uniformly in $i \in [J]$. For $k = 1$, we have $a_j(k) = \omega_j$ and

$$\int_0^h \sqrt{s(h-s)} \, ds = \frac{\pi h^2}{8},$$

which proves the second estimate. $\square$

Remark B.3 (Refined weight estimate). By Cauchy–Schwarz and $J/(J-1) \leq 2$, [Lemma B.2](#) gives

$$\max_{i \in [J]} \sum_{j \in [J]} |\omega_{i,j}| \leq \sqrt{\frac{5\pi^2 J}{128(J-1)}} h^2 \leq \frac{\pi\sqrt{5}}{8} h^2 \leq h^2.$$

The same bound holds with $\omega_{i,j}$ replaced by $\int_0^t (t-s)\ell_j(s) \, ds$, uniformly over $t \in [0, h]$: the proof above only uses $0 \leq (t-s)_+ \leq h-s$.

B.2 Interpolation error bound

We record the standard interpolation error bound.

Lemma B.4 (Banach-valued interpolation error bound). *Let E be a Banach space and let $f \in C^J([0, h]; E)$. Then*

$$\|f(t) - (\mathcal{I}_{J,h}f)(t)\|_E \leq \frac{1}{J!} \sup_{0 \leq s \leq h} \|f^{(J)}(s)\|_E \prod_{j=1}^J |t - t_j| \leq \frac{h^J}{J!} \sup_{0 \leq s \leq h} \|f^{(J)}(s)\|_E.$$

This is the Banach-valued form of the Genocchi–Hermite interpolation remainder; see [[dB05](#), Section 9]. Apply the scalar argument to $\ell \circ f$, where $\ell \in E^*$ has unit norm. Taking the supremum over all such ℓ concludes the proof.

C Gaussian analysis

This appendix collects the Gaussian analysis used throughout the paper.

C.1 Basic inequalities and heat smoothing

We recall the definition of a Wiener chaos. The n -th Wiener chaos is the subspace of $L^2(\gamma)$ consisting of polynomials of degree at most n that are orthogonal to every polynomial of degree less than n . The zeroth chaos consists of constants.

Derivative and divergence inequalities. Fix $d \geq 1$ and let γ be standard Gaussian measure on $\mathbb{R}^d$. We also let $\mathcal{H}$ be a finite-dimensional¹ Hilbert space, and let $V = (V_i)_{i \in [d]} : \mathbb{R}^d \rightarrow \mathcal{H}$ be smooth. Its Gaussian divergence is

$$\delta_\gamma V := \sum_{i \in [d]} (x_i V_i - \partial_i V_i).$$

It is the $L^2(\gamma)$ -adjoint of the gradient:

$$\mathbb{E}_\gamma \langle DF, V \rangle_{\text{HS}} = \mathbb{E}_\gamma \langle F, \delta_\gamma V \rangle_{\mathcal{H}}. \quad (\text{C.1})$$

Here, $D = \nabla$ is the usual gradient.

Componentwise Gaussian Poincaré and two applications of (C.1) give

$$\|F - \mathbb{E}F\|_{L^2(\gamma)} \leq \|DF\|_{L^2(\gamma; \text{HS})}, \quad \|\delta_\gamma V\|_{L^2(\gamma)}^2 \leq \|V\|_{L^2(\gamma; \text{HS})}^2 + \|DV\|_{L^2(\gamma; \text{HS})}^2. \quad (\text{C.2})$$

Indeed, the exact identity behind the second inequality is

$$\|\delta_\gamma V\|_{L^2(\gamma)}^2 = \|V\|_{L^2(\gamma; \text{HS})}^2 + \sum_{i, j \in [d]} \mathbb{E} \langle \partial_j V_i, \partial_i V_j \rangle_{\mathcal{H}};$$

the last sum is at most $\|DV\|_{L^2(\gamma; \text{HS})}^2$ by Cauchy–Schwarz. If $\gamma_t := \mathcal{N}(0, tI)$, then

$$\delta_{\gamma_t} V := \sum_{i \in [d]} (t^{-1} x_i V_i - \partial_i V_i),$$

and the change of variables $x = \sqrt{t} z$ gives

$$\|\delta_{\gamma_t} V\|_{L^2(\gamma_t)}^2 \leq t^{-1} \|V\|_{L^2(\gamma_t; \text{HS})}^2 + \|DV\|_{L^2(\gamma_t; \text{HS})}^2. \quad (\text{C.3})$$

For the L^q analogues, we use the following inequalities.

Lemma C.1 (Gaussian Sobolev inequalities). *Let $q \geq 2$. If $F \in W^{1,q}(\gamma; \mathcal{H})$, then*

$$\|F - \mathbb{E}F\|_{L^q(\gamma; \mathcal{H})} \leq C \sqrt{q} \|DF\|_{L^q(\gamma; \text{HS})}.$$

If $V \in W^{1,q}(\gamma; \mathcal{H}^d)$ and $\delta_\gamma V$ is its row divergence, then

$$\|\delta_\gamma V\|_{L^q(\gamma; \mathcal{H})} \leq C \sqrt{q} \|\mathbb{E}V\|_{\text{HS}} + Cq \|DV\|_{L^q(\gamma; \text{HS})}. \quad (\text{C.4})$$

If we replace γ by γ_t , the centered Gaussian measure of covariance tI , then the first term on the right side of (C.4) is multiplied by $t^{-1/2}$.

Proof. The first estimate follows from Pisier’s Gaussian convexity inequality [Pis86, Theorem 2.2], applied with the convex function $u \mapsto \|u\|_{\mathcal{H}}^2$. The divergence estimate is [CCLZ26b, Corollary 1.4], applied to the Hilbert target $\mathcal{H}$. The last sentence follows from rescaling. $\square$

¹We expect that the results can be extended to the infinite-dimensional setting, but this will not be needed here.

Dual Poincaré inequality. Let $\mathcal{L}_{\text{OU}} := \Delta - \langle x, \nabla \rangle = -\delta_\gamma D$ be the Ornstein–Uhlenbeck generator. On the n -th Wiener chaos, $\mathcal{L}_{\text{OU}} F = -nF$; its inverse below is taken on centered functions and extended by zero on constants.

For a centered, finite-dimensional Hilbert-valued $F \in L^2(\gamma; \mathcal{H})$, define

$$\|F\|_{\dot{H}^{-1}(\gamma; \mathcal{H})} := \sup \{ \mathbb{E} \langle F, H \rangle_{\mathcal{H}} : \mathbb{E} H = 0, \quad \|DH\|_{L^2(\gamma; \text{HS})} \leq 1 \}.$$

We then have

$$\|F\|_{\dot{H}^{-1}(\gamma; \mathcal{H})} = \|D(-\mathcal{L}_{\text{OU}})^{-1}F\|_{L^2(\gamma; \text{HS})}.$$

The following is the dual form of the Gaussian Poincaré inequality, see [Kla13].

Lemma C.2 (Dual Gaussian Poincaré inequality). *For all such F ,*

$$\|F\|_{\dot{H}^{-1}(\gamma; \mathcal{H})} \leq \|F\|_{L^2(\gamma; \mathcal{H})}.$$

Heat equation estimates. For later reference, let $(P_t)_{t \geq 0}$ denote the Euclidean heat semigroup,

$$(P_t F)(x) := \mathbb{E} F(x + \sqrt{t} G).$$

Thus $\partial_t P_t F = \frac{1}{2} \Delta P_t F$. If F is weakly differentiable, differentiation commutes with the heat semigroup, whereas Gaussian integration by parts gives the derivative formula

$$DP_t F(x) = P_t(DF)(x) = \frac{1}{\sqrt{t}} \mathbb{E}[F(x + \sqrt{t} G) \otimes G]. \quad (\text{C.5})$$

In particular, Jensen and Bessel's inequality give

$$\begin{aligned} \|DP_t F(x)\|_{\text{op}} &\leq \text{Lip}(F), \\ \|DP_t F(x)\|_{\text{HS}}^2 &\leq \frac{1}{t} \inf_{a \in \mathcal{H}} \mathbb{E} \|F(x + \sqrt{t} G) - a\|_{\mathcal{H}}^2. \end{aligned} \quad (\text{C.6})$$

To justify the second estimate, fix $a \in \mathcal{H}$ and put

$$Y := F(x + \sqrt{t} G) - a.$$

The orthogonal projection of $Y \in L^2(\gamma; \mathcal{H})$ onto the $\mathcal{H}$ -valued first Gaussian chaos is

$$\mathcal{P}_1 Y := \sum_{i \in [d]} G_i \mathbb{E}[G_i Y].$$

Since the coordinate functions $(G_i)_{i \in [d]}$ are orthonormal in $L^2(\gamma)$, Bessel's inequality gives

$$\sum_{i \in [d]} \|\mathbb{E}[G_i Y]\|_{\mathcal{H}}^2 = \mathbb{E} \|\mathcal{P}_1 Y\|_{\mathcal{H}}^2 \leq \mathbb{E} \|Y\|_{\mathcal{H}}^2.$$

Moreover, $\mathbb{E}[a \otimes G] = 0$, so (C.5) and the preceding display imply

$$t \|DP_t F(x)\|_{\text{HS}}^2 = \sum_{i \in [d]} \|\mathbb{E}[G_i Y]\|_{\mathcal{H}}^2 \leq \mathbb{E} \|F(x + \sqrt{t} G) - a\|_{\mathcal{H}}^2.$$

Taking the infimum over $a \in \mathcal{H}$ proves (C.6).

C.2 Cumulants and derivatives of smoothed potentials

For random variables $F_1, \dots, F_k$ whose joint moment-generating function is finite near the origin, define their joint cumulant by

$$\text{Cum}(F_1, \dots, F_k) := \partial_{t_1} \cdots \partial_{t_k} \log \mathbb{E} \exp \left\{ \sum_{j=1}^k t_j F_j \right\} \Big|_{t_1 = \dots = t_k = 0}.$$

If X is vector-valued, $\text{Cum}_k(X)$ denotes the symmetric k -tensor obtained by applying this definition to linear functionals of X .

We record the connection with derivatives of the smoothed potential. Let $X_y \sim R_{\eta, y}$, and set

$$Z_\eta(y) := \int \exp \left\{ -V(x) - \frac{\|x\|^2}{2\eta} + \frac{\langle y, x \rangle}{\eta} \right\} dx.$$

Up to an additive constant,

$$V_\eta(y) = \frac{\|y\|^2}{2\eta} - \log Z_\eta(y).$$

Since $\log Z_\eta$ is the cumulant-generating function of the exponential family $R_{\eta, y}$, differentiation gives, for every $k \geq 2$,

$$D^k V_\eta(y) = \mathbf{1}_{\{k=2\}} \eta^{-1} I - \eta^{-k} \text{Cum}_k(X_y). \quad (\text{C.7})$$

We need the following recursive representation of cumulants.

Lemma C.3 (Iterated Gaussian covariance). *Let $F_1, \dots, F_k$ be smooth scalar functions of a standard Gaussian, with bounded derivatives of every positive order. Define*

$$\begin{aligned} \Gamma_1(F_1) &:= F_1 - \mathbb{E} F_1, \\ \Gamma_j(F_1, \dots, F_j) &:= \langle DF_j, D(-\mathcal{L}_{\text{OU}})^{-1} \Gamma_{j-1}(F_1, \dots, F_{j-1}) \rangle, \quad j \geq 2. \end{aligned} \quad (\text{C.8})$$

Then

$$\text{Cum}(F_1, \dots, F_k) = \sum_{\pi \in \mathfrak{S}_{k-1}} \mathbb{E} \Gamma_k(F_1, F_{\pi(2)}, \dots, F_{\pi(k)}),$$

where each permutation is over $\{2, \dots, k\}$.

Proof. This is a symmetrized form of [NN11, Theorem 4.4]. $\square$

Lemma C.4 (Cumulants of a near-identity Gaussian image). *Suppose $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is locally Lipschitz and, almost everywhere,*

$$\|DT\|_{\text{op}} \leq 1, \quad \|DT - I\|_{\text{op}} \leq \varepsilon_T, \quad 0 \leq \varepsilon_T \leq 1.$$

If G is standard Gaussian, then, for every $k \geq 3$,

$$\begin{aligned} \sup_{\|v\|=1} \|\text{Cum}_k(T(G))[v, \cdot^{k-1}]\|_{\text{HS}} &\leq 2(k-1)! \varepsilon_T, \\ \|\text{Cum}_k(T(G))\|_{\text{HS}} &\leq 2(k-1)! \varepsilon_T \sqrt{d}. \end{aligned} \quad (\text{C.9})$$

Here, $A[v, \cdot^{k-1}]$ denotes the $(k-1)$ -tensor obtained by fixing the first argument of a k -tensor A to v , leaving the remaining $k-1$ arguments free. In particular,

$$\|A[v, \cdot^{k-1}]\|_{\text{HS}}^2 = \sum_{j_2, \dots, j_k \in [d]} |A[v, e_{j_2}, \dots, e_{j_k}]|^2,$$

where $(e_j)_{j \in [d]}$ is any orthonormal basis of $\mathbb{R}^d$.

Proof. First take T smooth. For fixed v , define the $(j-1)$ -tensor $\Gamma_j(v)$ by

$$[\Gamma_j(v)]_{i_2, \dots, i_j} := \Gamma_j(\langle v, T(G) \rangle, T_{i_2}(G), \dots, T_{i_j}(G)), \quad i_2, \dots, i_j \in [d],$$

using the scalar recursion (C.8). For $j=2$, its i -th entry is $\langle DT_i(G), D(-\mathcal{L}_{OU})^{-1}(\langle v, T(G) \rangle) \rangle$, so collecting the entries gives $\Gamma_2(v) = DT(G) D(-\mathcal{L}_{OU})^{-1}(\langle v, T(G) \rangle)$. Moreover, $\langle v, G \rangle$ belongs to the first Wiener chaos, so $D(-\mathcal{L}_{OU})^{-1}(\langle v, G \rangle) = v$. Subtracting this linear contribution yields

$$\begin{aligned} \|\Gamma_2(v) - v\|_{L^2(\gamma)} &\leq \|(DT - I)v\|_{L^2(\gamma)} + \|DT(D(-\mathcal{L}_{OU})^{-1}(\langle v, T \rangle) - v)\|_{L^2(\gamma)} \\ &\leq \varepsilon_T \|v\| + \|DT D(-\mathcal{L}_{OU})^{-1}(\langle v, T - I \rangle)\|_{L^2(\gamma)} \\ &\leq \varepsilon_T \|v\| + \|D(-\mathcal{L}_{OU})^{-1}(\langle v, T - I \rangle)\|_{L^2(\gamma)} \\ &= \varepsilon_T \|v\| + \|\langle v, T - I \rangle\|_{\dot{H}^{-1}(\gamma)} \leq \varepsilon_T \|v\| + \|D\langle v, T - I \rangle\|_{L^2(\gamma)} \leq 2\varepsilon_T \|v\|, \end{aligned}$$

where we applied Lemma C.2 and the Gaussian Poincaré inequality. This implies

$$\|\Gamma_2(v) - \mathbb{E}\Gamma_2(v)\|_{L^2(\gamma; \text{HS})} \leq 2\varepsilon_T \|v\|. \quad (\text{C.10})$$

For $j \geq 2$, collect the free indices into a tensor. Then, Γ_{j+1} is a contraction of DT with $D(-\mathcal{L}_{OU})^{-1}\Gamma_j$; hence, by $\|DT\|_{\text{op}} \leq 1$ and Lemma C.2,

$$\|\Gamma_{j+1}(v)\|_{L^2(\gamma; \text{HS})} \leq \|D(-\mathcal{L}_{OU})^{-1}\Gamma_j(v)\|_{L^2(\gamma; \text{HS})} = \|\Gamma_j(v)\|_{\dot{H}^{-1}(\gamma; \text{HS})} \leq \|\Gamma_j(v) - \mathbb{E}\Gamma_j(v)\|_{L^2(\gamma; \text{HS})}.$$

Starting from (C.10) thus gives $\|\Gamma_k(v)\|_{L^2(\gamma; \text{HS})} \leq 2\varepsilon_T \|v\|$ for every ordering. Thus Lemma C.3, Minkowski, and Jensen imply

$$\|\text{Cum}_k(T(G))[v, \cdot^{k-1}]\|_{\text{HS}} \leq \sum_{\pi \in \mathfrak{S}_{k-1}} \|\mathbb{E}\Gamma_k^\pi(v)\|_{\text{HS}} \leq 2(k-1)! \varepsilon_T \|v\|.$$

Squaring and summing over an orthonormal basis proves (C.9). A standard approximation argument extends the bound to locally Lipschitz T . $\square$

C.3 Proofs of the second-order Wasserstein estimates

Proof of Lemma 4.11. By approximation, it is enough to treat smooth f . Throughout the proof, for functions of (z, w) , super- or sub-scripts 1 and 2 indicate operations in the first and second arguments, respectively. Thus, $P_s^{(2)}$ is the heat semigroup in the second argument, D_1 and D_2 are the corresponding derivatives, δ_1 is Gaussian row divergence in the first argument under $\mathbf{N}(0, I_m)$, and $\delta_{2,t}$ is Gaussian row divergence in the second argument under $\mathbf{N}(0, tI_n)$.

Let $(W_t)_{0 \leq t \leq 1}$ be Brownian motion in $\mathbb{R}^n$, independent of Z , and define

$$u_t(z, w) := P_{1-t}^{(2)} f(z, \cdot)(w), \quad X_t := Z + u_t(Z, W_t).$$

Thus, $X_0 = Z + \bar{f}(Z)$, while X_1 has the law of $Z + f(Z, G)$. Writing $(\mathcal{F}_t^W)_{0 \leq t \leq 1}$ for the Brownian filtration, the Markov property gives

$$u_t(z, W_t) = \mathbb{E}[f(z, W_1) \mid \mathcal{F}_t^W], \quad (0 \leq t \leq 1).$$

Consequently, $(X_t)_{0 \leq t \leq 1}$ is a martingale with respect to the enlarged filtration $(\sigma(Z) \vee \mathcal{F}_t^W)_{0 \leq t \leq 1}$. Since u solves the backward heat equation with respect to w , Itô's formula identifies its martingale representation as

$$dX_t = D_2 u_t(Z, W_t) dW_t.$$

Define

$$\Sigma_t(z, w) := D_2 u_t(z, w) D_2 u_t(z, w)^\top.$$

A second application of Itô's formula gives, for any smooth scalar test function φ ,

$$\partial_t \mathbb{E} \varphi(X_t) = \frac{1}{2} \mathbb{E} \langle \Sigma_t(Z, W_t), D^2 \varphi(X_t) \rangle_{\text{HS}}.$$

To estimate the Wasserstein distance, we recast this as a continuity equation. The chain rule gives

$$\begin{aligned} D_1 \nabla \varphi(z + u_t(z, w)) &= D^2 \varphi(z + u_t(z, w)) (I + D_1 u_t(z, w)), \\ D_2 \nabla \varphi(z + u_t(z, w)) &= D^2 \varphi(z + u_t(z, w)) D_2 u_t(z, w). \end{aligned}$$

Since Σ_t and $D^2 \varphi$ are symmetric, it follows that

$$\begin{aligned} \langle \Sigma_t(z, w), D^2 \varphi(z + u_t(z, w)) \rangle \\ = \langle \Sigma_t(z, w), D_1 \nabla \varphi(z + u_t(z, w)) \rangle - \langle D_1 u_t(z, w) D_2 u_t(z, w), D_2 \nabla \varphi(z + u_t(z, w)) \rangle. \end{aligned}$$

Here and in the sequel, all derivatives of u_t and Σ_t are understood as evaluated at (Z, W_t) . Applying the divergence adjoint identity (C.1) in each Gaussian variable gives

$$\mathbb{E} \langle \Sigma_t, D^2 \varphi(X_t) \rangle_{\text{HS}} = \mathbb{E} \langle \delta_1 \Sigma_t - \delta_{2,t}(D_1 u_t D_2 u_t), \nabla \varphi(X_t) \rangle.$$

It follows that

$$\partial_t \mathbb{E} \varphi(X_t) = \mathbb{E} \langle V_t, \nabla \varphi(X_t) \rangle, \quad V_t := \frac{1}{2} \{ \delta_1 \Sigma_t - \delta_{2,t}(D_1 u_t D_2 u_t) \}.$$

Consequently, the evolution of $\text{Law}(LX_t)$ satisfies the continuity equation

$$\partial_t \text{Law}(LX_t) + \text{div}(v_t \text{Law}(LX_t)) = 0,$$

where the velocity field is given by

$$v_t(y) := \mathbb{E}[LV_t \mid LX_t = y], \quad \|v_t\|_{L^2} \leq \|LV_t\|_{L^2}.$$

It remains to estimate this explicit velocity. The heat semigroup preserves the first derivative bounds, while (C.6) applied to the weak derivatives of f gives the following bounds:

$$\begin{aligned} \|D_1 u_t\|_{\text{op}} &\leq A, & \|D_2 u_t\|_{\text{op}} &\leq B, \\ \|D_1 D_2 u_t\|_{\text{HS}} \vee \|D_2 D_1 u_t\|_{\text{HS}} &\leq \frac{A\sqrt{m}}{\sqrt{1-t}}, & \|D_2^2 u_t\|_{\text{HS}} &\leq \frac{B\sqrt{m}}{\sqrt{1-t}}. \end{aligned}$$

Here, we used $\|D_1 f\|_{\text{HS}} \leq \sqrt{m} A$ and $\|D_2 f\|_{\text{HS}} \leq \sqrt{m} B$. The product rule and $\|RS\|_{\text{HS}} \leq \|R\|_{\text{op}} \|S\|_{\text{HS}}$ therefore give

$$\begin{aligned} \|\Sigma_t\|_{\text{HS}} &\leq B^2 \sqrt{m}, & \|D_1 \Sigma_t\|_{\text{HS}} &\leq \frac{2AB\sqrt{m}}{\sqrt{1-t}}, \\ \|D_1 u_t D_2 u_t\|_{\text{HS}} &\leq AB\sqrt{m}, & \|D_2(D_1 u_t D_2 u_t)\|_{\text{HS}} &\leq \frac{2AB\sqrt{m}}{\sqrt{1-t}}. \end{aligned}$$

Every column of $D_2 u_t$ lies in E , so $\text{range}(\Sigma_t) \subseteq E$. Since E is a fixed linear subspace, differentiating the columns of Σ_t in z preserves membership in E , and hence $\delta_1 \Sigma_t \in E$.

We now apply the Gaussian divergence bounds conditionally in each variable. First, (C.2) in the z -variable gives

$$\begin{aligned} \|L\delta_1\Sigma_t\|_{L^2} &= \|\delta_1(L\Sigma_t)\|_{L^2} \leq \|L\Sigma_t\|_{L^2(\text{HS})} + \|LD_1\Sigma_t\|_{L^2(\text{HS})} \\ &\leq \sqrt{m} \left\{ \|L\Pi_E\|_{\text{op}} B^2 + 2\|L\|_{\text{op}} \frac{AB}{\sqrt{1-t}} \right\}. \end{aligned}$$

Here we used $L\Sigma_t = L\Pi_E\Sigma_t$ in the first term. Next, the estimate (C.3) in the w -variable gives

$$\begin{aligned} \|L\delta_{2,t}(D_1u_tD_2u_t)\|_{L^2} &= \|\delta_{2,t}(LD_1u_tD_2u_t)\|_{L^2} \\ &\leq t^{-1/2} \|LD_1u_tD_2u_t\|_{L^2(\text{HS})} + \|LD_2(D_1u_tD_2u_t)\|_{L^2(\text{HS})} \\ &\leq \sqrt{m} \|L\|_{\text{op}} AB \{t^{-1/2} + 2(1-t)^{-1/2}\}. \end{aligned}$$

Combining these two estimates in the definition of V_t yields

$$\|LV_t\|_{L^2} \leq \sqrt{m} \left\{ \frac{1}{2} \|L\Pi_E\|_{\text{op}} B^2 + \frac{1}{2} \|L\|_{\text{op}} AB t^{-1/2} + 2\|L\|_{\text{op}} AB (1-t)^{-1/2} \right\}.$$

The first singularity comes from rescaling Gaussian divergence under $\mathbf{N}(0, tI_n)$, and the second from differentiating the heat extension.

Since $\int_0^1 t^{-1/2} dt = \int_0^1 (1-t)^{-1/2} dt = 2$, we have

$$W_2(\text{Law}(LX_\varepsilon), \text{Law}(LX_{1-\varepsilon})) \leq \int_\varepsilon^{1-\varepsilon} \|v_t(LX_t)\|_{L^2} dt \leq \sqrt{m} \left\{ \frac{1}{2} \|L\Pi_E\|_{\text{op}} B^2 + 5\|L\|_{\text{op}} AB \right\}.$$

We can then let $\varepsilon \searrow 0$. This proves (4.21). $\square$

Proof of Lemma 5.5. Use the notation and repeat the proof of Lemma 4.11 through the construction of u_t, X_t, Σ_t, V_t , and v_t , including all the displayed pointwise derivative estimates. Conditional expectation is now used as an L^q contraction. Applying (C.4) conditionally in the z -variable, and retaining separately its mean and derivative terms, gives

$$\|L\delta_1\Sigma_t\|_{L^q} \leq C\sqrt{m} \left\{ \sqrt{q} \|L\Pi_E\|_{\text{op}} B^2 + q\|L\|_{\text{op}} AB (1-t)^{-1/2} \right\}.$$

Its rescaled version in the w -variable similarly gives

$$\|L\delta_{2,t}(D_1u_tD_2u_t)\|_{L^q} \leq C\sqrt{m} \|L\|_{\text{op}} AB \left\{ \sqrt{q} t^{-1/2} + q(1-t)^{-1/2} \right\}.$$

Integrating these bounds proves (5.4). $\square$

C.4 Gaussian polynomial bounds

For this subsection, D denotes the gradient in $p \in \mathbb{R}^d$, and the Gaussian divergence is the operator defined above with x replaced by p .

Lemma C.5 (Gaussian polynomial estimates). *Let $r \geq 0$ be an integer. If $F : \mathbb{R}^d \rightarrow \mathcal{H}$ is a polynomial of total degree at most r , then*

$$\|DF\|_{L^2(\gamma; \text{HS})} \leq \sqrt{r} \|F\|_{L^2(\gamma)}.$$

If $V : \mathbb{R}^d \rightarrow \mathcal{H}^d$ is a polynomial vector field of total degree at most r , then

$$\|\delta_\gamma V\|_{L^2(\gamma)} \leq \sqrt{r+1} \|V\|_{L^2(\gamma; \text{HS})}.$$

These estimates are standard; see [Nua06, Propositions 1.2.2 and 1.3.1], restricting to Wiener chaoses of degree at most r . We use the following consequences of the sharp Gaussian divergence and Bernstein–Markov inequalities in [CCLZ26b].

Lemma C.6 (L^q Gaussian polynomial estimates). *Let $q \geq 2$ and let $r \geq 0$ be an integer. If $F : \mathbb{R}^d \rightarrow \mathcal{H}$ is a polynomial of total degree at most r , then*

$$\|DF\|_{L^q(\gamma; \mathcal{HS})} \leq C\sqrt{r} \|F - \mathbb{E}F\|_{L^q(\gamma; \mathcal{H})}.$$

If $V : \mathbb{R}^d \rightarrow \mathcal{H}^d$ is a polynomial vector field of total degree at most r , then

$$\|\delta_\gamma V\|_{L^q(\gamma; \mathcal{H})} \leq C\sqrt{q} \|\mathbb{E}V\|_{\mathcal{HS}} + Cq\sqrt{r} \|V - \mathbb{E}V\|_{L^q(\gamma; \mathcal{HS})} \leq C(\sqrt{q} + q\sqrt{r}) \|V\|_{L^q(\gamma; \mathcal{HS})}.$$

Proof. The scalar Bernstein–Markov bound is [CCLZ26b, Theorem 1.2]. Its Hilbert-valued consequence is obtained by applying it to $\langle G_F, F - \mathbb{E}F \rangle$, where G_F is a standard Gaussian in the finite-dimensional span $\mathcal{H}_F$ of the coefficients of F . Jensen’s inequality bounds the L^q norm of DF by the joint L^q norm of $D\langle G_F, F - \mathbb{E}F \rangle$; the Gaussian moment factor on the right cancels the scalar theorem’s denominator. The divergence estimate then follows from [CCLZ26b, Corollary 1.4], exactly as in the proof of [CCLZ26b, Corollary 1.7], now with Hilbert target $\mathcal{H}$. For $r = 0$, the derivative vanishes and the divergence is a centered Gaussian with L^q norm at most $C\sqrt{q} \|\mathbb{E}V\|_{\mathcal{HS}}$. $\square$

D Replacing proximal queries by gradient queries

D.1 Approximate proximal queries

For $0 < \eta < 1$ and $\varepsilon_{\text{prox}} \geq 0$, an $\varepsilon_{\text{prox}}$ -accurate proximal query at y returns a measurable point $\tilde{x}_y$ satisfying

$$\|\tilde{x}_y - \text{prox}_{\eta V}(y)\| \leq \varepsilon_{\text{prox}}. \quad (\text{D.1})$$

Given such a query and $G \sim \mathcal{N}(0, I)$, define

$$\hat{g}_{\eta, \varepsilon_{\text{prox}}}(y; G) := \nabla V(\tilde{x}_y + \sqrt{\eta} G).$$

The approximate proximal variant of Algorithm 3.1 replaces every occurrence of $\hat{g}_\eta$ by $\hat{g}_{\eta, \varepsilon_{\text{prox}}}$.

The proof of Theorem 5.1 in the main text assumes exact proximal queries. We now record the modifications needed when these calls are replaced by the approximate proximal queries from (D.1).

For every queried center y , let $\tilde{x}_y$ be the output of an abstract $\varepsilon_{\text{prox}}$ -accurate proximal oracle, and let $\hat{K}^{\varepsilon_{\text{prox}}}$ denote the resulting kernel.

Lemma D.1 (Stability under approximate proximal centers). *If $h^2 \leq 1$, then*

$$W_q(\delta_z \hat{K}^{\varepsilon_{\text{prox}}}, \delta_z \hat{K}) \leq \mathcal{E}_{\text{prox}} := Ch\varepsilon_{\text{prox}}, \quad q \geq 1. \quad (\text{D.2})$$

Consequently, for $J \geq 2$ and $h \leq c/(\kappa q)$, and every phase-space law $\hat{\mu}$,

$$W_{q, M_\kappa}^2(\hat{\mu}(\hat{K}^{\varepsilon_{\text{prox}}})^N, \Pi_\eta) \leq e^{-cNh/\kappa} W_{q, M_\kappa}^2(\hat{\mu}, \Pi_\eta) + \frac{C\kappa q}{h} \mathcal{E}_{\text{var}, q}^2 + \frac{C\kappa^2}{h^2} (\mathcal{E}_{\text{bias}}^2 + \mathcal{E}_{\text{disc}, q}^2 + \mathcal{E}_{\text{prox}}^2). \quad (\text{D.3})$$

Proof. For the same Gaussian G in the stochastic gradient, 1-smoothness and (D.1) imply

$$\|\nabla V(\tilde{x}_y + \sqrt{\eta} G) - \nabla V(x_y^+ + \sqrt{\eta} G)\| \leq \varepsilon_{\text{prox}}.$$

The first stochastic gradient block therefore moves every Picard node by at most $h^2 \varepsilon_{\text{prox}}$. The error in the second block stochastic gradient is at most $\varepsilon_{\text{prox}} + h^2 \varepsilon_{\text{prox}}$. This yields position error at most $Ch^2 \varepsilon_{\text{prox}}$ and momentum error at most $Ch\varepsilon_{\text{prox}}$, proving (D.2). Then, (D.3) by applying the triangle inequality and Young’s inequality in the one-step recurrence (5.2). $\square$

For the gradient-only implementation, set

$$\varepsilon_{\text{prox}} := c\mathfrak{L}_q^{-4}\sqrt{d}h^3. \quad (\text{D.4})$$

Since $\mathcal{E}_{\text{prox}} \leq Ch\varepsilon_{\text{prox}}$, we have

$$\frac{\kappa}{h}\mathcal{E}_{\text{prox}} \leq C\kappa\varepsilon_{\text{prox}} \leq C\mathfrak{L}_q^{9/2}\kappa\sqrt{d+q}h^3.$$

Thus (D.3) gives the same conclusion as the exact proximal proof of Theorem 5.1.

D.2 Gradient-only implementation

The Picard HMC algorithm in Algorithm 3.1 is stated using exact proximal queries, and Section 5.1 introduces its $\varepsilon_{\text{prox}}$ -accurate variant. This subsection implements every such approximate query using only evaluations of the gradient.

Fix a potential V satisfying $\kappa^{-1}I \preceq \nabla^2 V \preceq I$, a smoothing variance $0 < \eta \leq c_0 < 1$, and a center y . Write $x_y^+ := \text{prox}_{\eta V}(y)$.

Algorithm D.2 (Gradient descent). Given $y \in \mathbb{R}^d$ and $\varepsilon_{\text{prox}} > 0$, set $x^{(0)} := y$ and iterate

$$x^{(k+1)} := y - \eta \nabla V(x^{(k)})$$

until

$$\|x^{(k+1)} - x^{(k)}\| \leq (1 - \eta)\varepsilon_{\text{prox}}.$$

Return $\tilde{x}_y := x^{(k)}$.

Lemma D.3 (Certificate and pointwise work). *The iteration in Algorithm D.2 stops after finitely many steps and satisfies*

$$\|\tilde{x}_y - x_y^+\| \leq \varepsilon_{\text{prox}}. \quad (\text{D.5})$$

Its number $N_{\text{prox}}(y)$ of gradient evaluations obeys

$$N_{\text{prox}}(y) \leq C_{c_0} \left[1 + \log \left(1 + \frac{\|\nabla V(y)\|}{\varepsilon_{\text{prox}}} \right) \right]. \quad (\text{D.6})$$

Proof. The map $T_y := y - \eta \nabla V$ is an η -contraction, and its unique fixed point is x_y^+ . Therefore

$$\|x^{(k)} - x_y^+\| \leq \sum_{j=k}^{\infty} \|x^{(j+1)} - x^{(j)}\| \leq \frac{\|x^{(k+1)} - x^{(k)}\|}{1 - \eta},$$

which proves the certificate is valid. Moreover, $\|x^{(k+1)} - x^{(k)}\| \leq \eta^k \|x^{(1)} - x^{(0)}\| = \eta^{k+1} \|\nabla V(y)\|$. Using $\eta \leq c_0 < 1$ gives (D.6). $\square$

The pointwise bound depends on the queried center. To obtain the expected complexity claimed by the gradient-only sampler, we record that the centers encountered along the realized run have controlled second moments.

Lemma D.4 (Expected work along the Picard HMC run). *Under the parameter choices in (5.12) and (D.4), assume also that $h\mathfrak{L}_q \leq c/\kappa$ and $\eta \leq c_0$, and implement every $\varepsilon_{\text{prox}}$ -accurate proximal query in Algorithm 3.1 using Algorithm D.2. Every argument Y supplied to this routine obeys*

$$\mathbb{E}\|\nabla V(Y)\|^2 \leq C\kappa d\mathfrak{L}_q^{18}. \quad (\text{D.7})$$

Consequently the expected total number of gradient evaluations is at most

$$\mathbb{E}N_{\nabla V} \leq C \frac{\kappa}{h} \mathfrak{L}_q^3. \quad (\text{D.8})$$

Proof. Let $x_\star$ minimize V . By (D.3) with $q = 2$ and Lemma 4.16,

$$\sup_{j \leq N} \mathbb{E}[\|X_j - x_\star\|^2 + \|P_j\|^2] \leq C\kappa d \mathfrak{L}_q^{18}.$$

This also controls every layer 1 node. Since $x_{x_\star}^+ = x_\star$, non-expansiveness and (D.5) give $\|\tilde{x}_Y - x_\star\| \leq \|Y - x_\star\| + \varepsilon_{\text{prox}}$; hence 1-smoothness and Gaussian second moments control the gradients used to form layer 2. The bound $\max_{i \in [J]} \sum_{j \in [J]} |\omega_{i,j}| \leq h^2$ then controls every layer 2 node. Finally, $\|\nabla V(Y)\| \leq \|Y - x_\star\|$ at either layer, proving (D.7).

By Lemma D.3 and Jensen's inequality,

$$\mathbb{E}N_{\text{prox}}(Y) \leq C \left[1 + \log \left(1 + \frac{(\mathbb{E}\|\nabla V(Y)\|^2)^{1/2}}{\varepsilon_{\text{prox}}} \right) \right] \leq C \mathfrak{L}_q.$$

There are $2J$ proximal and $2J$ direct gradient calls per phase. Using $J \leq C \mathfrak{L}_q$ and $N \leq C\kappa \mathfrak{L}_q/h$ proves (D.8). $\square$

Proof of Theorem 5.1. Replace every proximal query in Algorithm 3.1 by Algorithm D.2 at the tolerance (D.4). The certificate in Lemma D.3 makes every call a valid $\varepsilon_{\text{prox}}$ -accurate proximal query. Then, Section D.1 gives the claimed W_q bound, and the query bound follows from Lemma D.4 after substituting the step size chosen in the main text proof. $\square$

E Extension to arbitrary Picard depth

The algorithm used in the main results takes Picard depth 2. In this appendix, we record the discretization error of Picard HMC using the exact smoothed gradient, at arbitrary depth.

Retain the Chebyshev–Lobatto nodes and coefficients from (3.6); their properties are summarized in Section B. Let $g_\eta := \nabla V_\eta$, let $\Pi_\eta := \pi_\eta \otimes \mathcal{N}(0, I)$, set $\Omega := (\omega_{i,j})_{i,j=1}^J$. Recall that

$$\max_{i \in [J]} \sum_{j \in [J]} |\omega_{i,j}| \leq h^2.$$

Algorithm E.1 (Depth- K exact Picard HMC phase). Given $(X_{\text{init}}, P_{\text{init}})$, draw independent $\zeta_0, \zeta_1 \sim \mathcal{N}(0, I)$, set $a_h := e^{-h/2}$, $\sigma_h := (1 - e^{-h})^{1/2}$, and define

$$\begin{aligned} P_0 &:= a_h P_{\text{init}} + \sigma_h \zeta_0, & X_{t_i}^{[0]} &:= X_{\text{init}} + t_i P_0, \\ X_{t_i}^{[\ell+1]} &:= X_{\text{init}} + t_i P_0 - \sum_{j \in [J]} \omega_{i,j} g_\eta(X_{t_j}^{[\ell]}), & 0 &\leq \ell < K-1. \end{aligned} \quad (\text{E.1})$$

Set

$$P_h^{[K]} := P_0 - \sum_{j \in [J]} \omega_j g_\eta(X_{t_j}^{[K-1]}), \quad X_h^{[K]} := X_{\text{init}} + h P_0 - \sum_{j \in [J]} \omega_{J,j} g_\eta(X_{t_j}^{[K-1]}),$$

and write the resulting next-phase initialization as

$$X_{\text{init}}^+ := X_h^{[K]}, \quad P_{\text{init}}^+ := a_h P_h^{[K]} + \sigma_h \zeta_1.$$

For $z = (X_{\text{init}}, P_{\text{init}})$ and $\zeta = (\zeta_0, \zeta_1)$, write $\Phi_z^{[J,K]}(\zeta) := (X_{\text{init}}^+, P_{\text{init}}^+)$ and $K_{\eta,h}^{[J,K]}$ for its kernel.

Theorem E.2 (Depth- K Picard HMC). *Suppose $\kappa \geq 1$, $0 < \eta \leq 1$, $V \in C^2(\mathbb{R}^d)$, and $\kappa^{-1}I \preceq \nabla^2 V \preceq I$. There are universal $c, C > 0$ such that, if*

$$h^2 \leq \frac{1}{4}, \quad h \leq \frac{c}{\kappa}, \quad (\text{E.2})$$

then, for every integer $K \geq 1$, every $q \geq 2$, every $z, z' \in \mathbb{R}^{2d}$, and every $\zeta \in \mathbb{R}^{2d}$,

$$W_{q, M_\kappa}(\delta_z K_{\eta, h}^{[J, K]}, \delta_{z'} K_{\eta, h}^{[J, K]}) \leq (1 - ch/\kappa) \|z - z'\|_{M_\kappa}, \quad (\text{E.3})$$

$$\|\{\Phi_z^{[J, K]}(\zeta) - z\} - \{\Phi_{z'}^{[J, K]}(\zeta) - z'\}\|_{M_\kappa} \leq Ch \|z - z'\|_{M_\kappa}, \quad (\text{E.4})$$

$$W_{q, M_\kappa}(\Pi_\eta K_{\eta, h}^{[J, K]}, \Pi_\eta) \leq C \sqrt{d+q} \left\{ B_{J, q} h \left(\frac{h}{\sqrt{\eta}} \right)^J + h^{2K+1} \right\}, \quad (\text{E.5})$$

where $B_{J, q} := C_0^{J+1} q^J \sqrt{J!}$, and C_0 is universal. In particular, $K = 2$ recovers [Proposition 5.9](#).

Proof. Contraction and coupling. As in the proof of [Proposition 4.7](#), couple inputs $z = (x_{\text{init}}, p_{\text{init}})$ and $z' = (x'_{\text{init}}, p'_{\text{init}})$ using the same Gaussian variables, and write $\Delta_0 := z - z' = (\Delta x, \Delta p)$, $\Delta_1 := \Phi_z^{[J, K]}(\zeta) - \Phi_{z'}^{[J, K]}(\zeta)$. Then,

$$\Delta X_{t_i}^{[0]} = \Delta x + a_h t_i \Delta p, \quad \Delta X_{t_i}^{[\ell+1]} = \Delta X_{t_i}^{[0]} - \sum_{j \in [J]} \omega_{i, j} H_j^{[\ell]} \Delta X_{t_j}^{[\ell]},$$

where

$$H_j^{[\ell]} := H(X_{t_j}^{[\ell]}, X_{t_j}'^{[\ell]}), \quad (2\kappa)^{-1}I \preceq H_j^{[\ell]} \preceq I.$$

Since $\max_{i \in [J]} \sum_{j \in [J]} |\omega_{i, j}| \leq h^2 \leq 1/4$, induction in ℓ gives

$$\max_{j \in [J]} \|\Delta X_{t_j}^{[K-1]} - \Delta X_{t_j}^{[0]}\| \leq Ch^2 \|\Delta_0\|.$$

Now set

$$H_j := H(X_{t_j}^{[K-1]}, X_{t_j}'^{[K-1]}), \quad H := h^{-1} \sum_{j \in [J]} \omega_j H_j.$$

The endpoint calculation in [Lemma 4.6](#) therefore applies unchanged and yields

$$\Delta_1 = (I + hA_H)\Delta_0 + \mathcal{R}_h \Delta_0, \quad \|\mathcal{R}_h\|_{\text{op}} \leq Ch^2.$$

The matrix inequality in the proof of [Proposition 4.7](#), followed by $h \leq c/\kappa$, proves (E.3). Moreover, subtracting Δ_0 from the expansion and using $\|A_H\|_{\text{op}} \leq C$ gives

$$\|\Delta_1 - \Delta_0\|_{M_\kappa} \leq C(h + h^2) \|\Delta_0\|_{M_\kappa} \leq Ch \|\Delta_0\|_{M_\kappa},$$

which is (E.4).

Discretization error. Start the phase from Π_η . The first OU half-refresh preserves this law. Let X_t , $0 \leq t \leq h$, be the position along the exact Hamiltonian trajectory, and put $g(t) := g_\eta(X_t)$. The interpolation estimate already proved in (5.11) gives

$$\mathcal{E}_{J, q} := \sup_{0 \leq t \leq h} \|g(t) - \mathcal{I}_{J, h} g(t)\|_{L^q} \leq B_{J, q} \sqrt{d+q} \left(\frac{h}{\sqrt{\eta}} \right)^J. \quad (\text{E.6})$$

Let $\mathbf{X} := (X_{t_j})_{j \in [J]}$ and $\mathbf{X}^{[0]} := (X_{t_j}^{[0]})_{j \in [J]}$, and use the norm given by the maximum over nodes of the L^q -norm. The exact Volterra equation and polynomial interpolation give

$$\mathbf{X} = \mathbf{X}^{[0]} - \Omega g_\eta(\mathbf{X}) + \varrho, \quad \|\varrho\| \leq \frac{h^2}{2} \mathcal{E}_{J,q}. \quad (\text{E.7})$$

The map $\mathcal{T} : y \mapsto \mathbf{X}^{[0]} - \Omega g_\eta(y)$ has Lipschitz constant at most h^2 , because g_η is 1-Lipschitz. Let $\bar{\mathbf{X}}$ be its fixed point. From (E.7),

$$\|\bar{\mathbf{X}} - \mathbf{X}\| \leq \frac{h^2}{2(1-h^2)} \mathcal{E}_{J,q}. \quad (\text{E.8})$$

The proof of Proposition 5.9 gives $\sup_{0 \leq t \leq h} \|g(t)\|_{L^q} \leq C\sqrt{d+q}$. Combining this with (E.8) and the fixed-point equation yields

$$\|\mathbf{X}^{[0]} - \bar{\mathbf{X}}\| \leq Ch^2\sqrt{d+q} + Ch^2\mathcal{E}_{J,q}.$$

The recursion (E.1) is precisely the iteration of $\mathcal{T}$, starting from $\mathbf{X}^{[0]}$. Hence

$$\|\mathbf{X}^{[K-1]} - \bar{\mathbf{X}}\| \leq Ch^{2K} \{\sqrt{d+q} + \mathcal{E}_{J,q}\}. \quad (\text{E.9})$$

The exact Hamiltonian endpoint satisfies

$$X_h = X_{\text{init}} + hP_0 - \int_0^h (h-t)g(t) dt, \quad P_h = P_0 - \int_0^h g(t) dt.$$

The quadrature rules integrate $\mathcal{I}_{J,h}g$ exactly. Use (E.8)–(E.9), 1-Lipschitzness of g_η , and the coefficient bounds in (B.2). The momentum estimate dominates and gives

$$\|(X_{\text{init}}^+, P_{\text{init}}^+) - (X_h, P_h^+)\|_{L^q, M_\kappa} \leq C \{h\mathcal{E}_{J,q} + h^{2K+1}\sqrt{d+q}\}. \quad (\text{E.10})$$

Here P_h^+ includes the common final OU half-refresh. The terms in (E.9) containing $\mathcal{E}_{J,q}$ are absorbed into the first term because $h^2 \leq 1/4$ and $h \leq 1$. Substituting (E.6) into (E.10) proves (E.5). $\square$

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